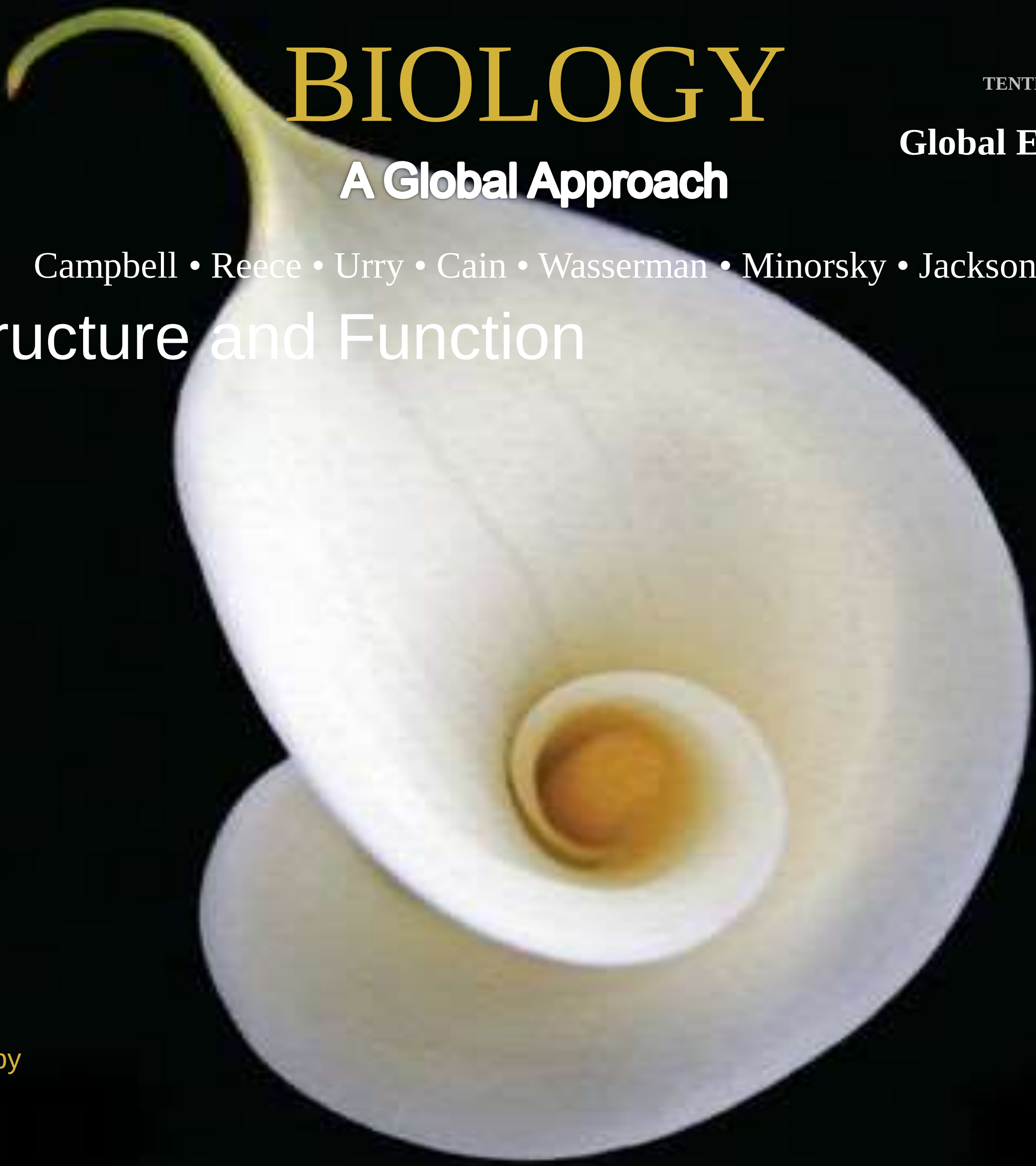


BIOLOGY

TENTH EDITION

Global Edition

A Global Approach

Campbell • Reece • Urry • Cain • Wasserman • Minorsky • Jackson

7 Cell Structure and Function

Lecture Presentation by
Nicole Tunbridge and
Kathleen Fitzpatrick

The Fundamental Units of Life

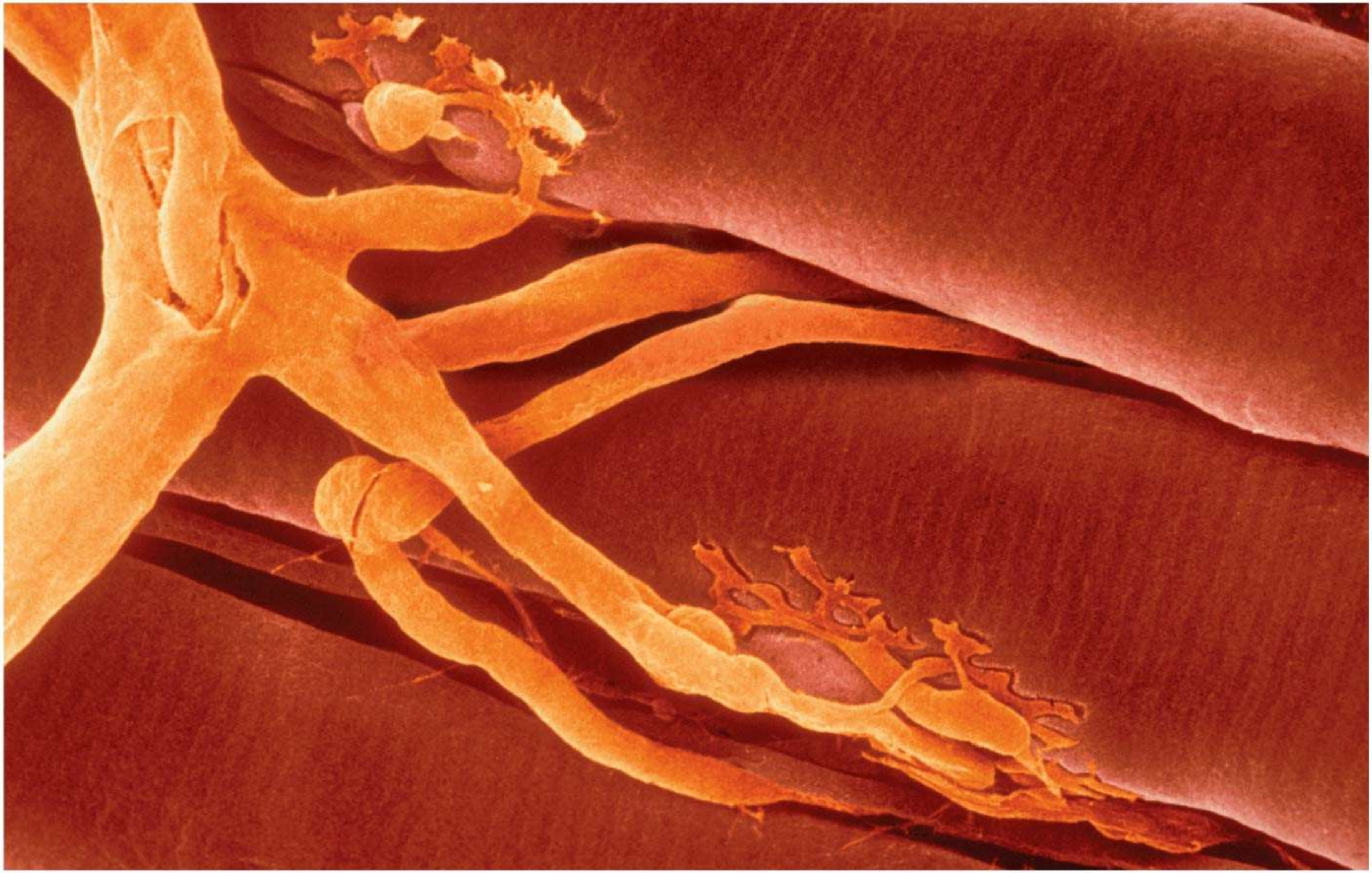
a) All organisms are made of cells

b) The cell is the simplest collection of matter that can be alive

c) All cells are related by their descent from earlier cells

d) Cells can differ substantially from one another but share common features

Figure 6.1



Concept 6.1: Biologists use microscopes and the tools of biochemistry to study cells

a) Cells are usually too small to be seen by the naked eye

Microscopy

a) Microscopes are used to visualize cells

b) In a **light microscope (LM)**, visible light is passed through a specimen and then through glass lenses

c) Lenses refract (bend) the light, so that the image is magnified

a) Three important parameters of microscopy

   a) *Magnification*, the ratio of an object's image size to its real size

   b) *Resolution*, the measure of the clarity of the image, or the minimum distance of two distinguishable points

   c) *Contrast*, visible differences in brightness between parts of the sample

Figure 6.2

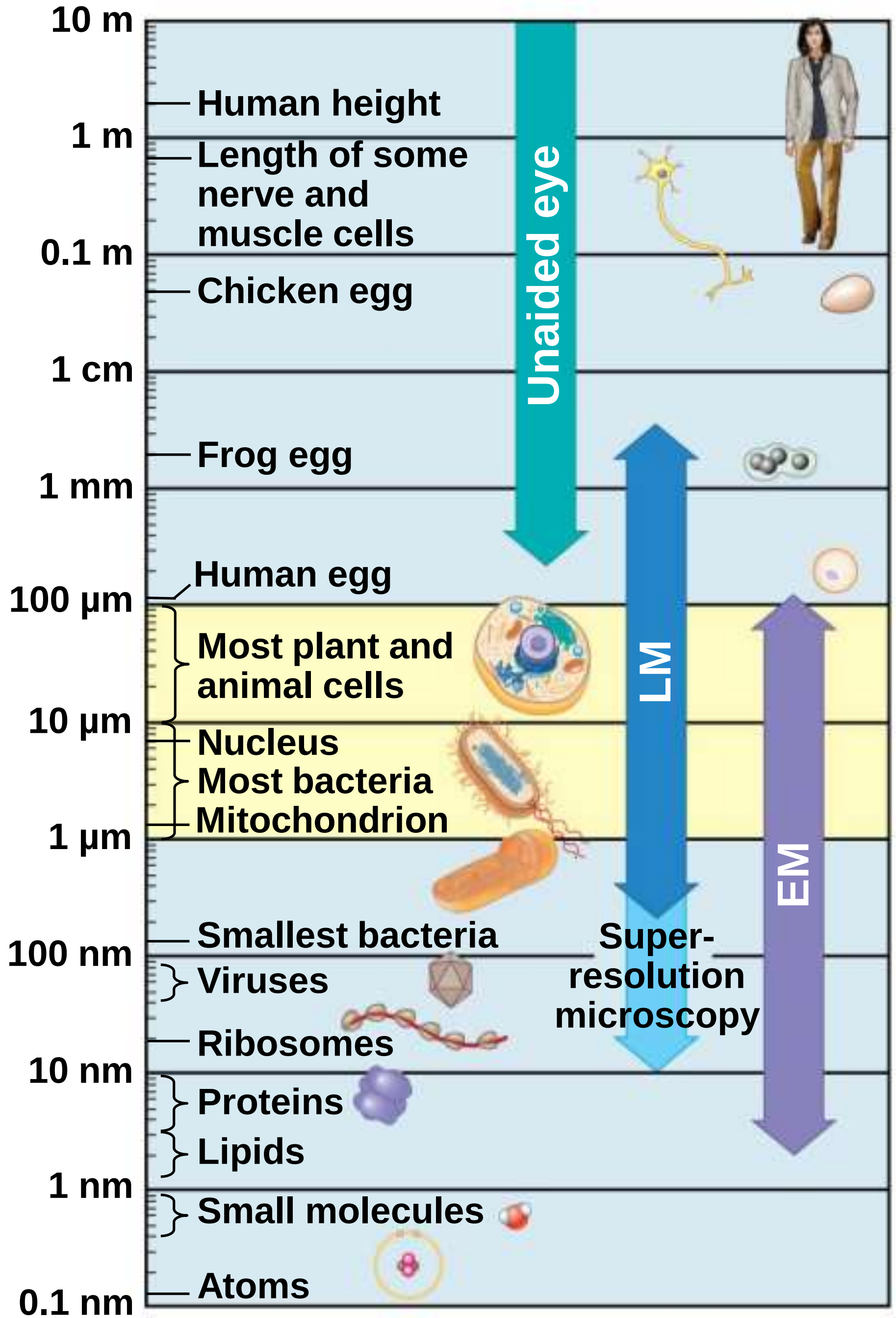
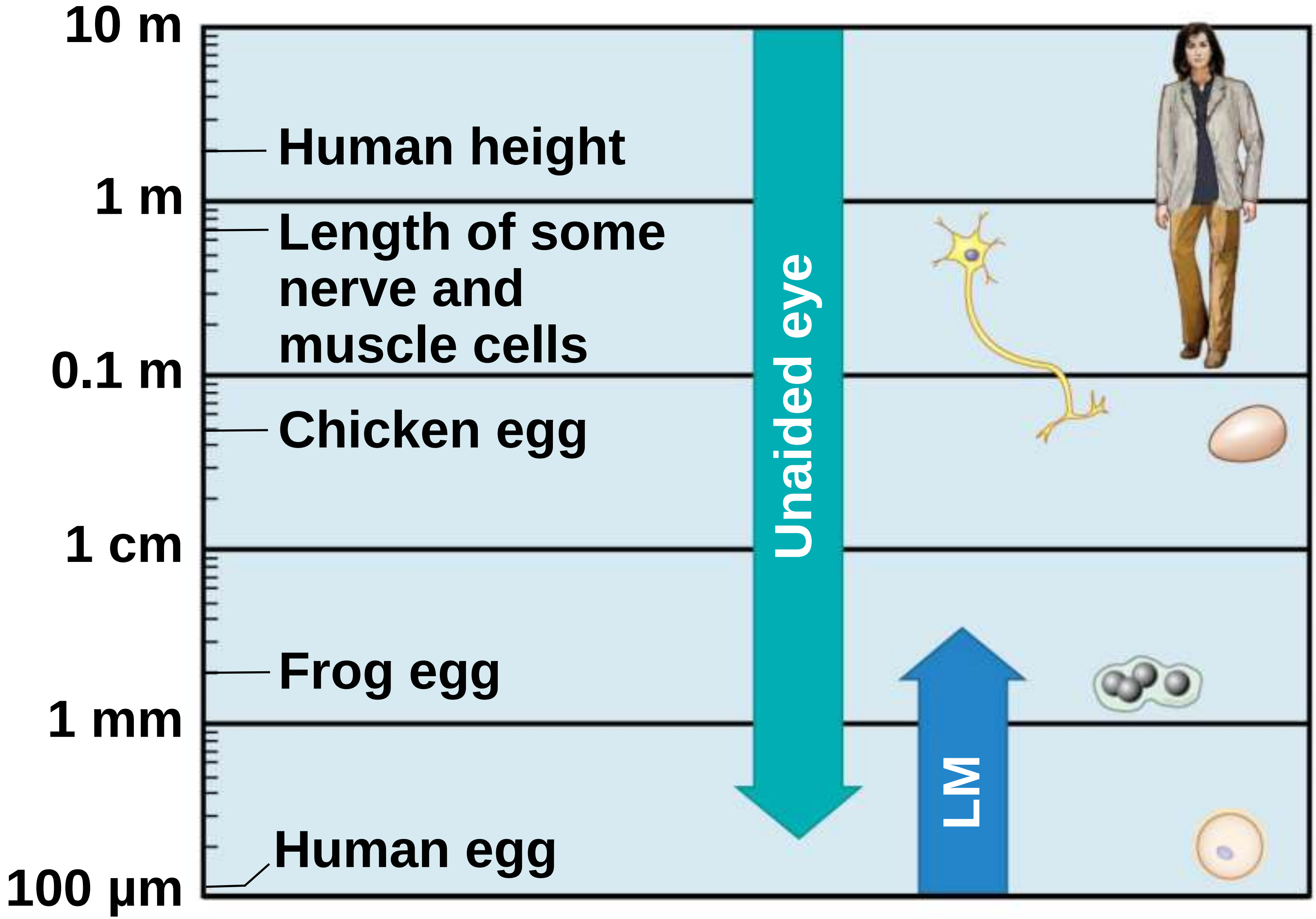

10 m
1 m
0.1 m
1 cm
1 mm
100 µm
10 µm
1 µm
100 nm
10 nm
1 nm
0.1 nm
Human height
Length of some nerve and muscle cells
Chicken egg
Frog egg
Human egg
Most plant and animal cells
Nucleus
Most bacteria
Mitochondrion
Smallest bacteria
Viruses
Ribosomes
Proteins
Lipids
Small molecules
Atoms
Unaided eye
LM
EM
Super-resolution microscopy

Figure 6.2a



10 m
1 m
0.1 m
1 cm
1 mm
100 µm
Human height
Length of some nerve and muscle cells
Chicken egg
Frog egg
Human egg
Unaided eye
LM

Figure 6.2b

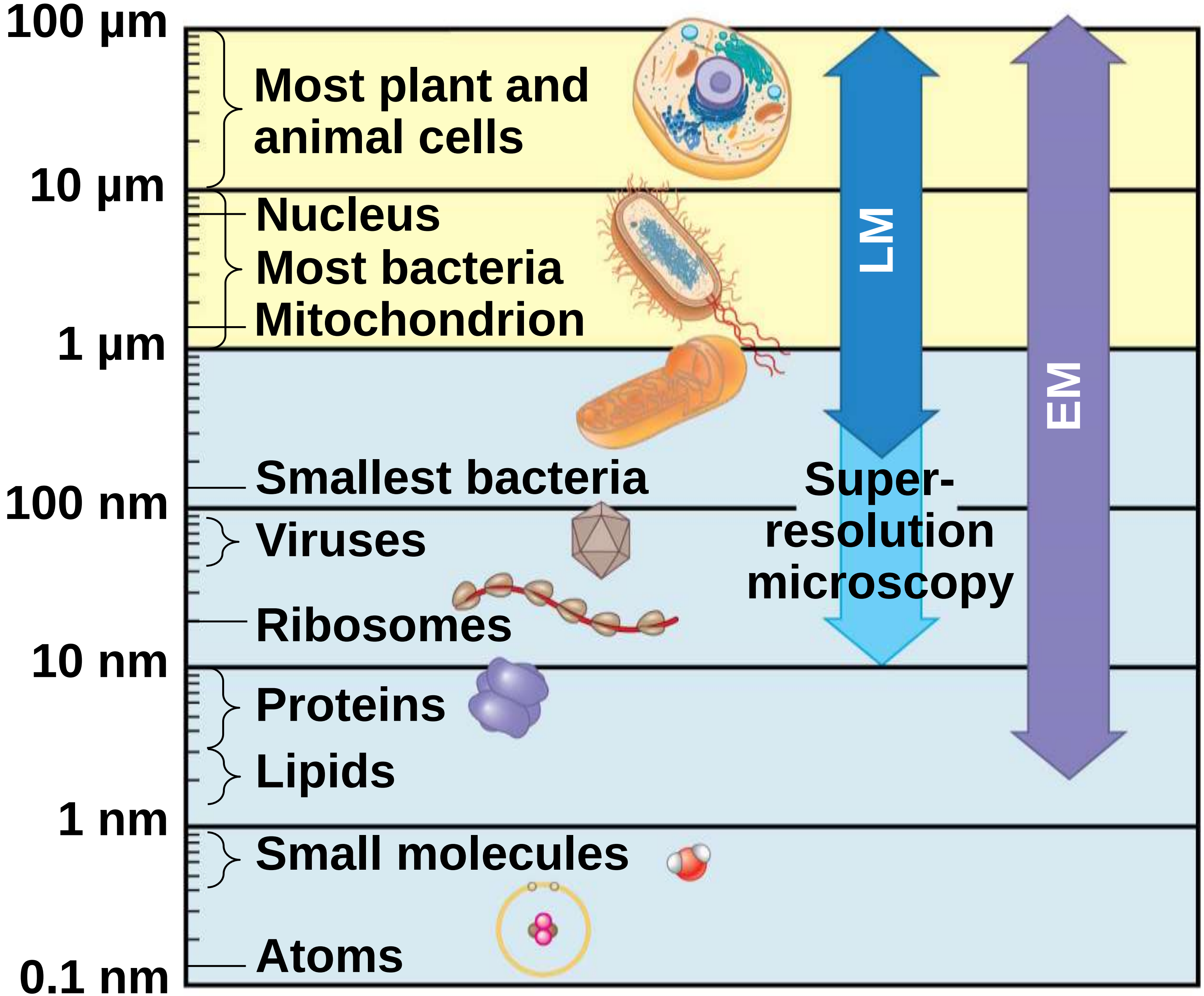
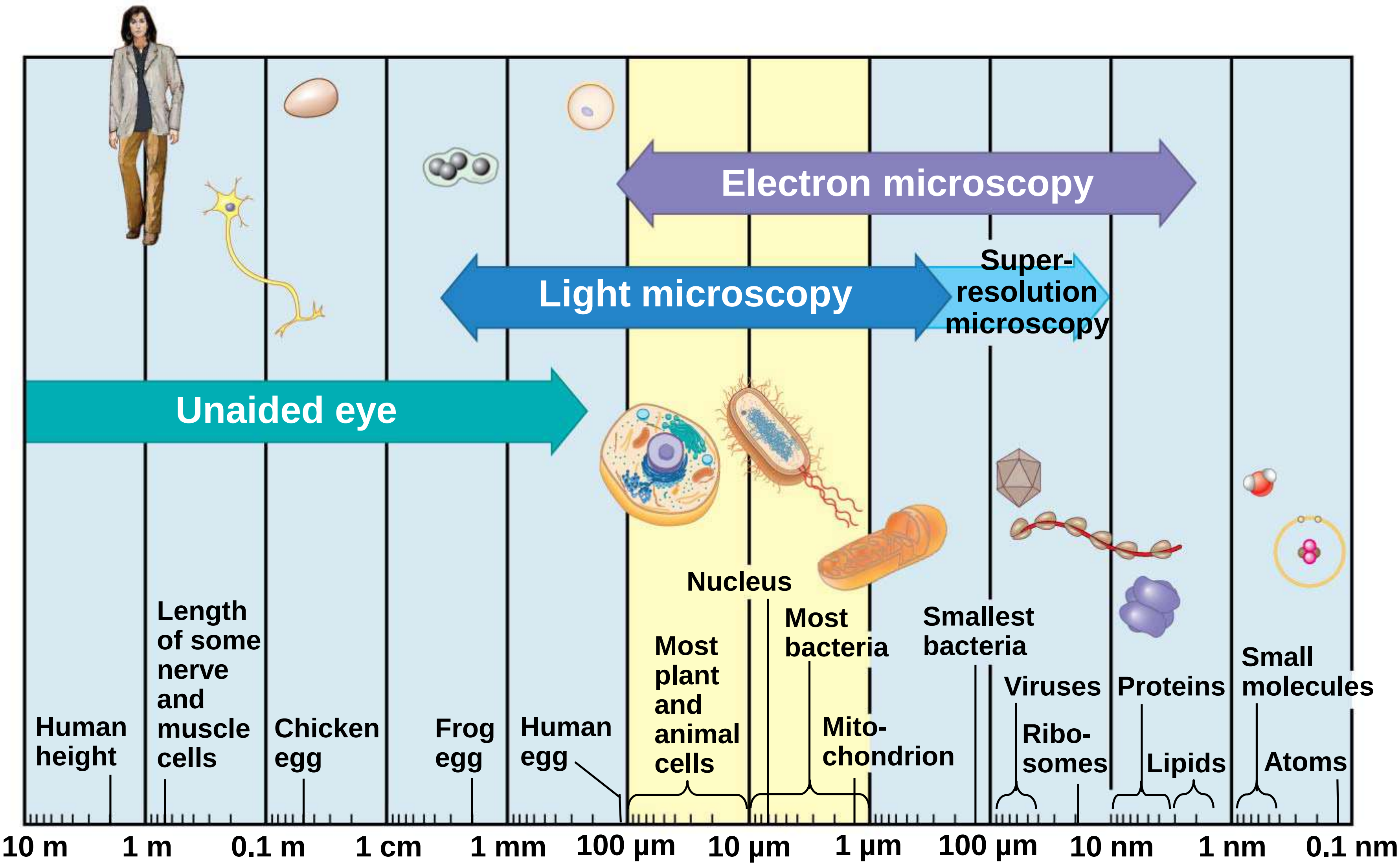

100 µm
Most plant and animal cells
10 µm
Nucleus
Most bacteria
Mitochondrion
1 µm
Smallest bacteria
100 nm
Viruses
Ribosomes
10 nm
Proteins
Lipids
1 nm
Small molecules
Atoms
0.1 nm
LM
EM
Super-resolution microscopy

Figure 6.2c



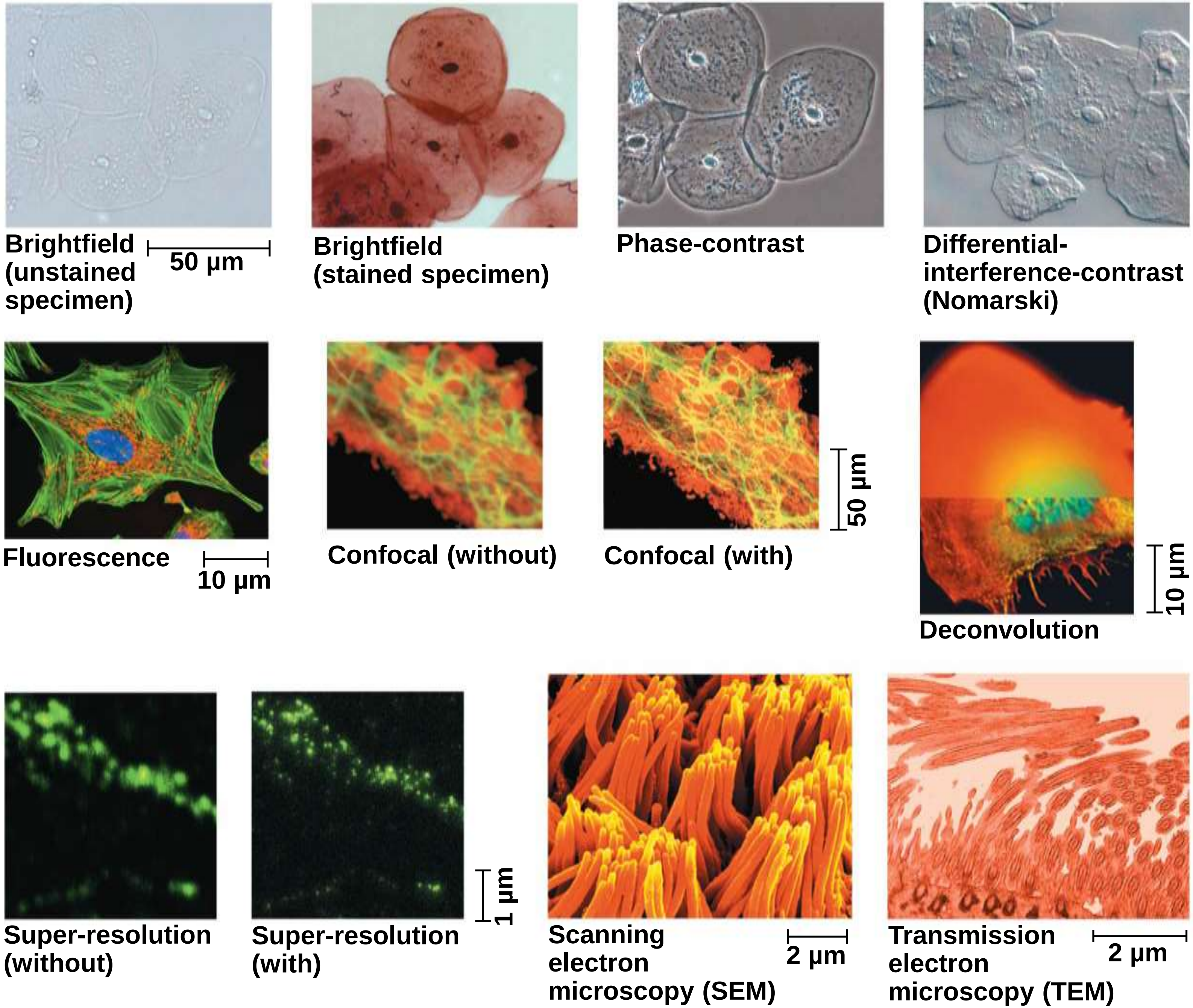
Electron microscopy
Light microscopy
Super-resolution microscopy
Unaided eye
Nucleus
Length of some nerve and muscle cells
Most plant and animal cells
Most bacteria
Smallest bacteria
Small molecules
Viruses
Proteins
Human height
Chicken egg
Frog egg
Human egg
Mito-chondrion
Ribo-somes
Lipids
Atoms
10 m
1 m
0.1 m
1 cm
1 mm
100 μm
10 μm
1 μm
100 μm
10 nm
1 nm
0.1 nm

a) Light microscopes can magnify effectively to about 1,000 times the size of the actual specimen with a resolution ~ 200nm

b) Various techniques enhance contrast and enable cell components to be stained or labeled

c) The resolution of standard light microscopy is too low to study **organelles**, the membrane-enclosed structures in eukaryotic cells

Figure 6.3


Brightfield (unstained specimen)
50 µm
Brightfield (stained specimen)
Phase-contrast
Differential-interference-contrast (Nomarski)
Fluorescence
10 µm
Confocal (without)
Confocal (with)
50 µm
Deconvolution
10 µm
Super-resolution (without)
Super-resolution (with)
1 µm
Scanning electron microscopy (SEM)
2 µm
Transmission electron microscopy (TEM)
2 µm

a) Two basic types of **electron microscopes (EMs)** are used to study subcellular structures; with a practical resolution of ~ 2nm

b) Scanning electron microscopes (SEMs) focus a beam of electrons onto the surface of a specimen, providing images that look 3-D

c) Transmission electron microscopes (TEMs) focus a beam of electrons through a specimen

d) TEMs are used mainly to study the internal structure of cells

a) Recent advances in light microscopy

    a) Confocal microscopy and deconvolution microscopy provide sharper images of three-dimensional tissues and cells

    b) New techniques for labeling cells improve resolution

Cell Fractionation

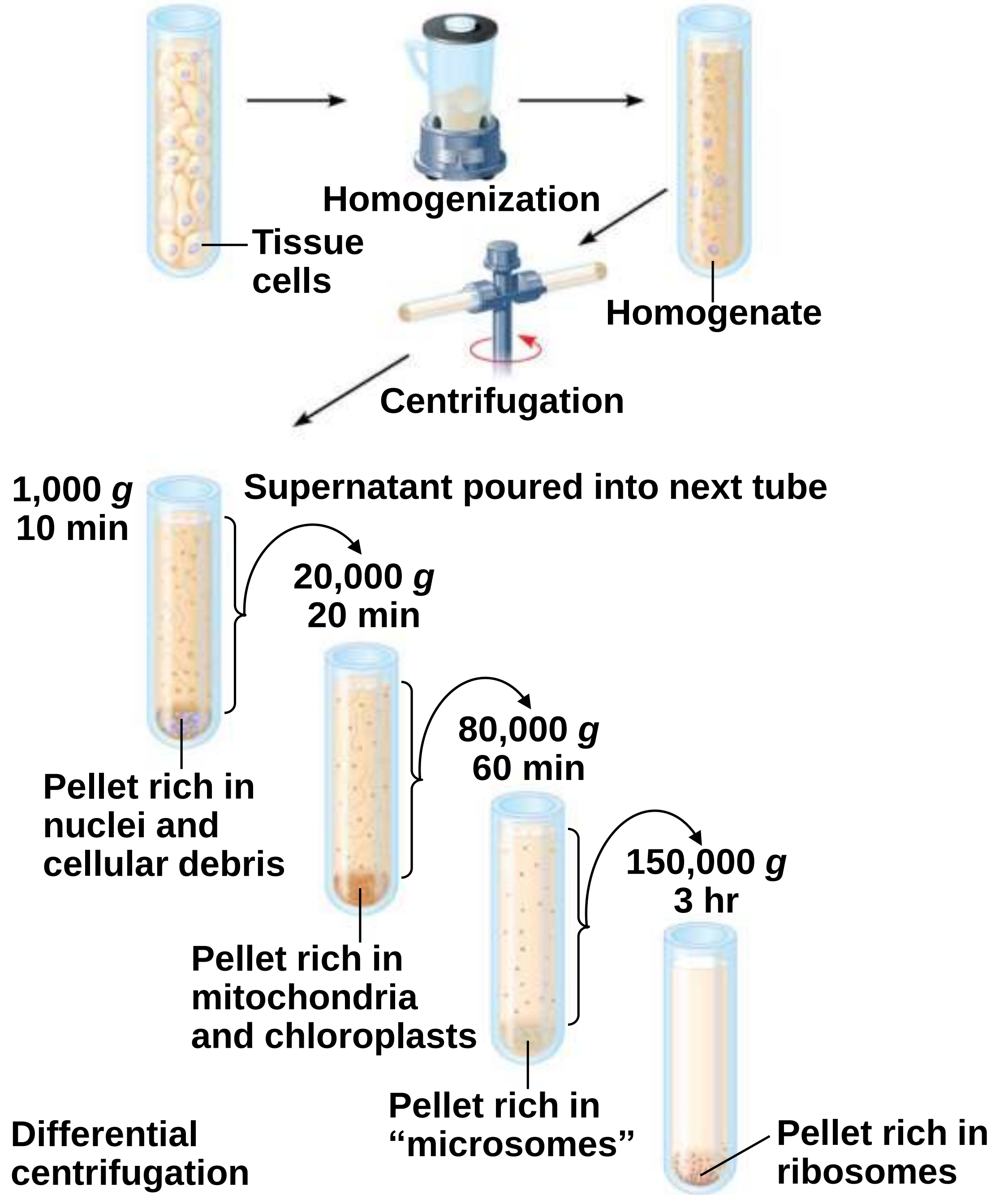
a) **Cell fractionation** takes cells apart and separates the major organelles from one another

b) Centrifuges fractionate cells into their component parts

c) Cell fractionation enables scientists to determine the functions of organelles

d) Biochemistry and cytology help correlate cell function with structure

Figure 6.4



Homogenization
Tissue cells
Homogenate
Centrifugation
1,000 g
10 min
Supernatant poured into next tube
20,000 g
20 min
80,000 g
60 min
150,000 g
3 hr
Pellet rich in nuclei and cellular debris
Pellet rich in mitochondria and chloroplasts
Pellet rich in "microsomes"
Pellet rich in ribosomes
Differential centrifugation

Figure 6.4a

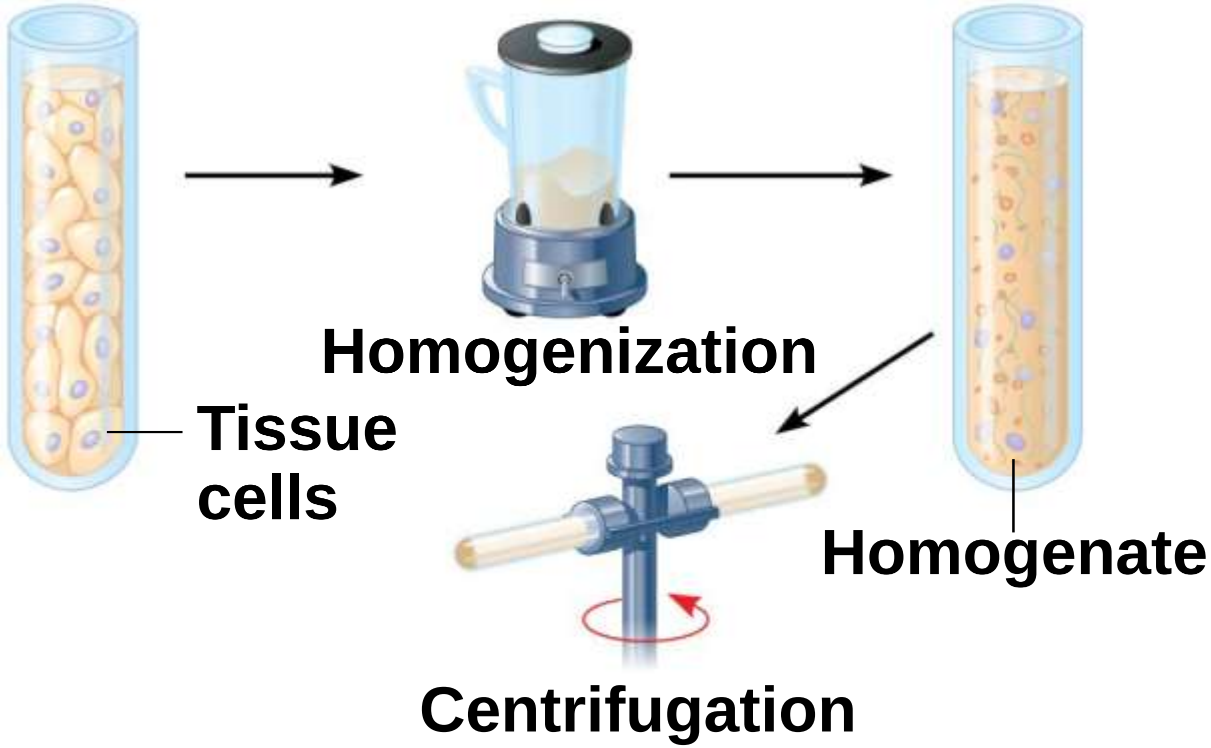
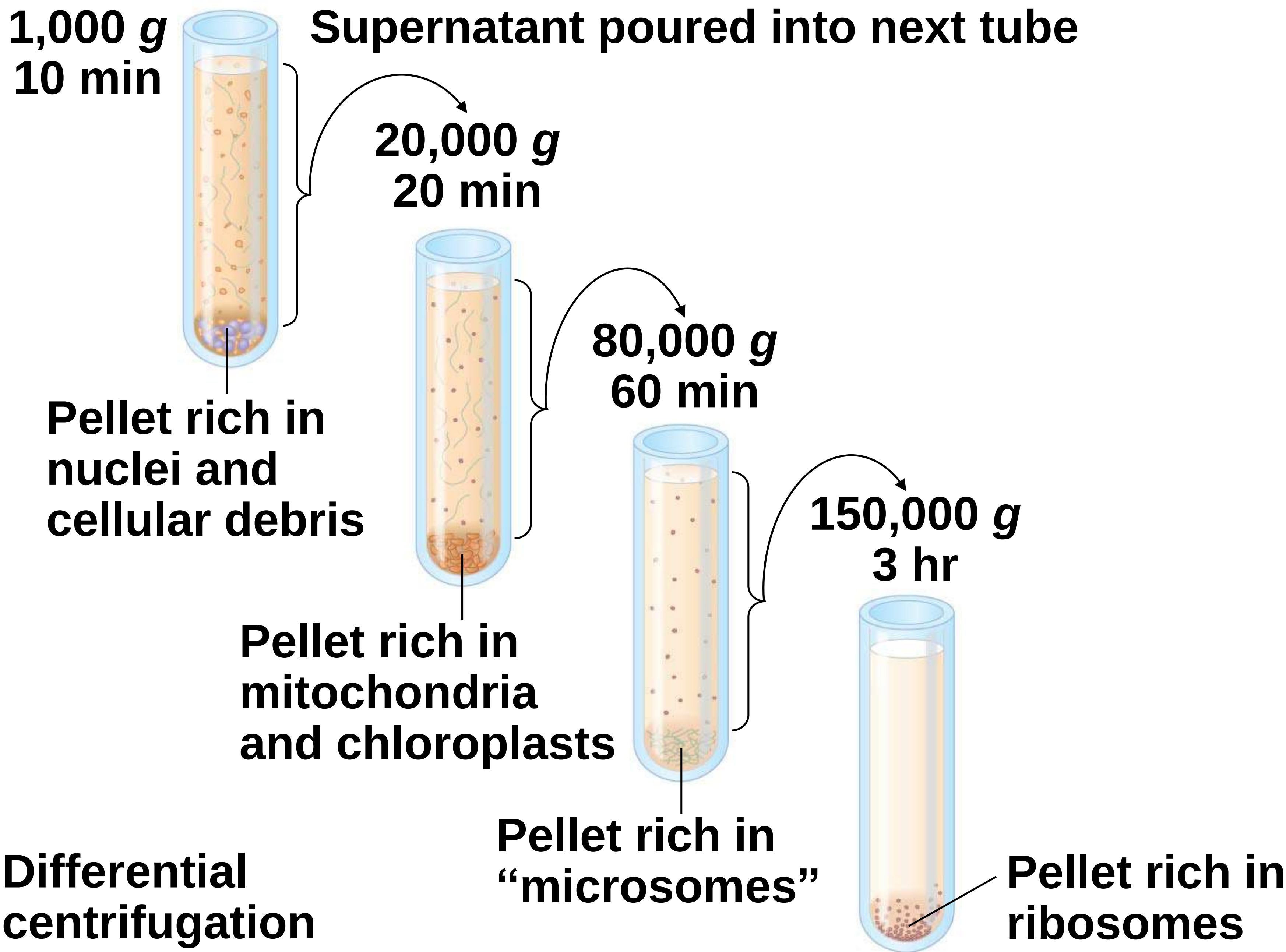


Figure 6.4b



1,000 *g*
10 min
Supernatant poured into next tube
20,000 *g*
20 min
80,000 *g*
60 min
150,000 *g*
3 hr
Pellet rich in nuclei and cellular debris
Pellet rich in mitochondria and chloroplasts
Pellet rich in “microsomes”
Pellet rich in ribosomes
Differential centrifugation

Concept 6.2: Eukaryotic cells have internal membranes that compartmentalize their functions

a) The basic structural and functional unit of every organism is one of two types of cells: prokaryotic or eukaryotic

b) Only organisms of the domains Bacteria and Archaea consist of prokaryotic cells

c) Protists, fungi, animals, and plants all consist of eukaryotic cells

Comparing Prokaryotic and Eukaryotic Cells

a) Basic features of all cells
   a) Plasma membrane
   b) Semifluid substance called **cytosol**
   c) Chromosomes (carry genes)
   d) Ribosomes (make proteins)

a) **Prokaryotic cells** are characterized by having

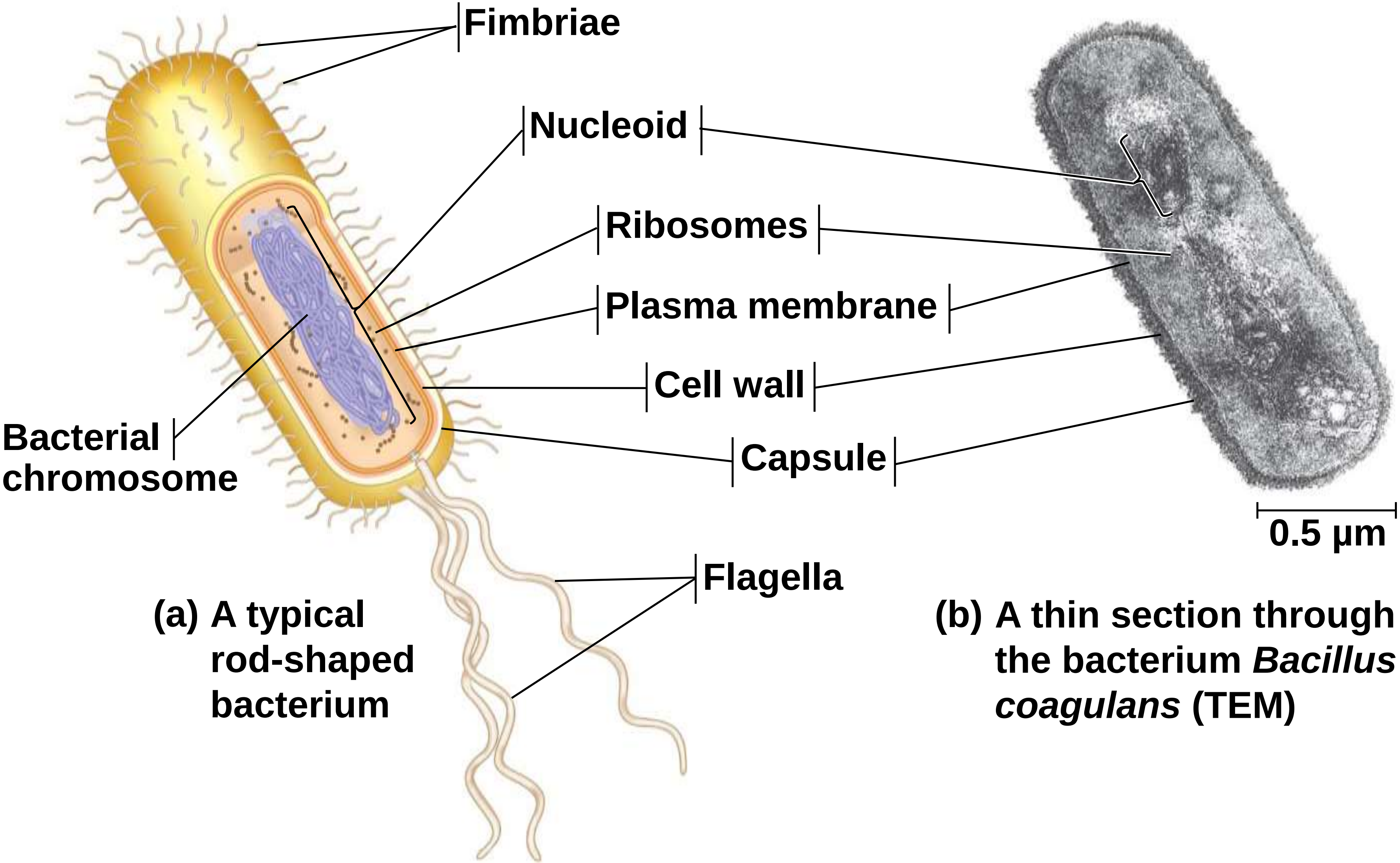
   a) No nucleus

   b) DNA in an unbound region called the **nucleoid**

   c) No membrane-bound organelles

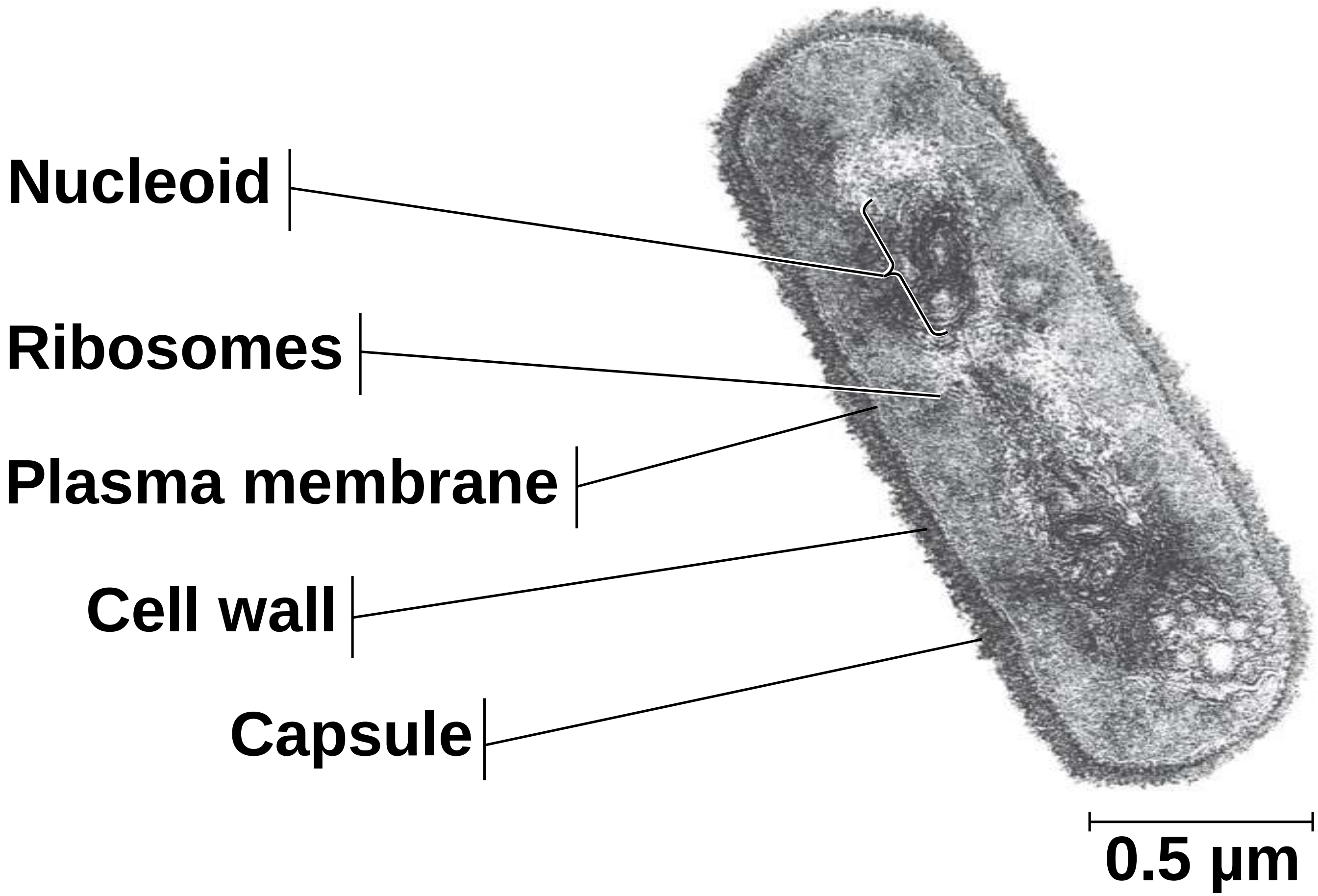
   d) Cytoplasm bound by the plasma membrane

Figure 6.5



(a) A typical rod-shaped bacterium

(b) A thin section through the bacterium *Bacillus coagulans* (TEM)

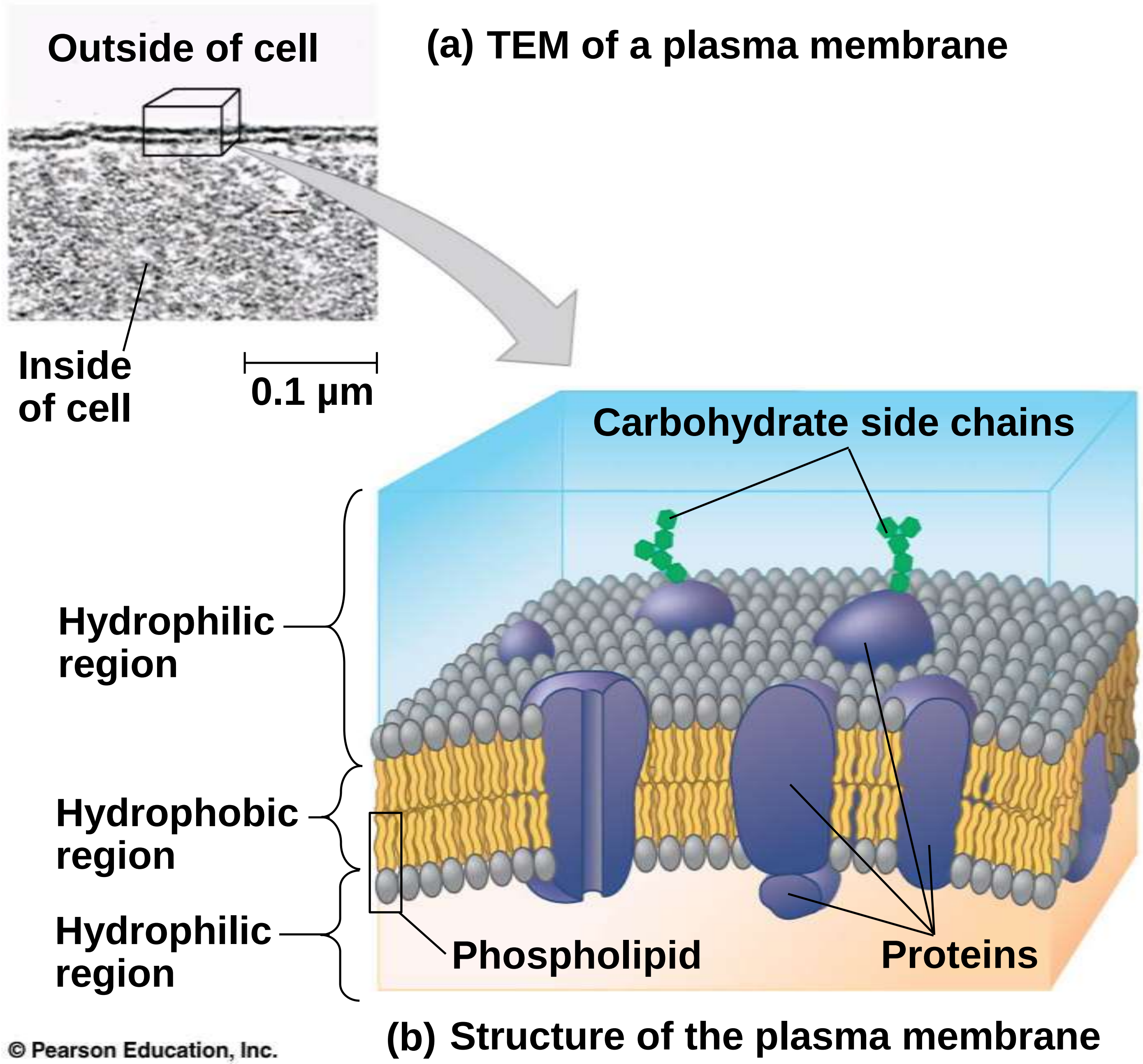


(b) A thin section through the bacterium *Bacillus coagulans* (TEM)

a) **Eukaryotic cells** are characterized by having
   a) DNA in a nucleus that is bounded by a membranous nuclear envelope
   b) Membrane-bound organelles
   c) Cytoplasm in the region between the plasma membrane and nucleus
b) Eukaryotic cells are generally much larger than prokaryotic cells

a) The **plasma membrane** is a selective barrier that allows sufficient passage of oxygen, nutrients, and waste to service the volume of every cell

Figure 6.6



Outside of cell
(a) TEM of a plasma membrane
Inside of cell
0.1 µm
Carbohydrate side chains
Hydrophilic region
Hydrophobic region
Hydrophilic region
Phospholipid
Proteins
(b) Structure of the plasma membrane
© Pearson Education, Inc.

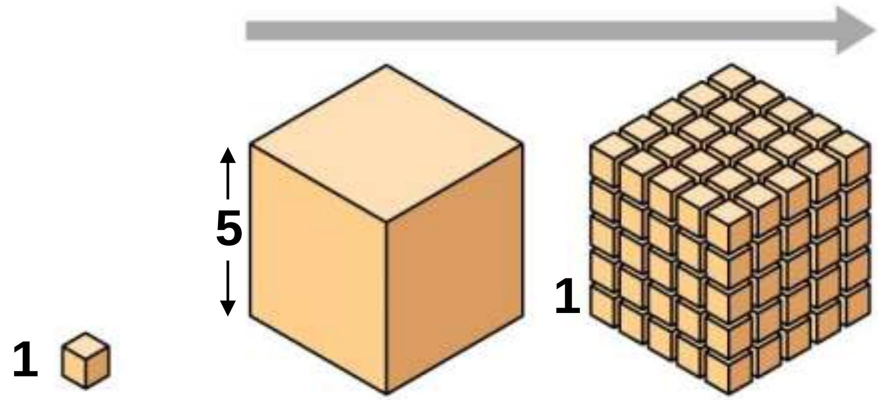
a) Metabolic requirements set upper limits on the size of cells

b) The surface area to volume ratio of a cell is critical

c) As a cell increases in size, its volume grows proportionately more than its surface area

Figure 6.7

Surface area increases while total volume remains constant



Total surface area [sum of the surface areas (height × width) of all box sides × number of boxes]	6	150	750
Total volume [height × width × length × number of boxes]	1	125	125
Surface-to-volume (S-to-V) ratio [surface area ÷ volume]	6	1.2	6

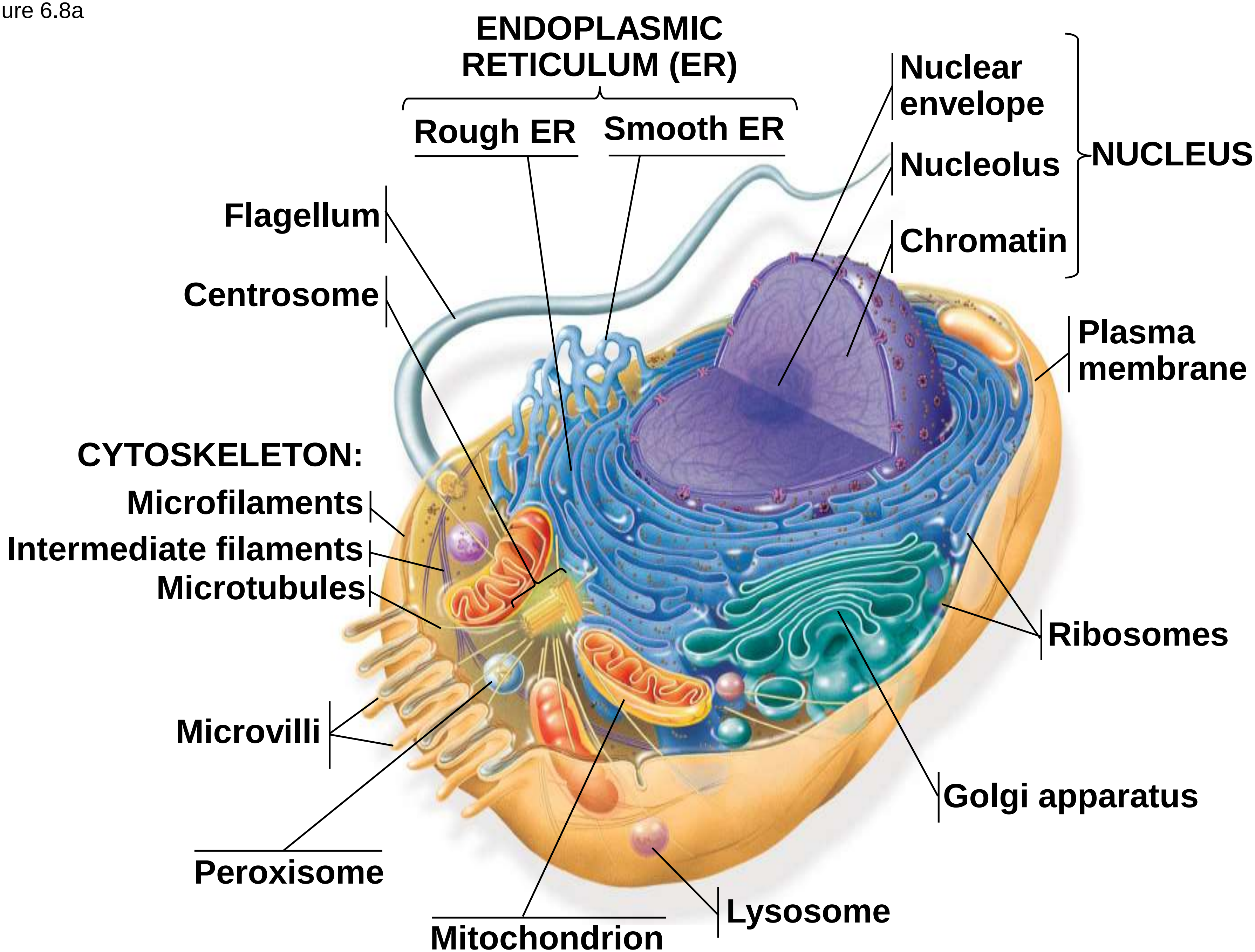
A Panoramic View of the Eukaryotic Cell

a) A eukaryotic cell has internal membranes that partition the cell into organelles

b) The basic fabric of biological membranes is a double layer of phospholipids and other lipids

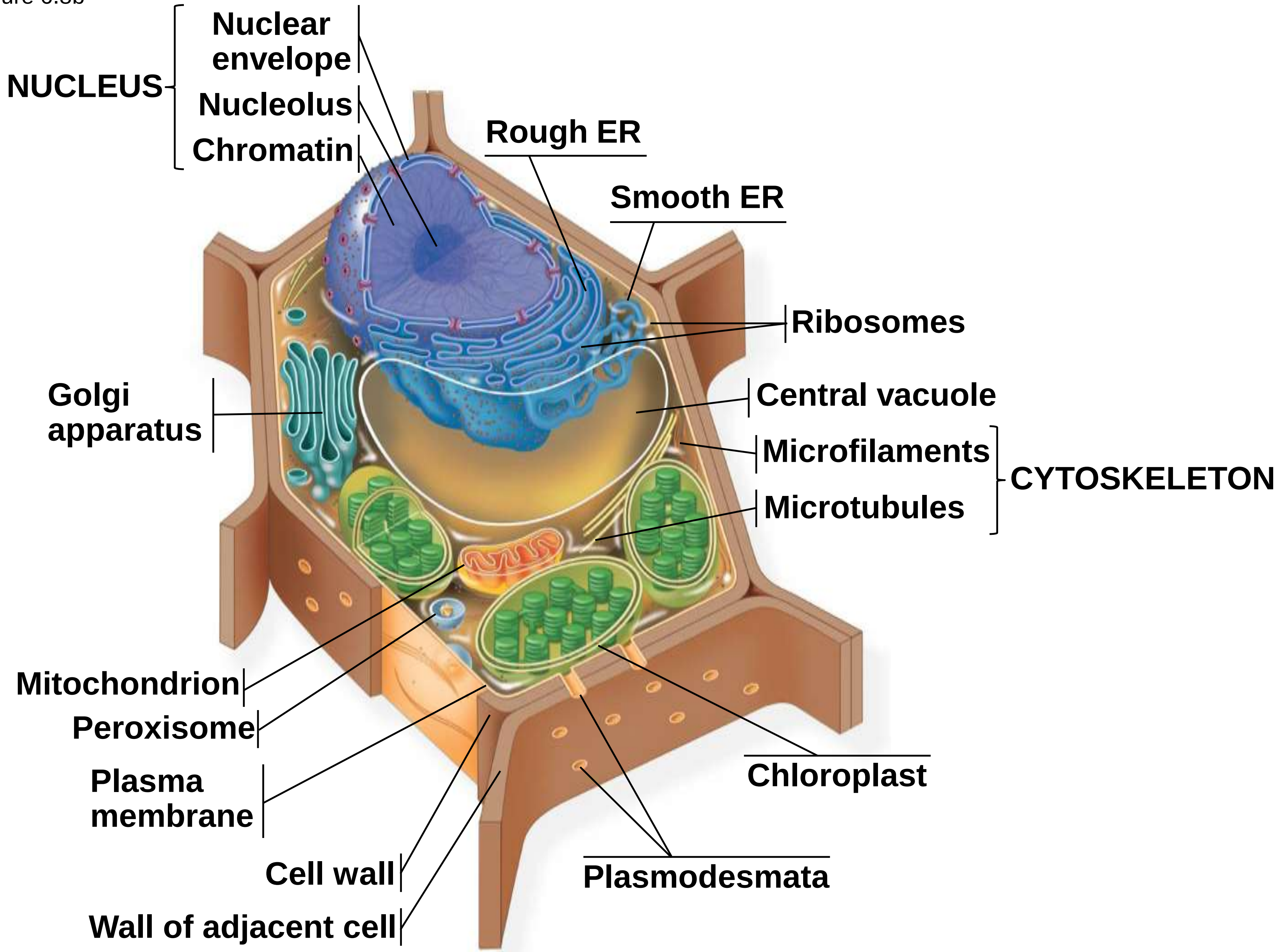
c) Plant and animal cells have most of the same organelles

Figure 6.8a



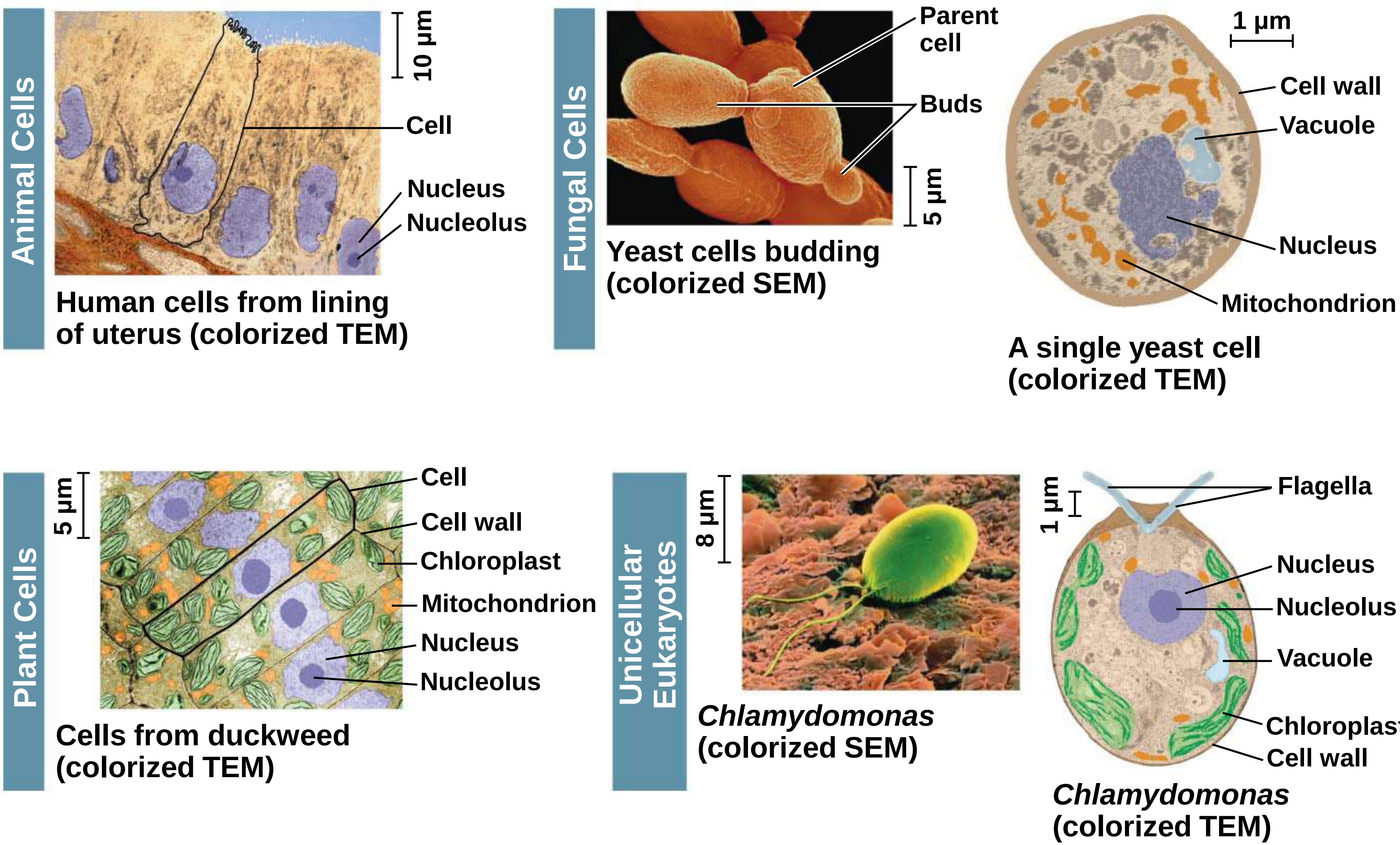
ENDOPLASMIC RETICULUM (ER)
Rough ER
Smooth ER
Nuclear envelope
Nucleolus
Chromatin
NUCLEUS
Flagellum
Centrosome
Plasma membrane
CYTOSKELETON:
Microfilaments
Intermediate filaments
Microtubules
Ribosomes
Microvilli
Golgi apparatus
Peroxisome
Lysosome
Mitochondrion

Figure 6.8b



NUCLEUS
Nuclear envelope
Nucleolus
Chromatin
Rough ER
Smooth ER
Ribosomes
Golgi apparatus
Central vacuole
Microfilaments
Microtubules
CYTOSKELETON
Mitochondrion
Peroxisome
Plasma membrane
Cell wall
Wall of adjacent cell
Chloroplast
Plasmodesmata

Figure 6.8c


Animal Cells
10 µm
Cell
Nucleus
Nucleolus
Human cells from lining of uterus (colorized TEM)
Fungal Cells
Parent cell
Buds
5 µm
Yeast cells budding (colorized SEM)
1 µm
Cell wall
Vacuole
Nucleus
Mitochondrion
A single yeast cell (colorized TEM)
Plant Cells
5 µm
Cell
Cell wall
Chloroplast
Mitochondrion
Nucleus
Nucleolus
Cells from duckweed (colorized TEM)
Unicellular Eukaryotes
8 µm
Chlamydomonas (colorized SEM)
1 µm
Flagella
Nucleus
Nucleolus
Vacuole
Chloroplast
Cell wall
Chlamydomonas (colorized TEM)

Concept 6.3: The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes

a) The nucleus contains most of the DNA in a eukaryotic cell

b) Ribosomes use the information from the DNA to make proteins

The Nucleus: Information Central

a) The **nucleus** contains most of the cell's genes and is usually the most conspicuous organelle

b) The **nuclear envelope** encloses the nucleus, separating it from the cytoplasm

c) The nuclear membrane is a double membrane; each membrane consists of a lipid bilayer

Figure 6.9

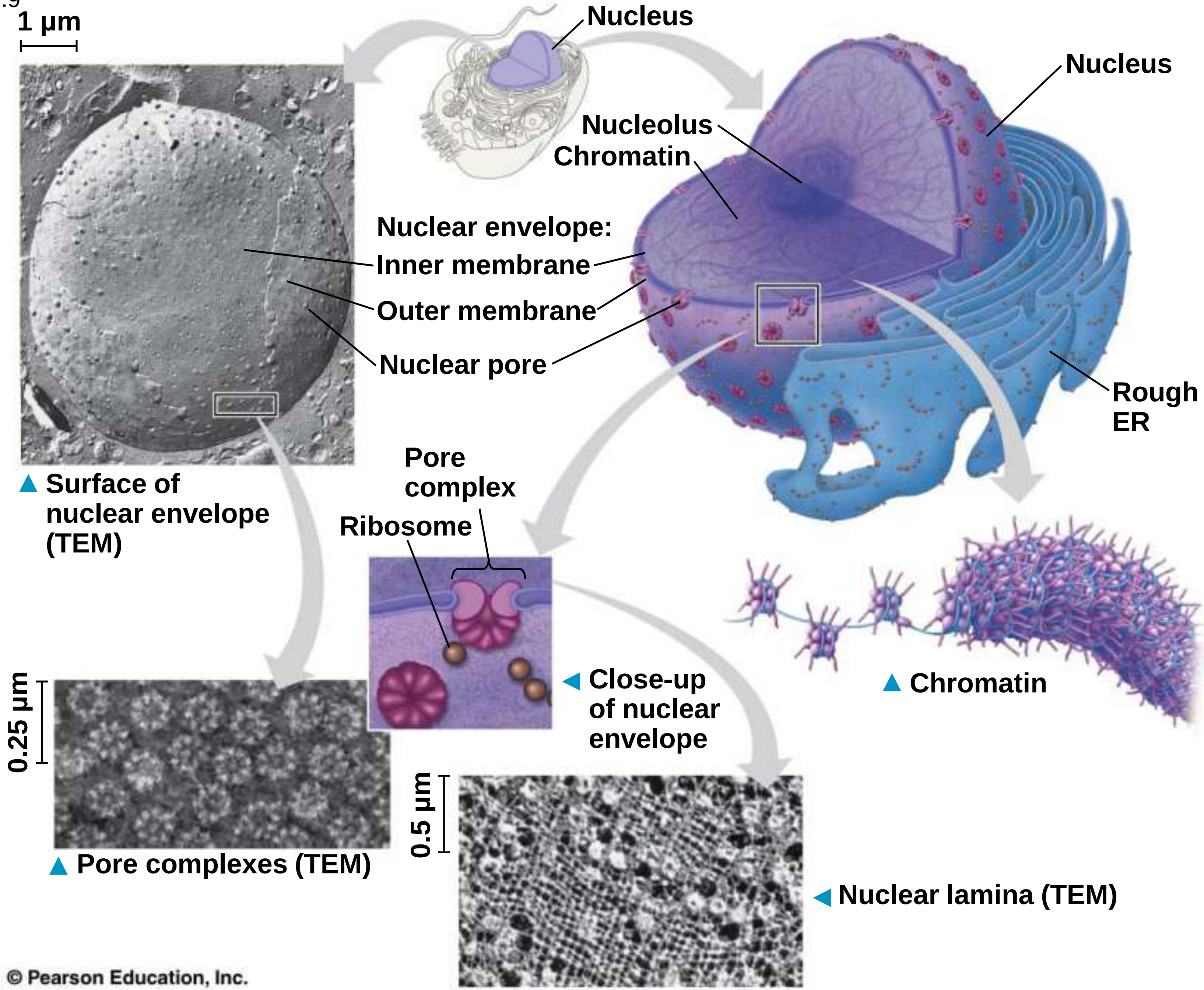
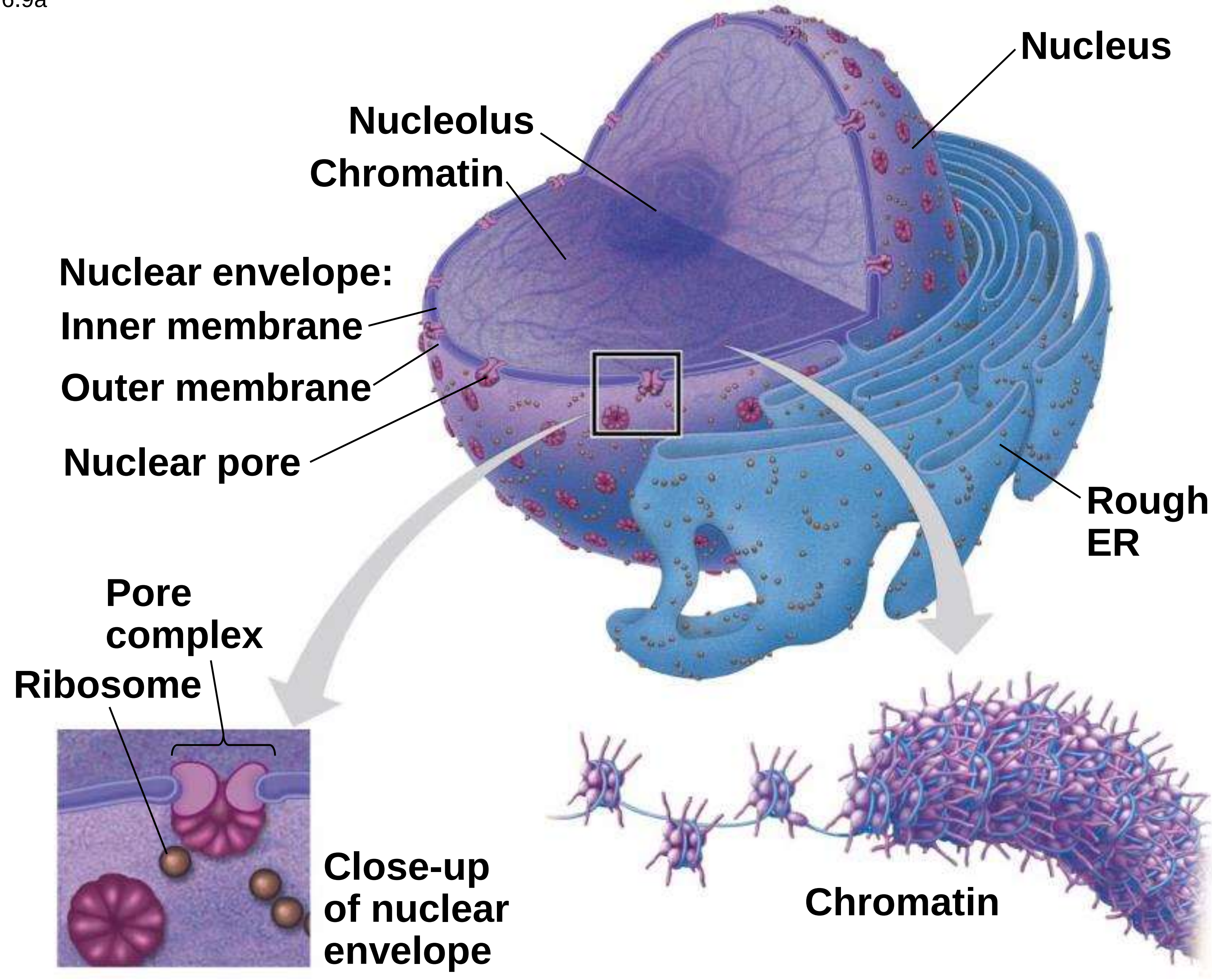
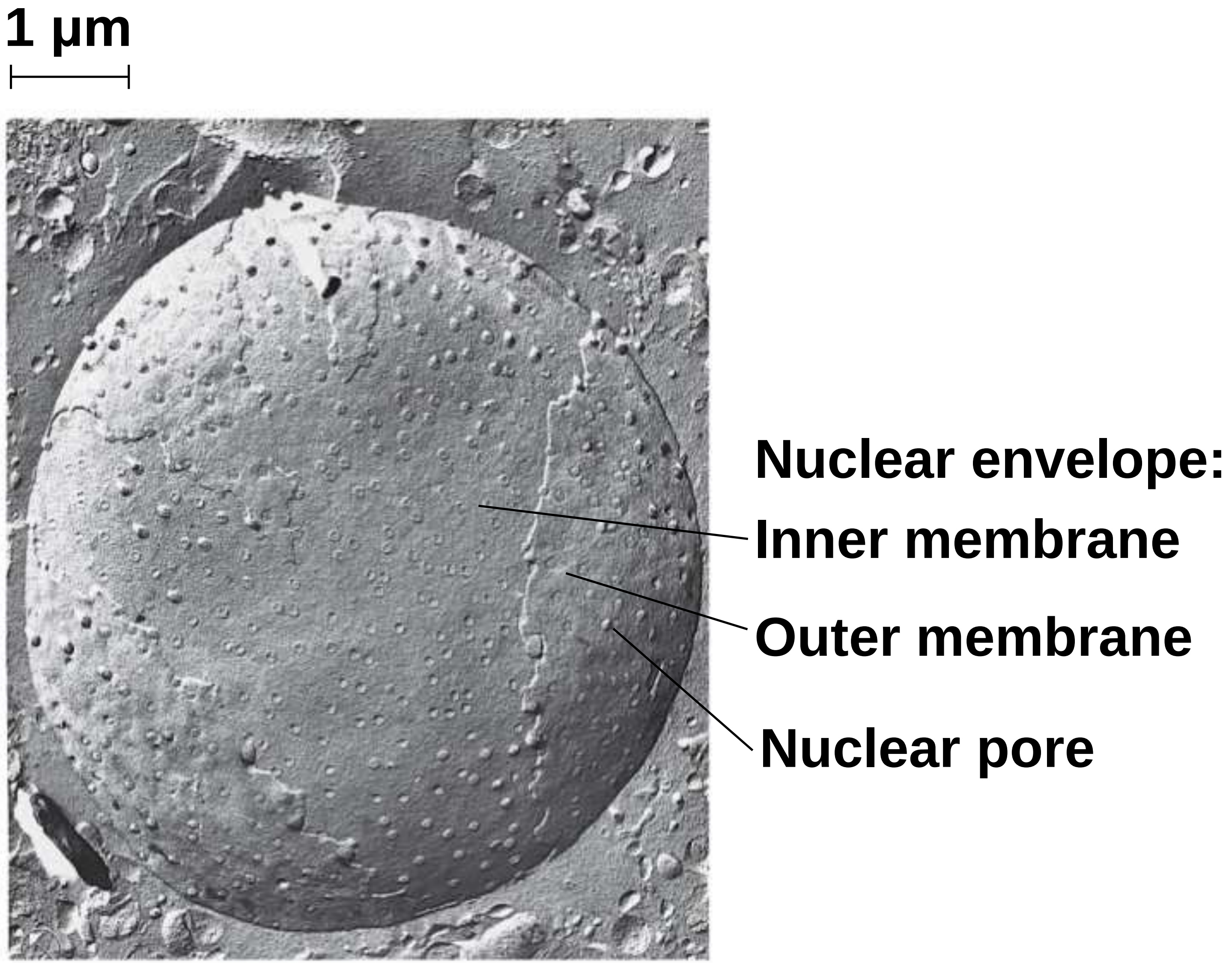

1 µm
Nucleus
Nucleus
Nucleolus
Chromatin
Nuclear envelope:
Inner membrane
Outer membrane
Nuclear pore
Rough ER
Surface of nuclear envelope (TEM)
Pore complex
Ribosome
Close-up of nuclear envelope
Chromatin
0.25 µm
Pore complexes (TEM)
0.5 µm
Nuclear lamina (TEM)
© Pearson Education, Inc.

Figure 6.9a



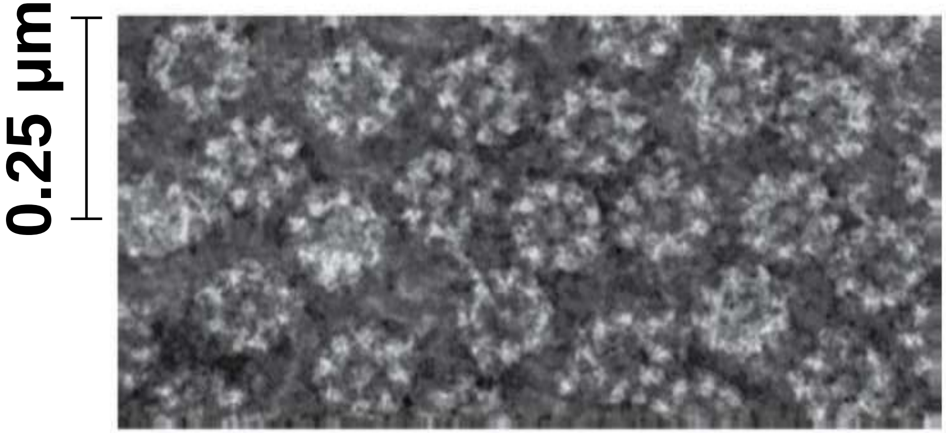
Nucleus
Nucleolus
Chromatin
Nuclear envelope:
Inner membrane
Outer membrane
Nuclear pore
Rough ER
Pore complex
Ribosome
Close-up of nuclear envelope
Chromatin

Figure 6.9b



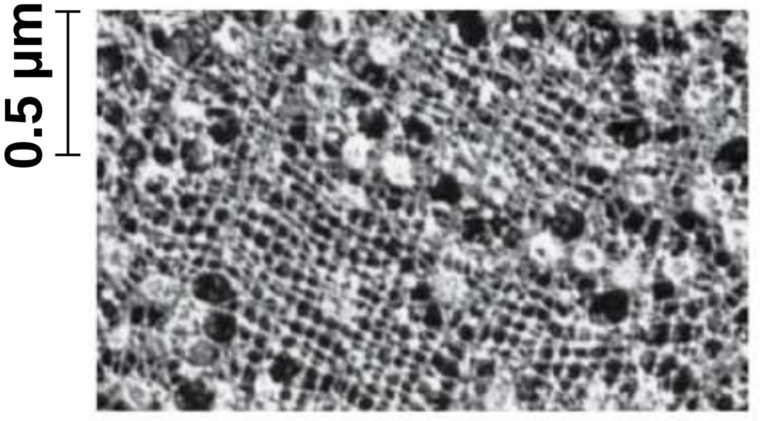
Surface of nuclear envelope (TEM)

Figure 6.9c



Pore complexes (TEM)

Figure 6.9d



Nuclear lamina (TEM)

a) Pores regulate the entry and exit of molecules from the nucleus

b) The nuclear size of the envelop is lined by the **nuclear lamina**, which is composed of proteins and maintains the shape of the nucleus

a) In the nucleus, DNA is organized into discrete units called **chromosomes**

b) Each chromosome is composed of a single DNA molecule associated with proteins

c) The DNA and proteins of chromosomes are together called **chromatin**

d) Chromatin condenses to form discrete **chromosomes** as a cell prepares to divide

e) The **nucleolus** is located within the nucleus and is the site of ribosomal RNA (rRNA) synthesis

Ribosomes: Protein Factories

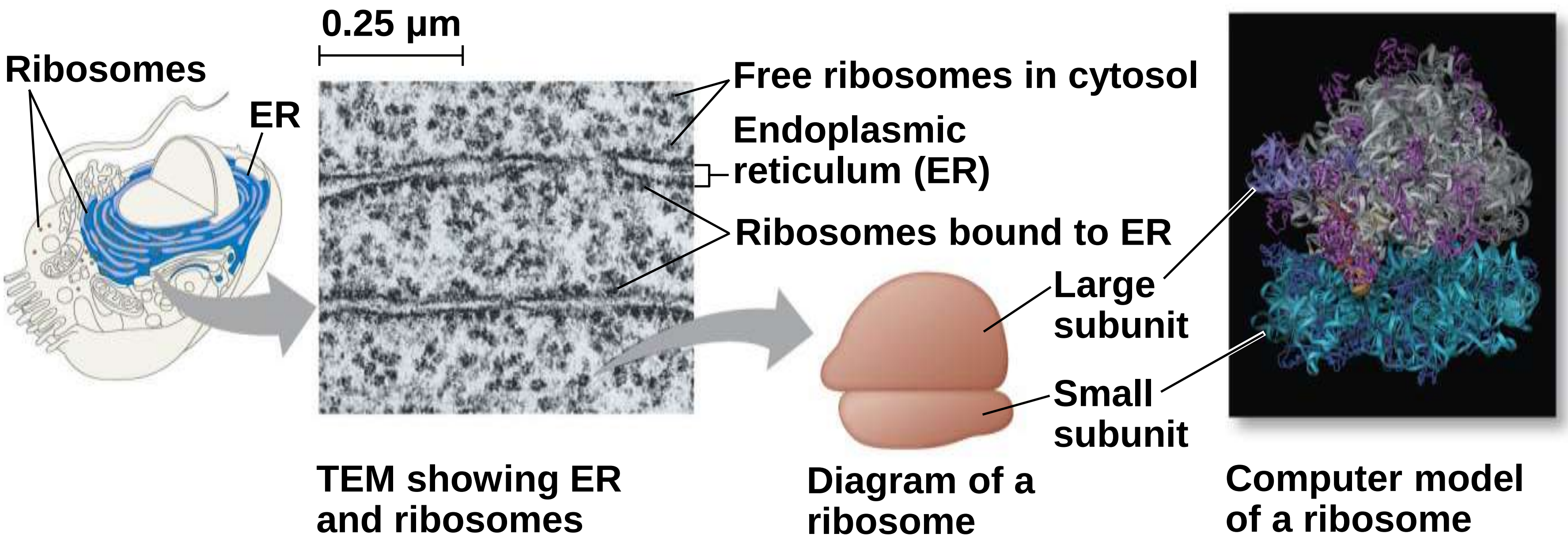
a) **Ribosomes** are complexes made of ribosomal RNA and protein

b) Ribosomes carry out protein synthesis in two locations

   a) In the cytosol (free ribosomes)

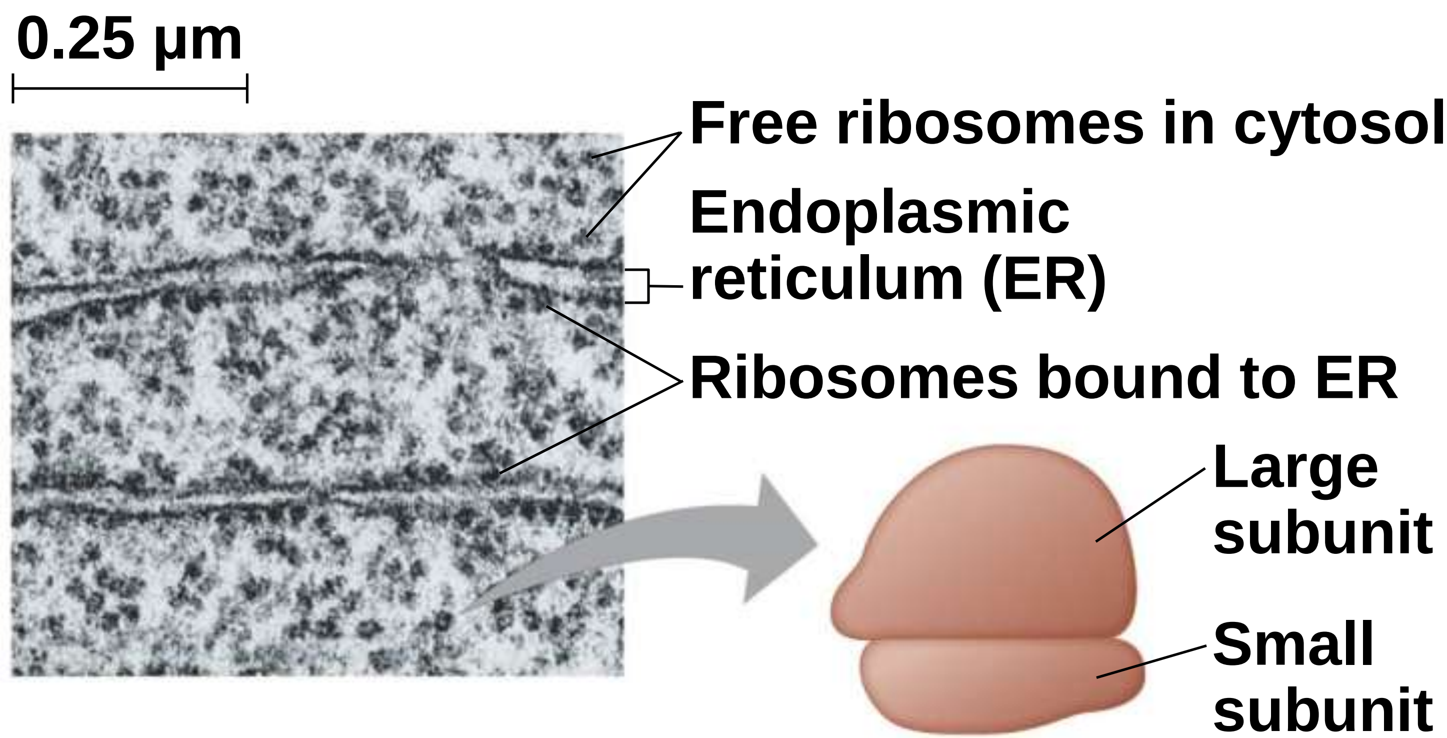
   b) On the outside of the endoplasmic reticulum or the nuclear envelope (bound ribosomes)

Figure 6.10



Ribosomes
ER
0.25 μm
Free ribosomes in cytosol
Endoplasmic reticulum (ER)
Ribosomes bound to ER
Large subunit
Small subunit
TEM showing ER and ribosomes
Diagram of a ribosome
Computer model of a ribosome

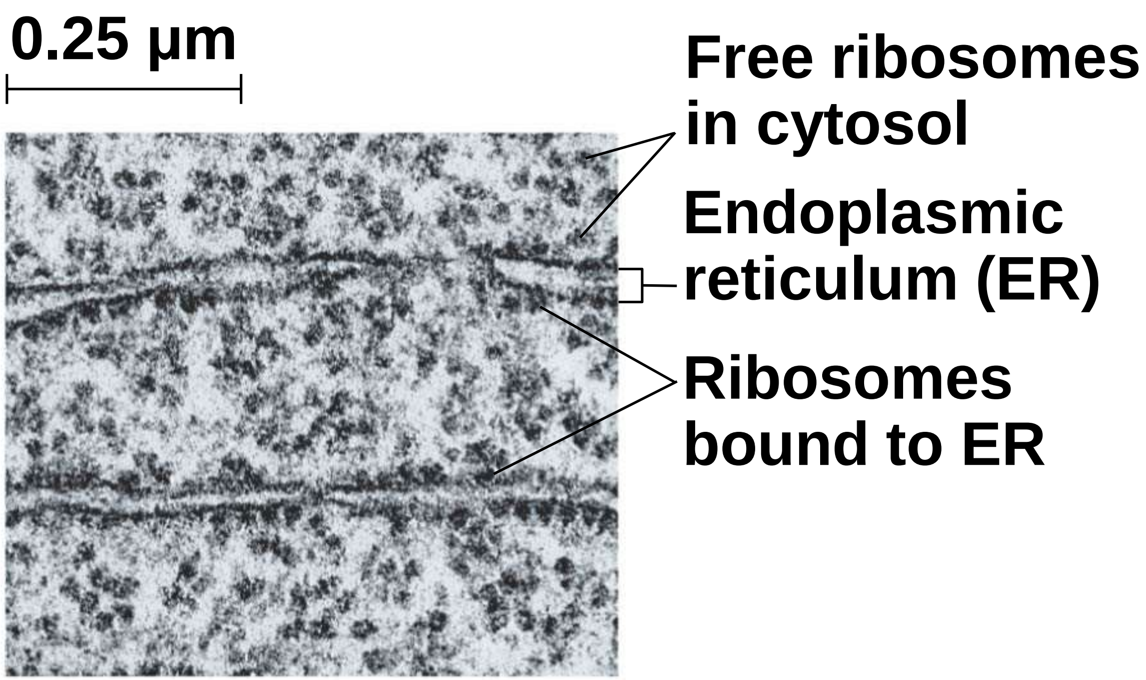
Figure 6.10a



TEM showing ER and ribosomes

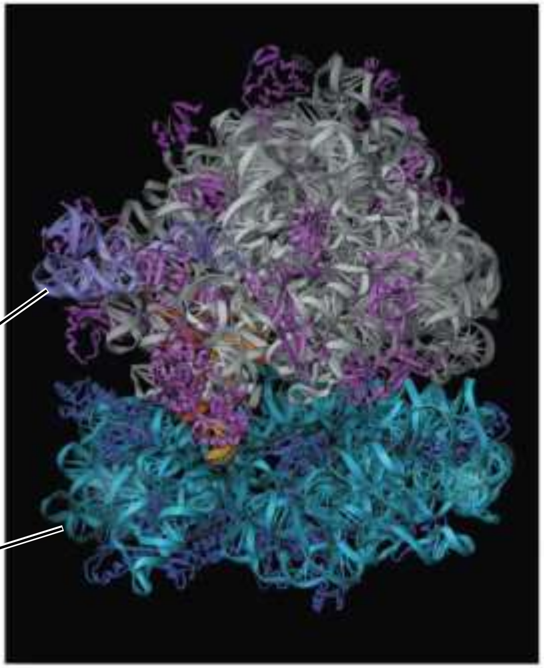
Diagram of a ribosome

Figure 6.10b



TEM showing ER and ribosomes

Figure 6.10c



Computer model of a ribosome

Concept 6.4: The endomembrane system regulates protein traffic and performs metabolic functions in the cell

a) The **endomembrane system** consists of
    a) Nuclear envelope
    b) Endoplasmic reticulum
    c) Golgi apparatus
    d) Lysosomes
    e) Vacuoles
    f) Plasma membrane
b) These components are either continuous or connected via transfer by **vesicles**

The Endoplasmic Reticulum: Biosynthetic Factory

a) The **endoplasmic reticulum (ER)** accounts for more than half of the total membrane in many eukaryotic cells

b) The ER membrane is continuous with the nuclear envelope

c) There are two distinct regions of ER

   a) **Smooth ER**, which lacks ribosomes

   b) **Rough ER**, whose surface is studded with ribosomes

Figure 6.11

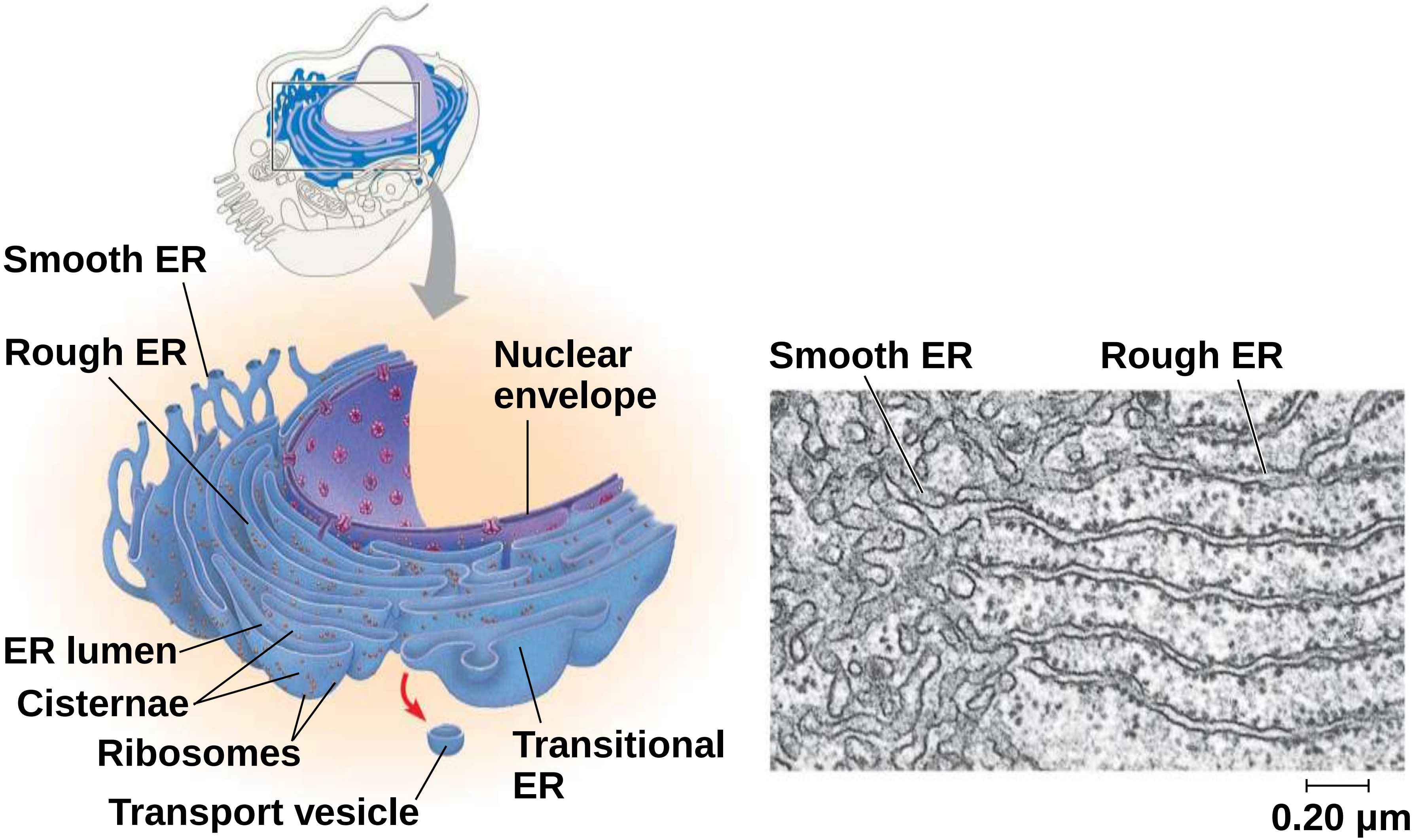
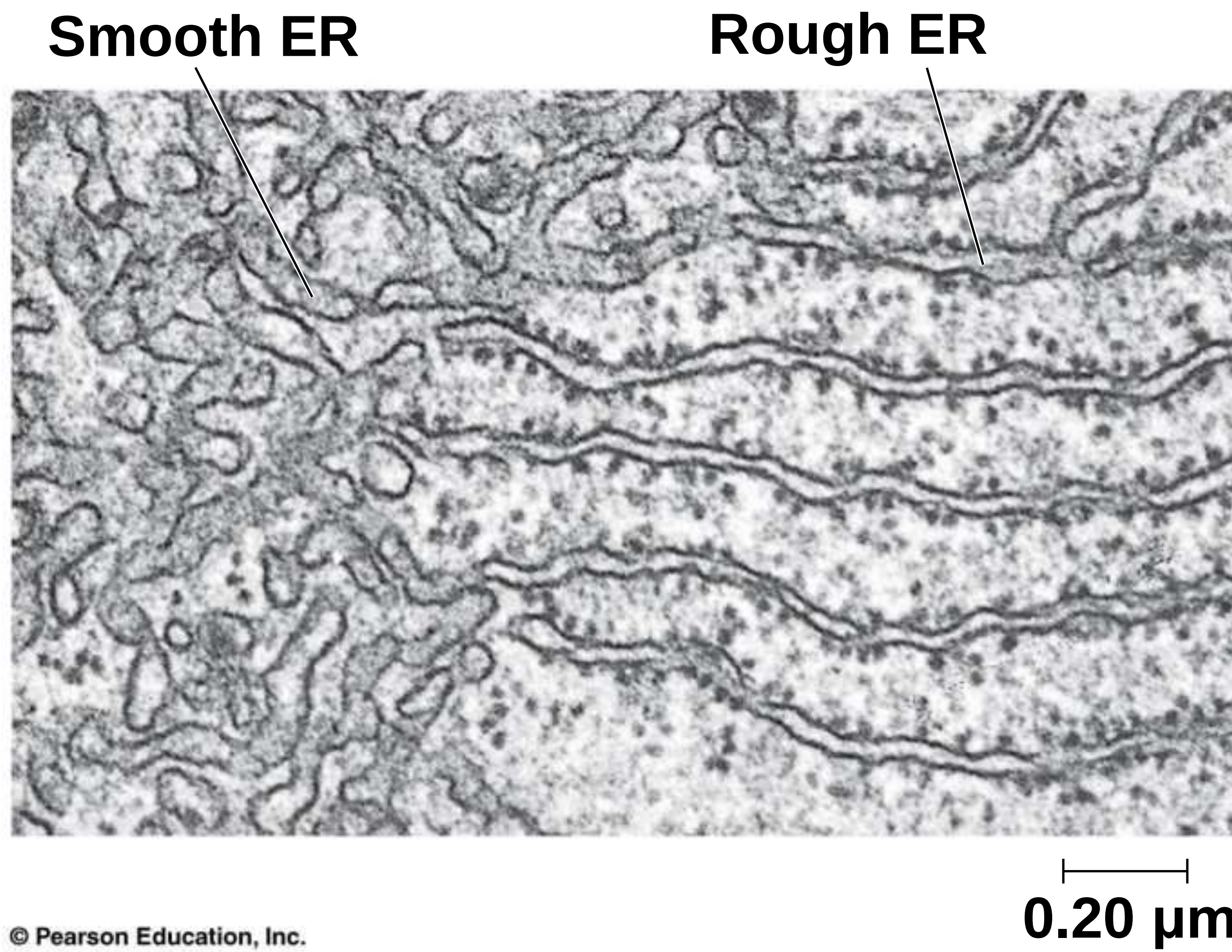

Smooth ER
Rough ER
Nuclear envelope
ER lumen
Cisternae
Ribosomes
Transport vesicle
Transitional ER
Smooth ER
Rough ER
0.20 μm

Figure 6.11a



Functions of Smooth ER

a) The smooth ER
   a) Synthesizes lipids
   b) Metabolizes carbohydrates
   c) Detoxifies drugs and poisons
   d) Stores calcium ions

Functions of Rough ER

a) The rough ER

   a) Has bound ribosomes, which secrete **glycoproteins** (proteins covalently bonded to carbohydrates)

   b) Distributes **transport vesicles**, secretory proteins surrounded by membranes

   c) Is a membrane factory for the cell

The Golgi Apparatus: Shipping and Receiving Center

a) The **Golgi apparatus** consists of flattened membranous sacs called cisternae

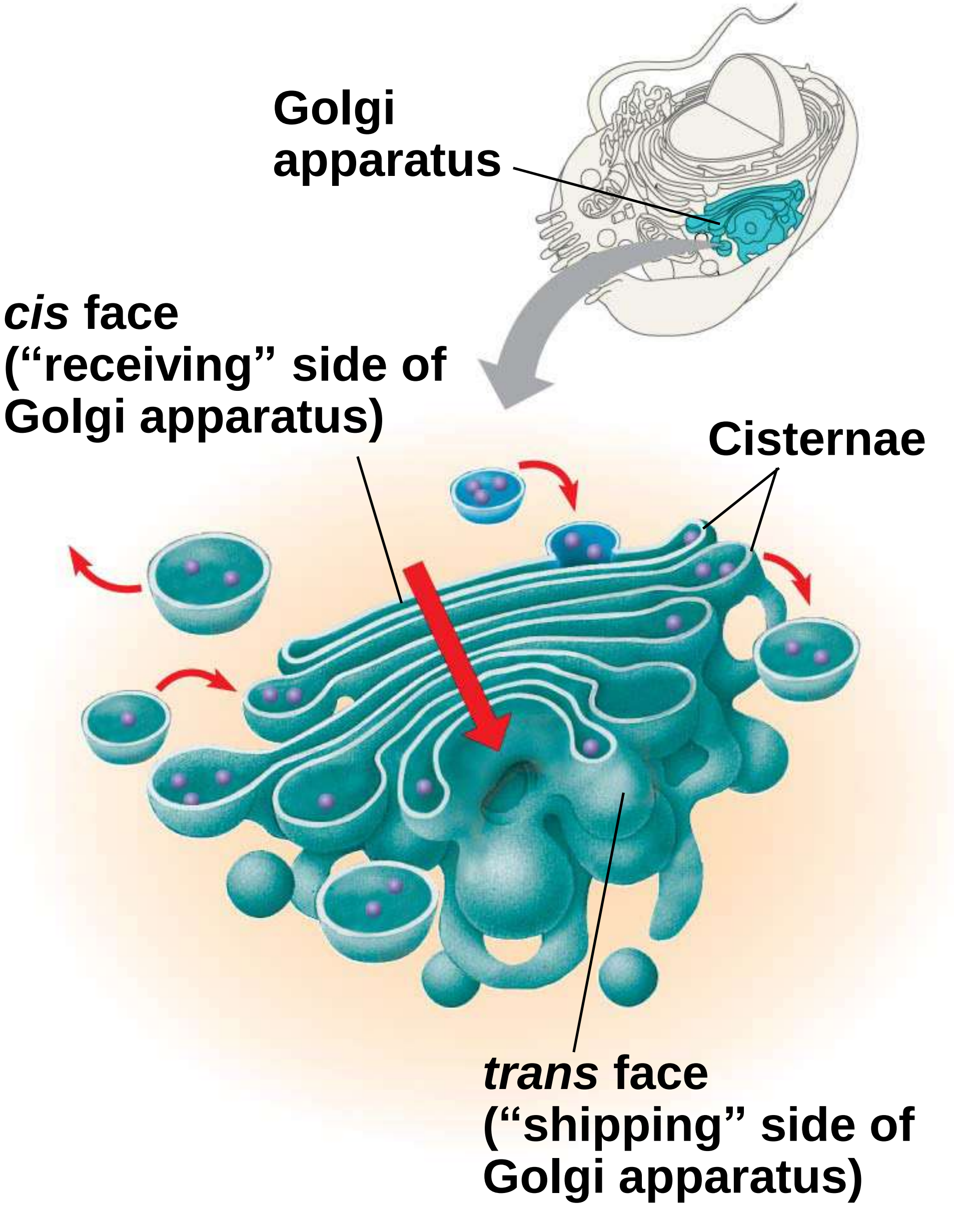
b) Functions of the Golgi apparatus

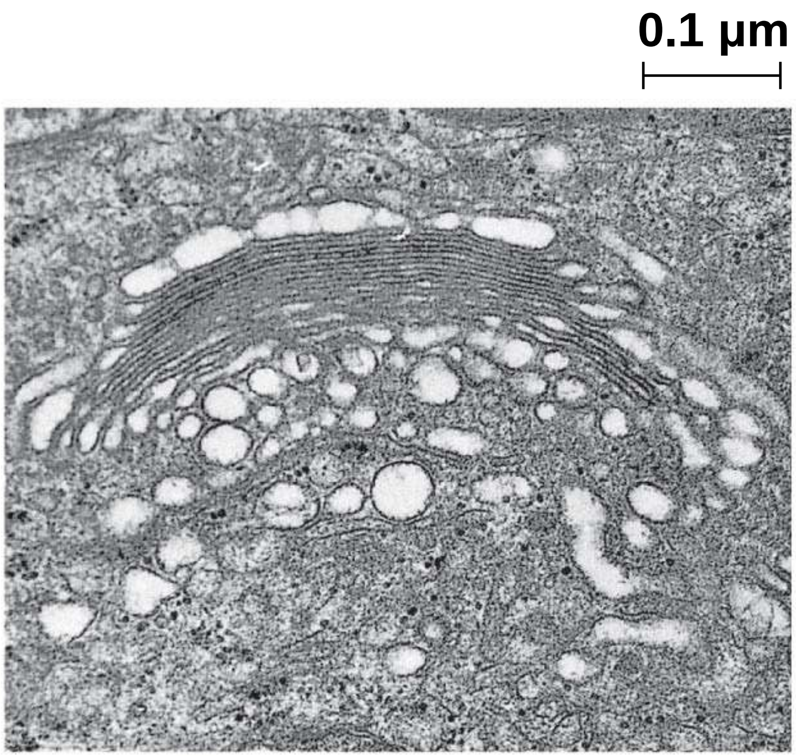
   a) Modifies products of the ER

   b) Manufactures certain macromolecules

   c) Sorts and packages materials into transport vesicles

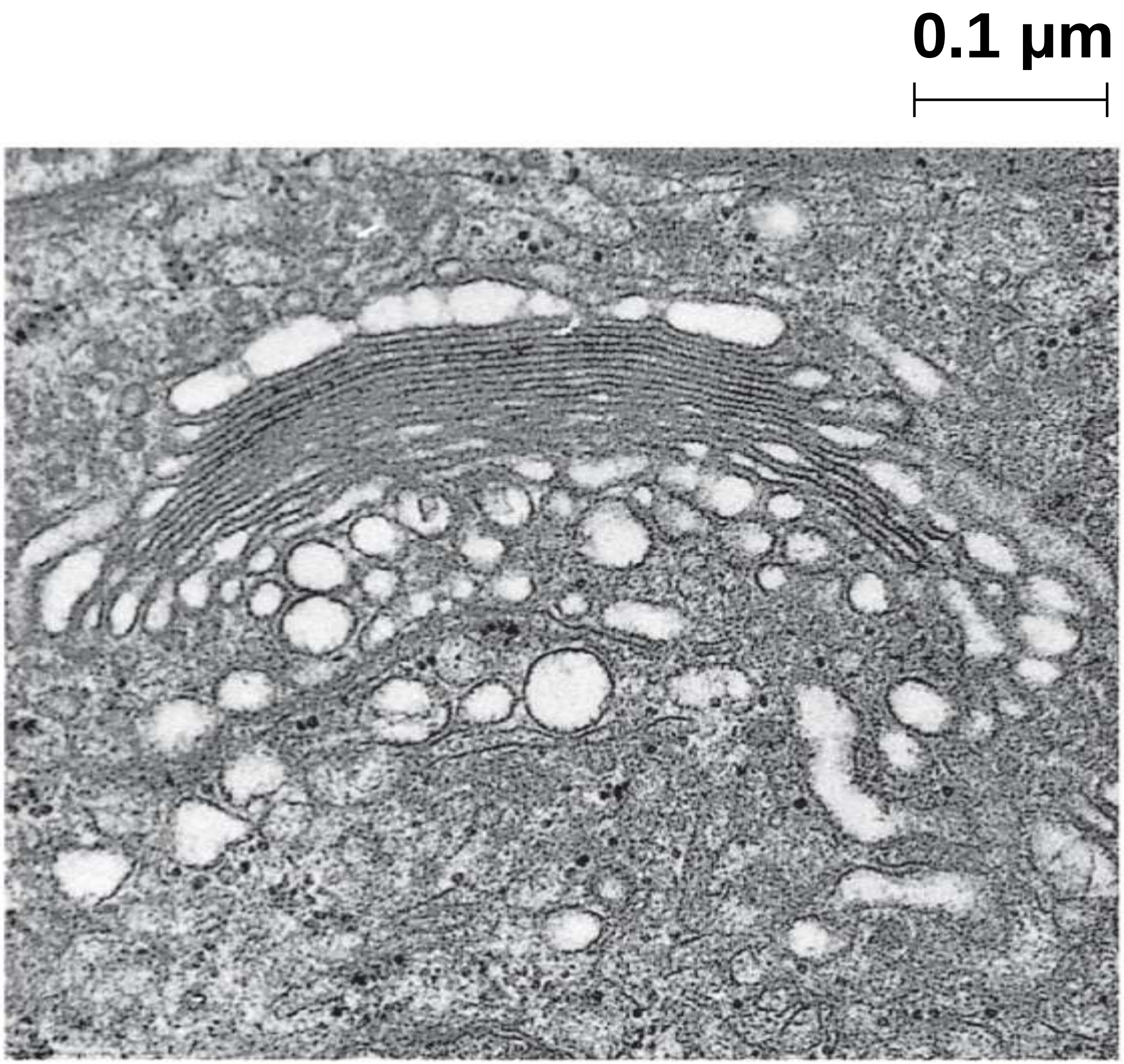
Figure 6.12





TEM of Golgi apparatus

Figure 6.12a



TEM of Golgi apparatus

Lysosomes: Digestive Compartments

a) A **lysosome** is a membranous sac of hydrolytic enzymes that can digest macromolecules

b) Lysosomal enzymes work best in the acidic environment inside the lysosome

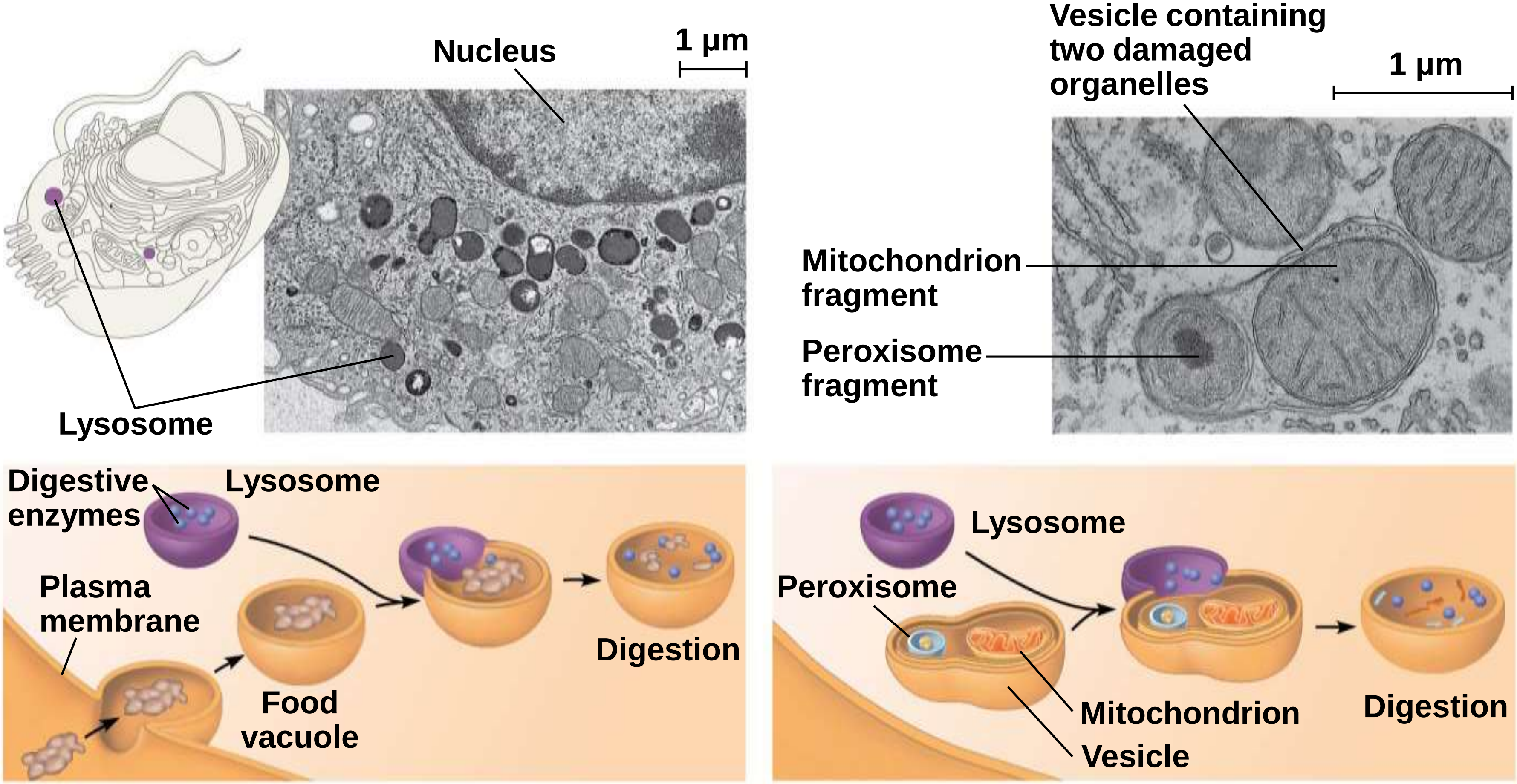
c) Hydrolytic enzymes and lysosomal membranes are made by rough ER and then transferred to the Golgi apparatus for further processing

a) Some types of cell can engulf another cell by **phagocytosis**; this forms a food vacuole

b) A lysosome fuses with the food vacuole and digests the molecules

c) Lysosomes also use enzymes to recycle the cell’s own organelles and macromolecules, a process called autophagy

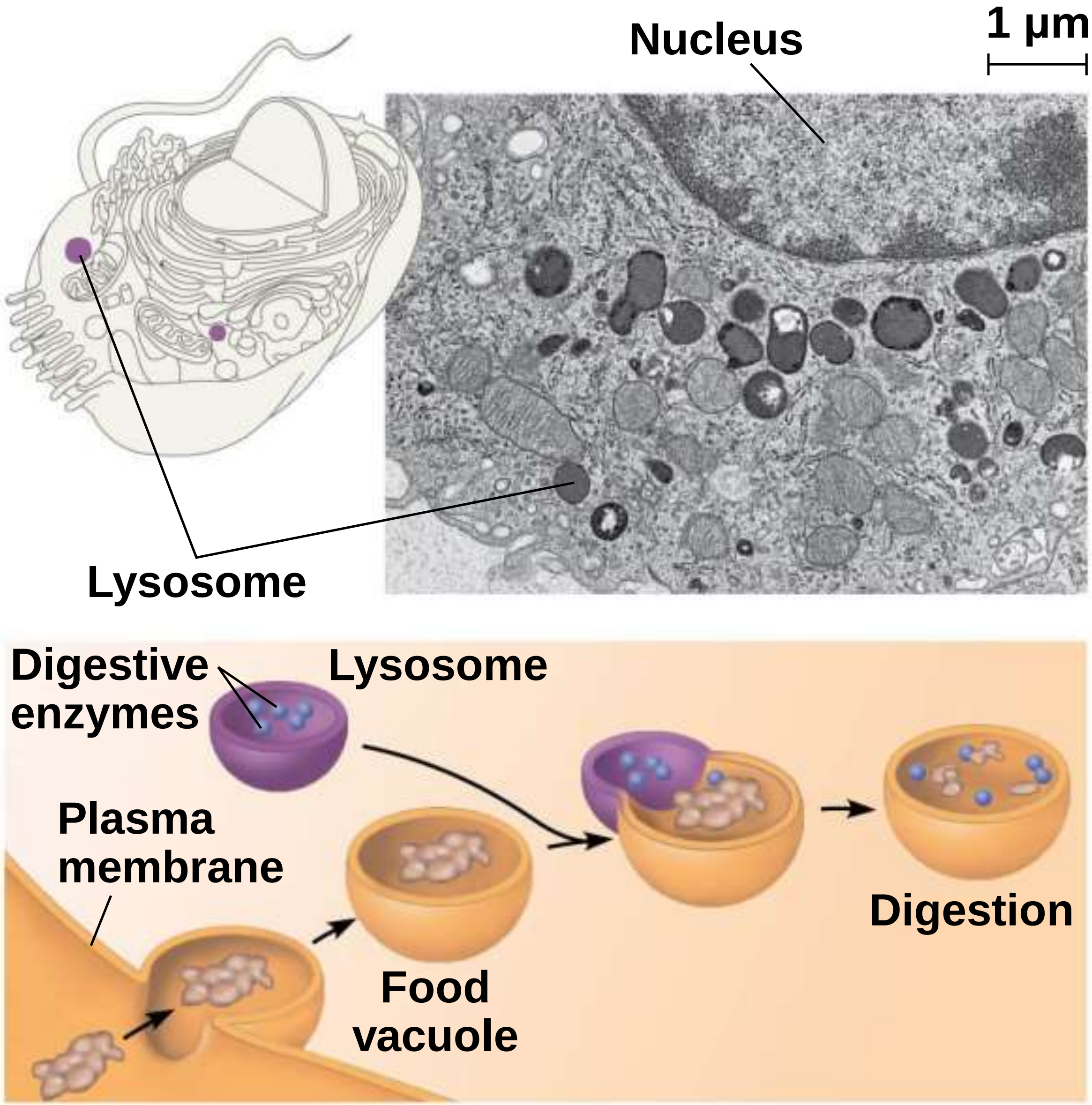
Figure 6.13



(a) Phagocytosis: lysosome digesting food

(b) Autophagy: lysosome breaking down damaged organelles

Figure 6.13a



(a) Phagocytosis: lysosome digesting food

Figure 6.13aa

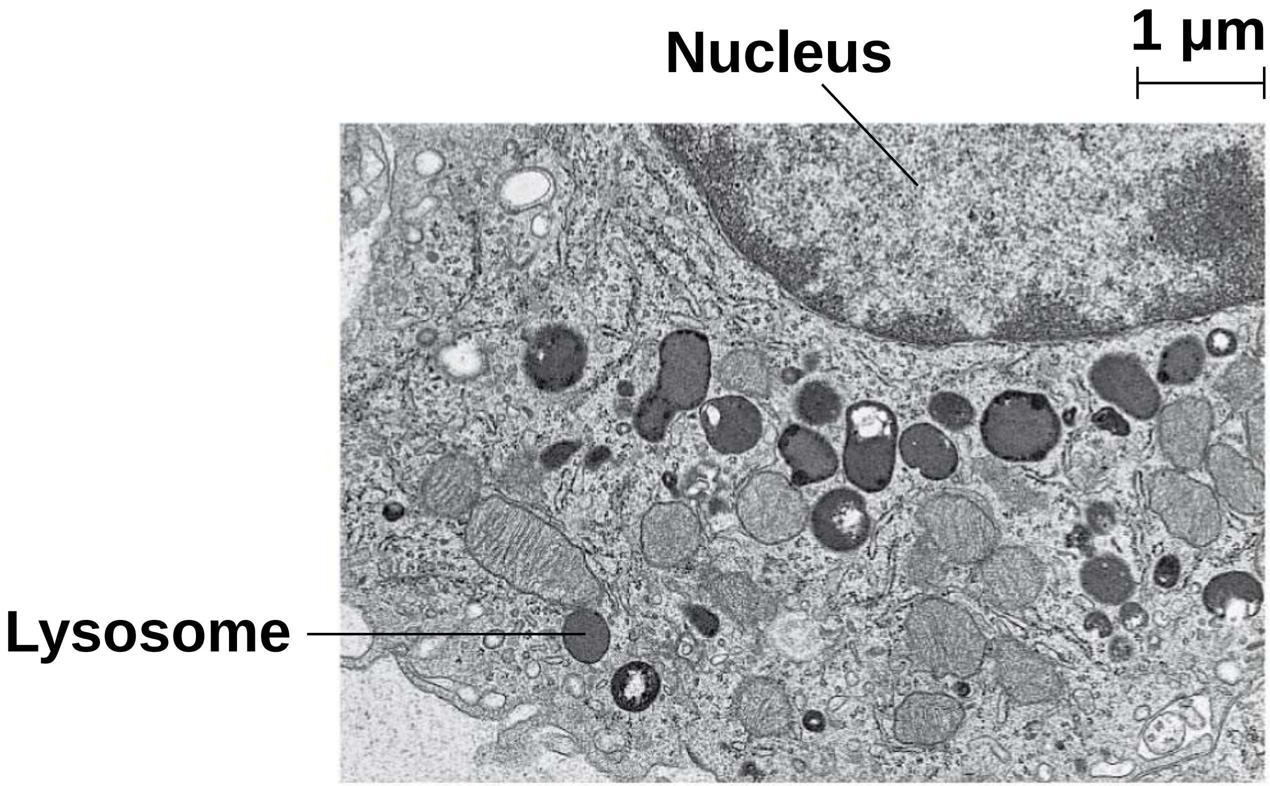
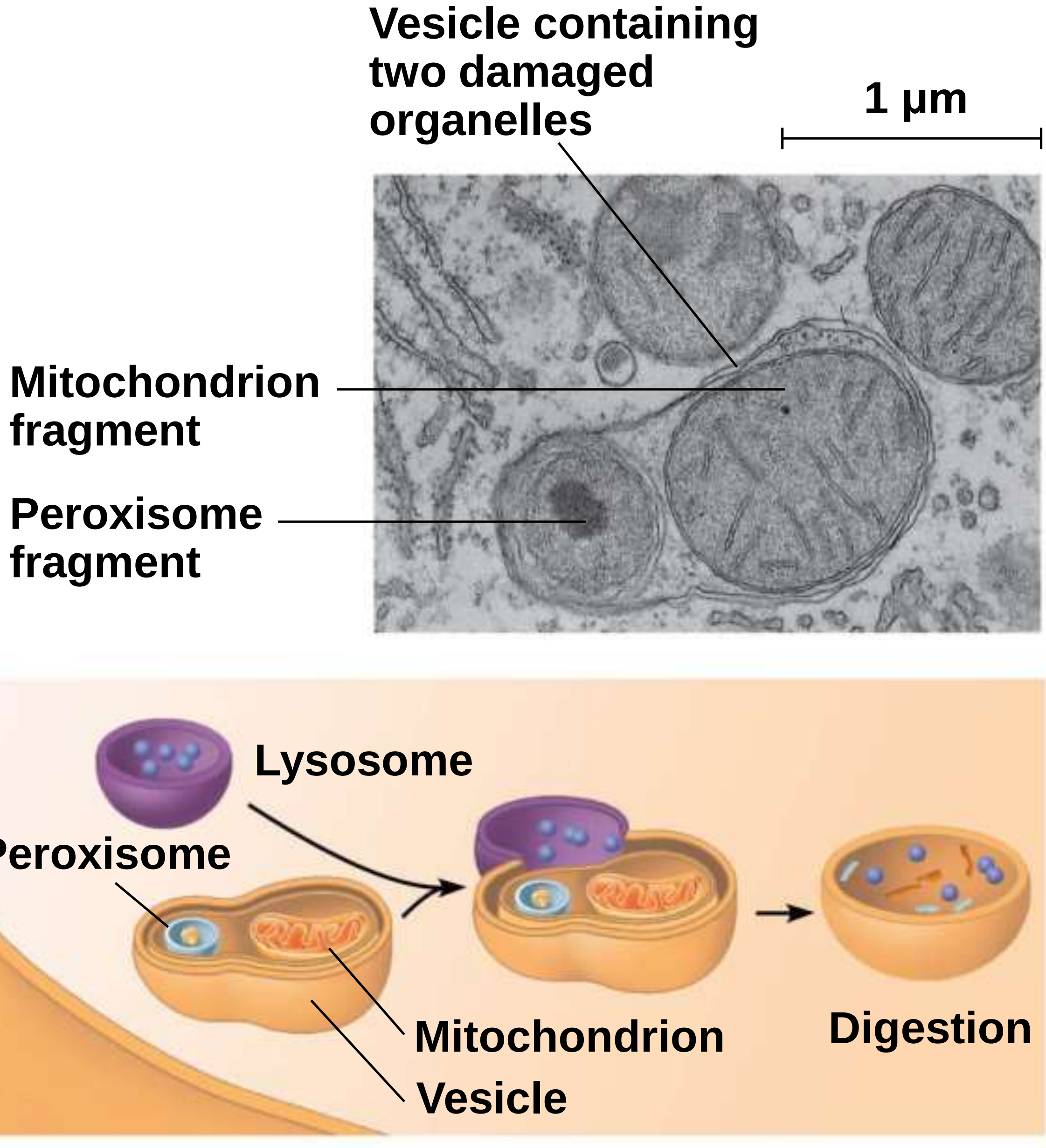
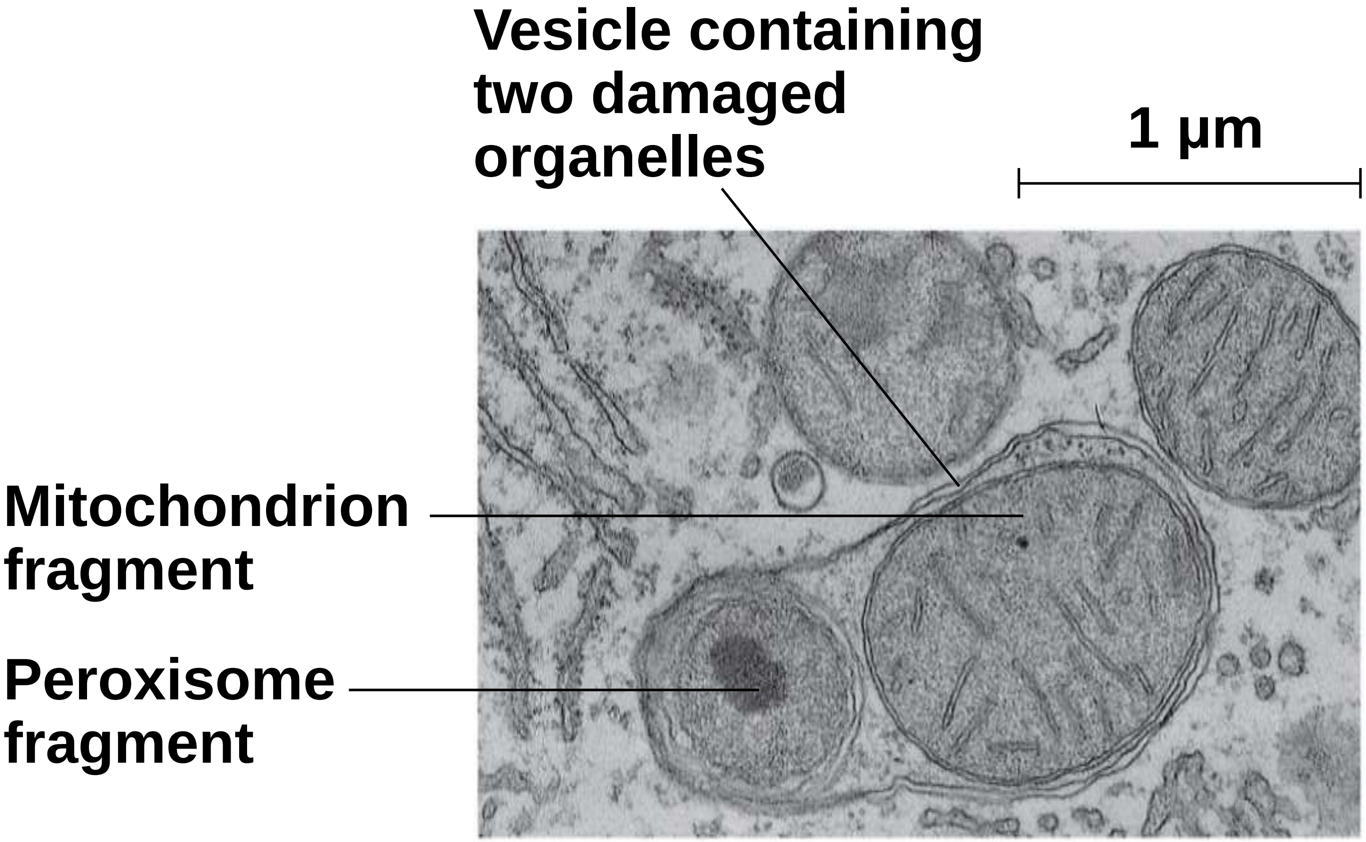


Figure 6.13b



(b) Autophagy: lysosome breaking down damaged organelles

Figure 6.13ba



Vacuoles: Diverse Maintenance Compartments

a) **Vacuoles** are large vesicles derived from the ER and Golgi apparatus

b) Vacuoles perform a variety of functions in different kinds of cells

a) **Food vacuoles** are formed by phagocytosis

b) **Contractile vacuoles**, found in many freshwater protists, pump excess water out of cells

c) **Central vacuoles**, found in many mature plant cells, hold organic compounds and water

Figure 6.14

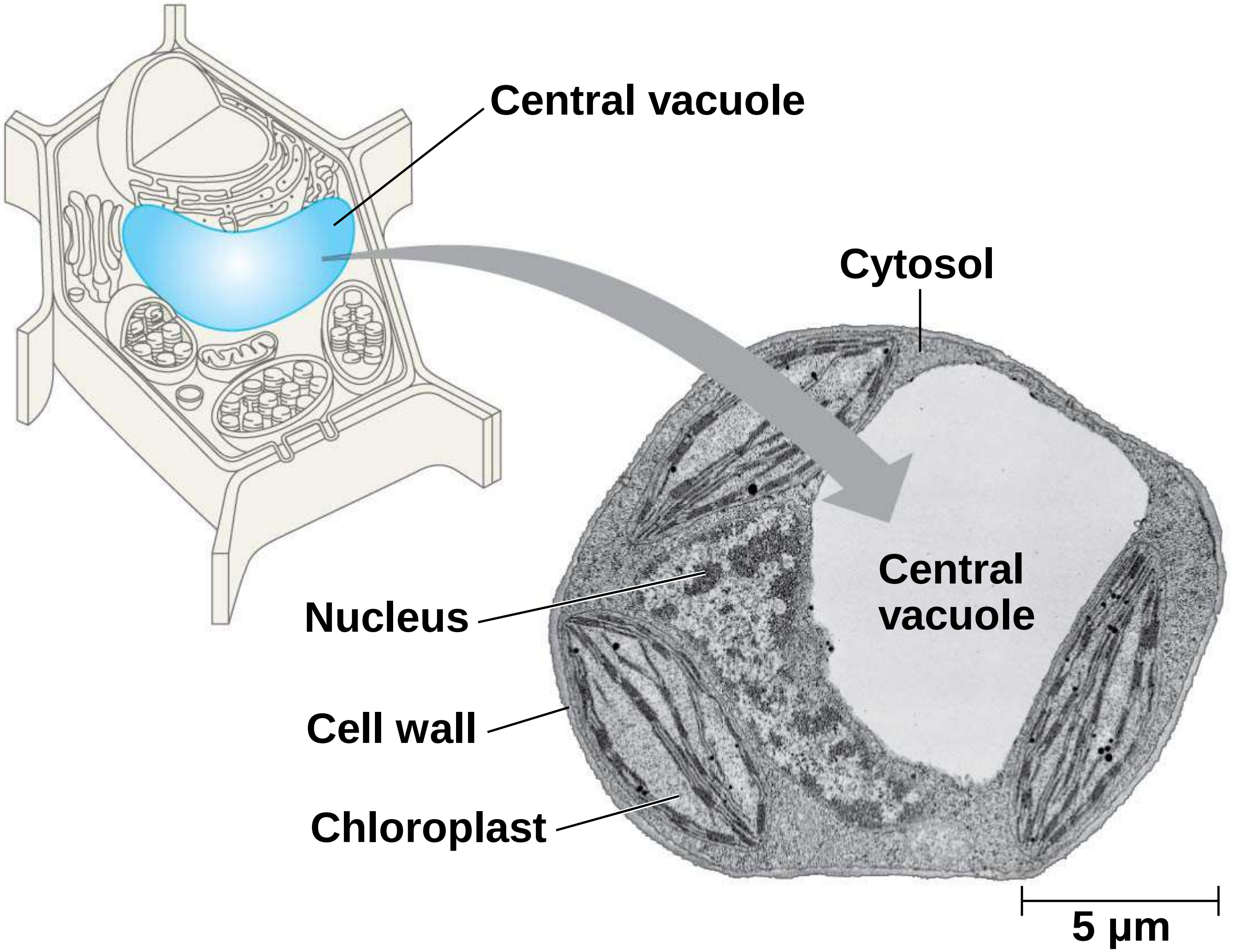
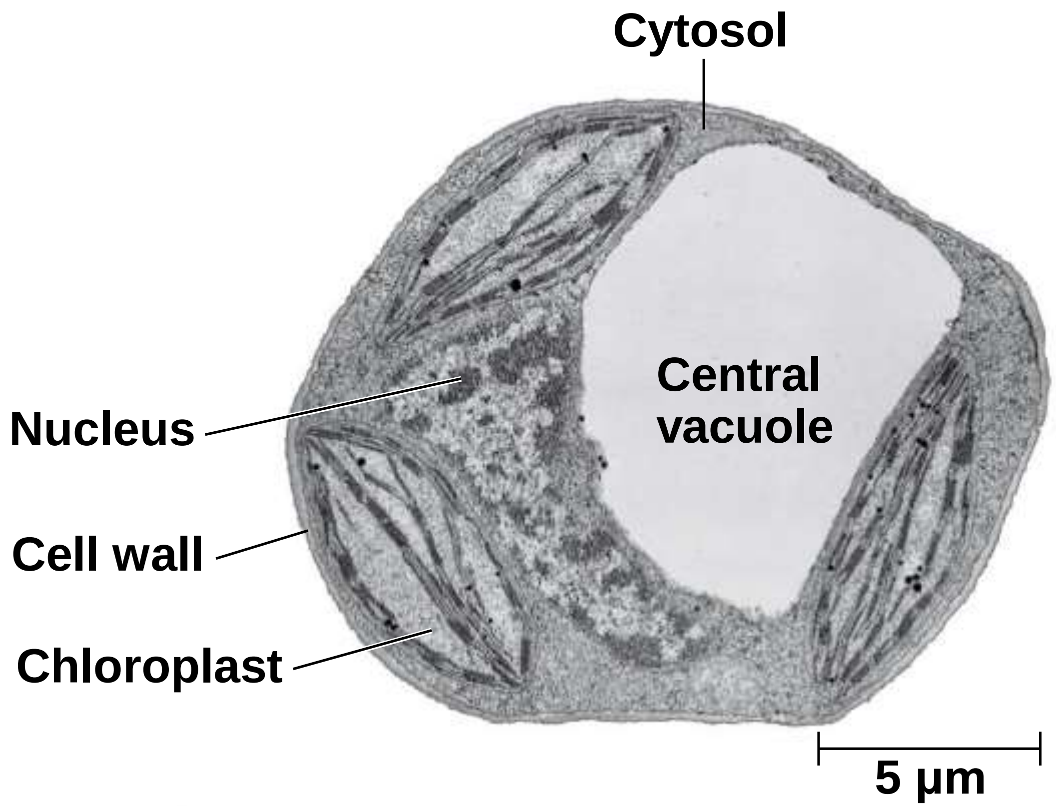
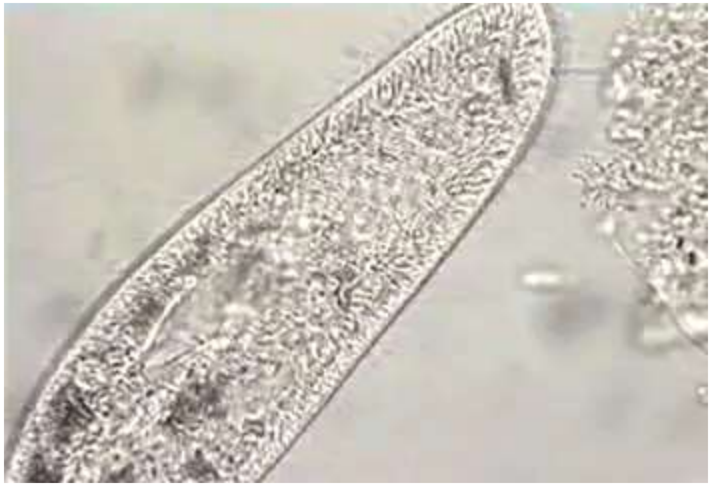

Central vacuole
Cytosol
Central
vacuole
Nucleus
Cell wall
Chloroplast
5 µm

Figure 6.14a



Cytosol
Central
vacuole
Nucleus
Cell wall
Chloroplast
5 µm

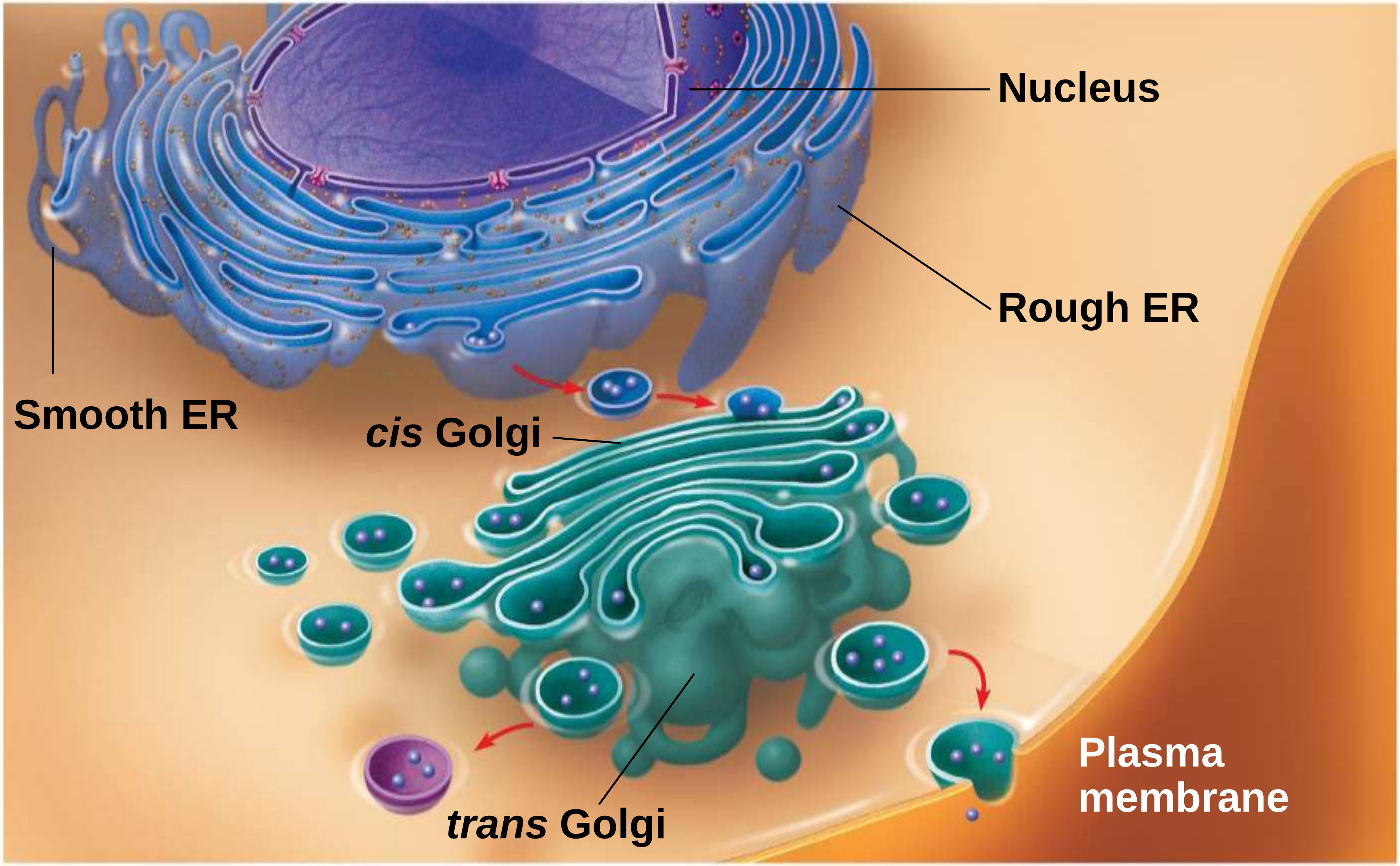
Video: *Paramecium* Vacuole



The Endomembrane System: *A Review*

a) The endomembrane system is a complex and dynamic player in the cell’s compartmental organization

Figure 6.15


Nucleus
Rough ER
Smooth ER
cis Golgi
trans Golgi
Plasma membrane
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Concept 6.5: Mitochondria and chloroplasts change energy from one form to another

a) **Mitochondria** are the sites of cellular respiration, a metabolic process that uses oxygen to generate ATP

b) **Chloroplasts**, found in plants and algae, are the sites of photosynthesis

c) Peroxisomes are oxidative organelles

The Evolutionary Origins of Mitochondria and Chloroplasts

a) Mitochondria and chloroplasts have similarities with bacteria
    a) Enveloped by a double membrane
    b) Contain free ribosomes and circular DNA molecules
    c) Grow and reproduce somewhat independently in cells
b) These similarities led to the **endosymbiont theory**

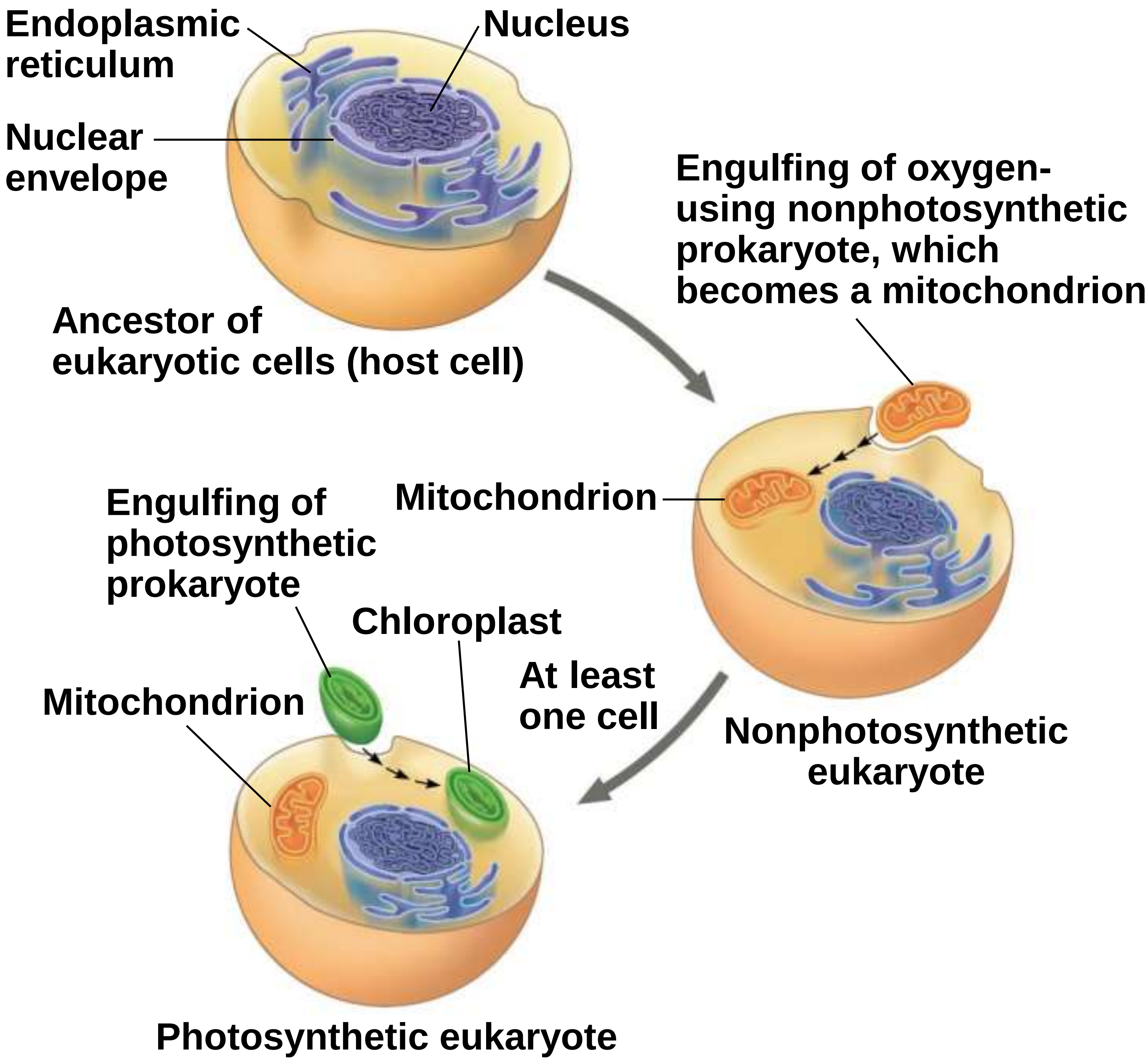
a) The endosymbiont theory suggests that an early ancestor of eukaryotes engulfed an oxygen-using nonphotosynthetic prokaryotic cell

b) The engulfed cell formed a relationship with the host cell, becoming an endosymbiont

c) The endosymbionts evolved into mitochondria

d) At least one of these cells may have then taken up a photosynthetic prokaryote, which evolved into a chloroplast

Figure 6.16



Endoplasmic reticulum
Nucleus
Nuclear envelope
Engulfing of oxygen-using nonphotosynthetic prokaryote, which becomes a mitochondrion
Ancestor of eukaryotic cells (host cell)
Mitochondrion
Engulfing of photosynthetic prokaryote
Chloroplast
At least one cell
Mitochondrion
Nonphotosynthetic eukaryote
Photosynthetic eukaryote

Mitochondria: Chemical Energy Conversion

a) Mitochondria are in nearly all eukaryotic cells

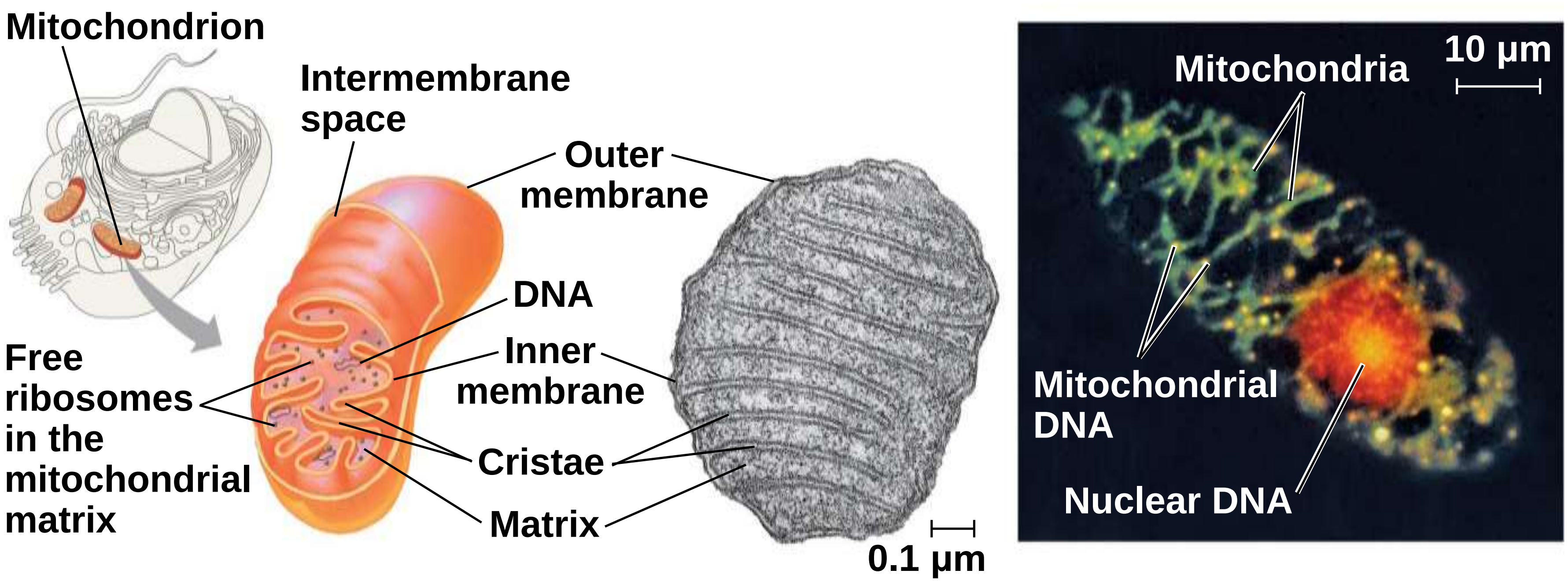
b) They have a smooth outer membrane and an inner membrane folded into **cristae**

c) The inner membrane creates two compartments: intermembrane space and **mitochondrial matrix**

d) Some metabolic steps of cellular respiration are catalyzed in the mitochondrial matrix

e) Cristae present a large surface area for enzymes that synthesize ATP

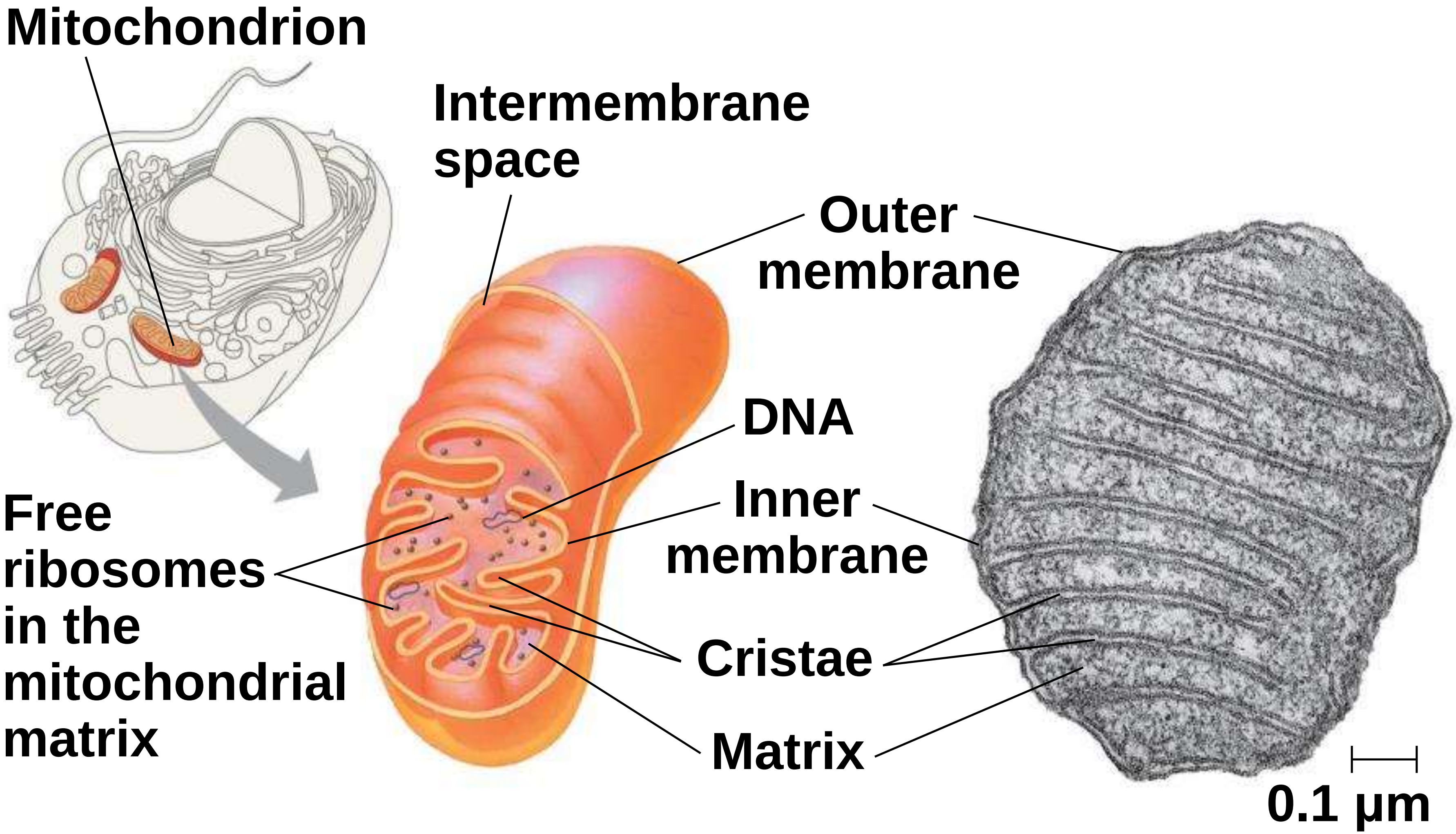
Figure 6.17



(a) Diagram and TEM of mitochondrion

(b) Network of mitochondria in *Euglena* (LM)

Figure 6.17a



(a) Diagram and TEM of mitochondrion

Figure 6.17aa

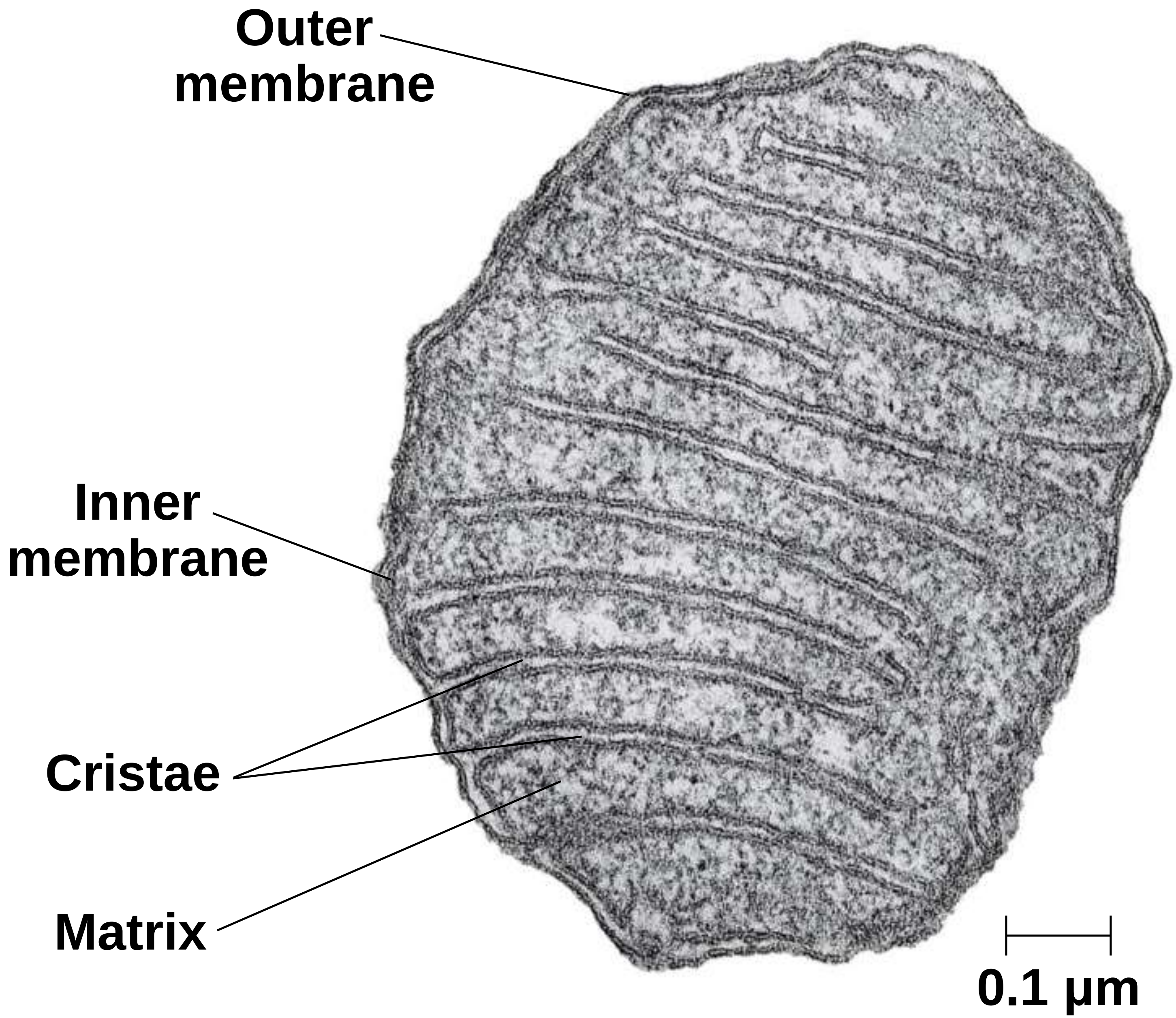
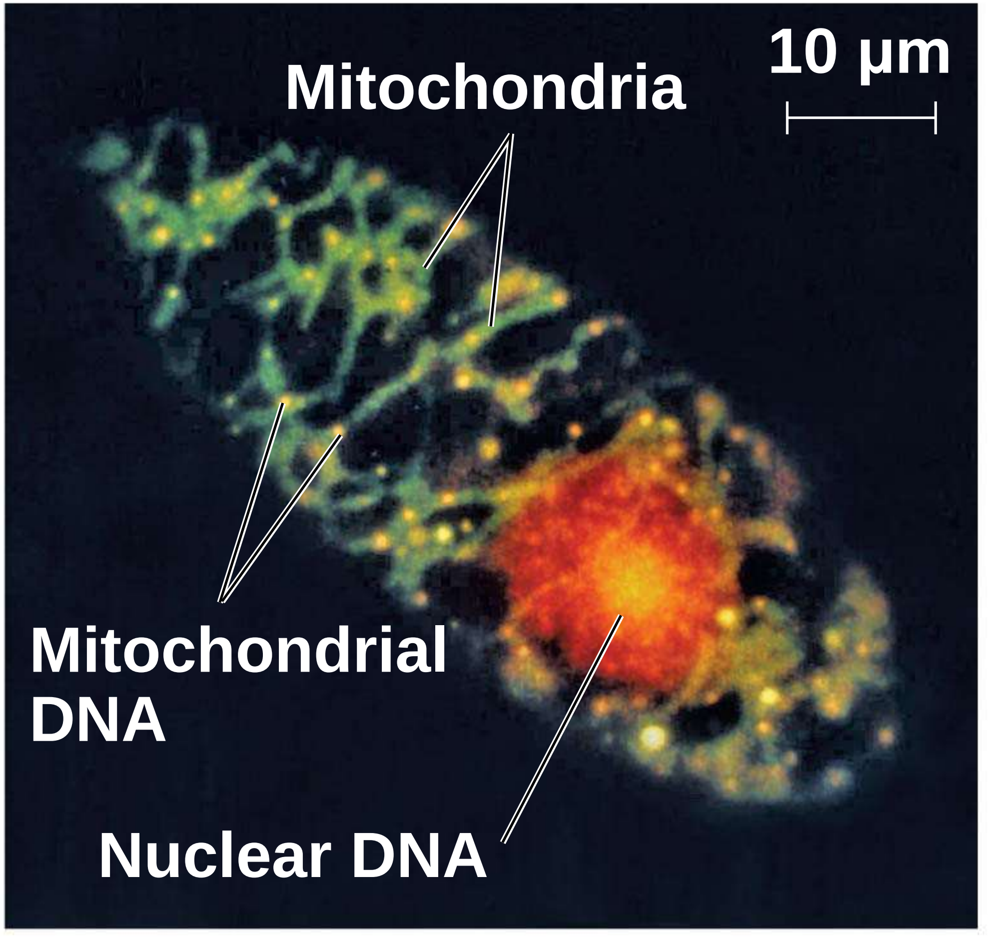


Figure 6.17b



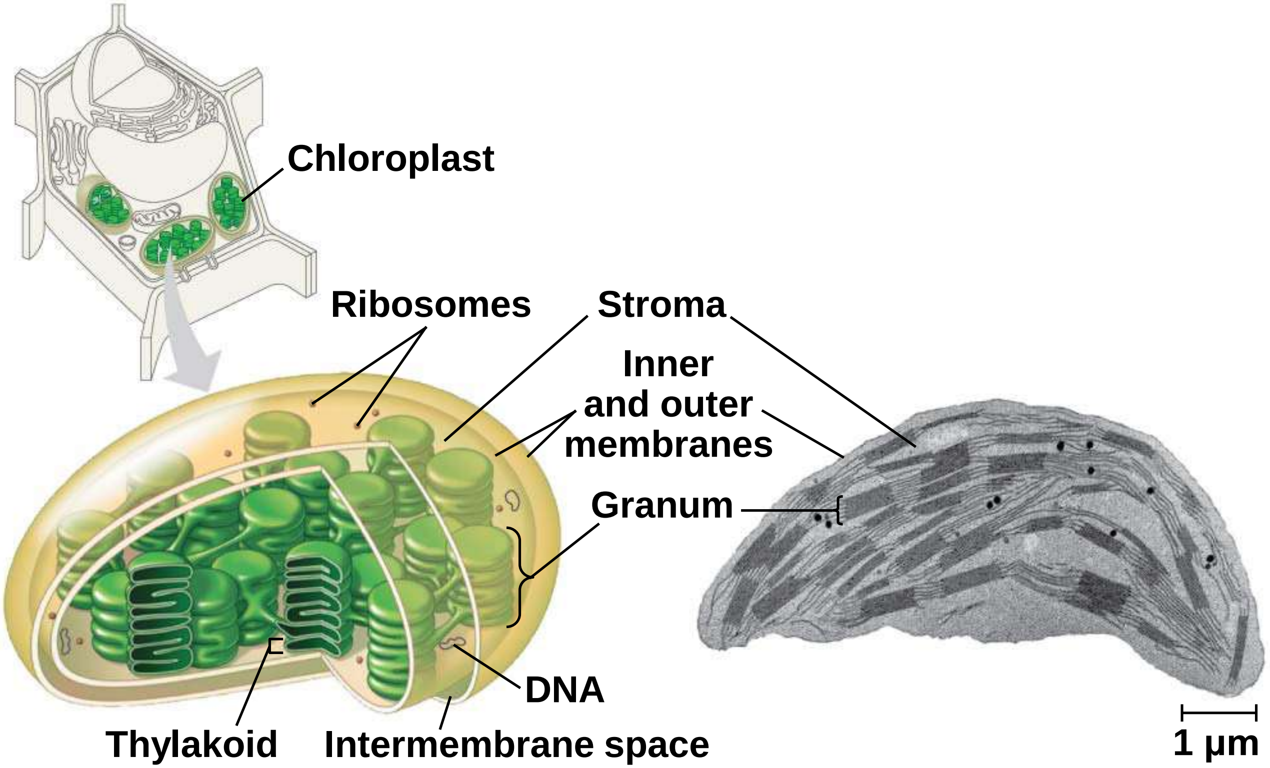
(b) Network of mitochondria in *Euglena* (LM)

Chloroplasts: Capture of Light Energy

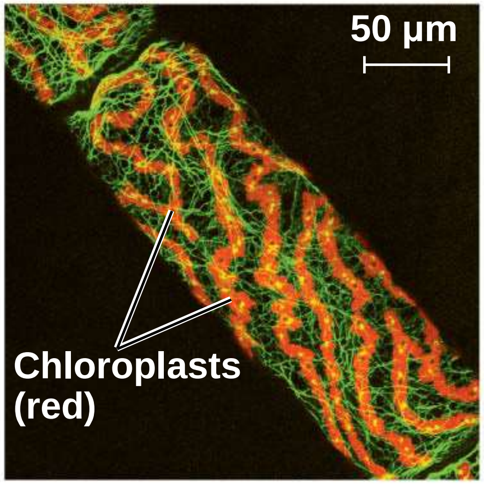
a) Chloroplasts contain the green pigment chlorophyll, as well as enzymes and other molecules that function in photosynthesis

b) Chloroplasts are found in leaves and other green organs of plants and in algae

Figure 6.18

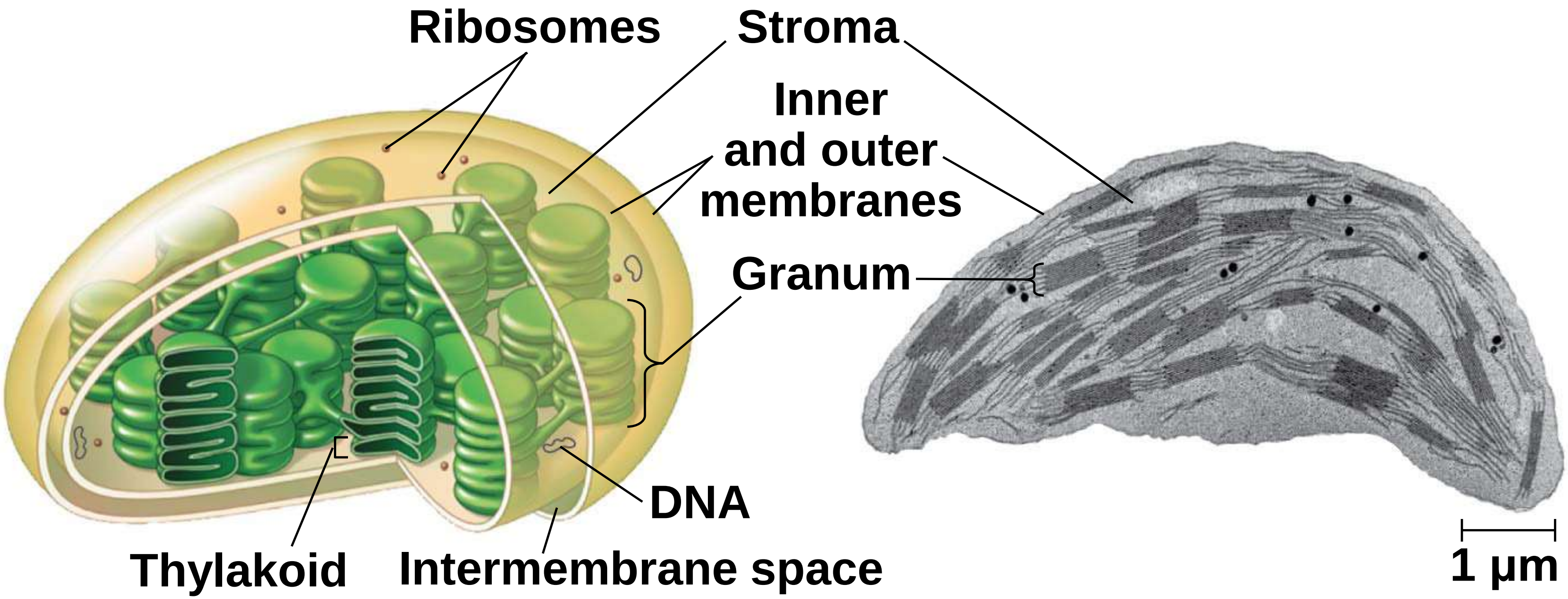


(a) Diagram and TEM of chloroplast



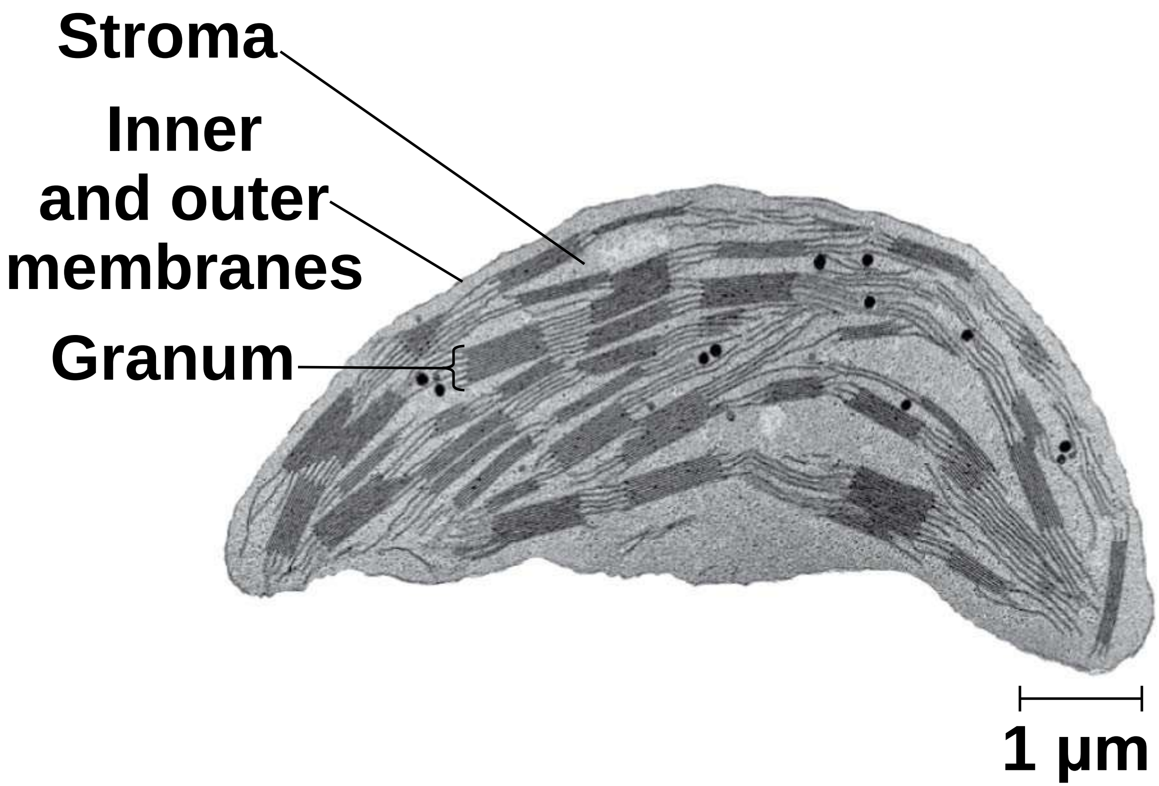
(b) Chloroplasts in an algal cell

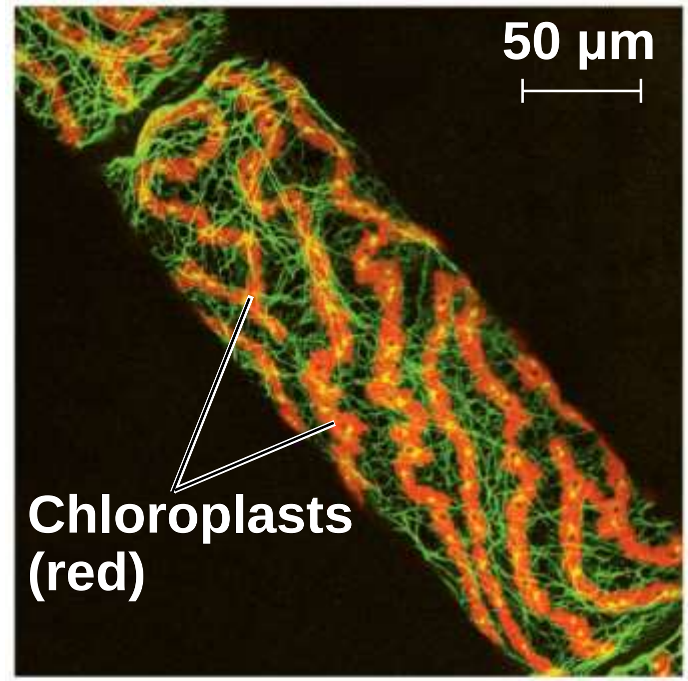
Figure 6.18a



(a) Diagram and TEM of chloroplast

Figure 6.18aa





(b) Chloroplasts in an algal cell

a) Chloroplast structure includes

   a) **Thylakoids**, membranous sacs, stacked to form a **granum**

   b) **Stroma**, the internal fluid

b) The chloroplast is one of a group of plant organelles, called **plastids**

Peroxisomes: Oxidation

a) **Peroxisomes** are specialized metabolic compartments bounded by a **single membrane**

b) Peroxisomes produce hydrogen peroxide and convert it to water

c) Peroxisomes perform reactions with many different functions

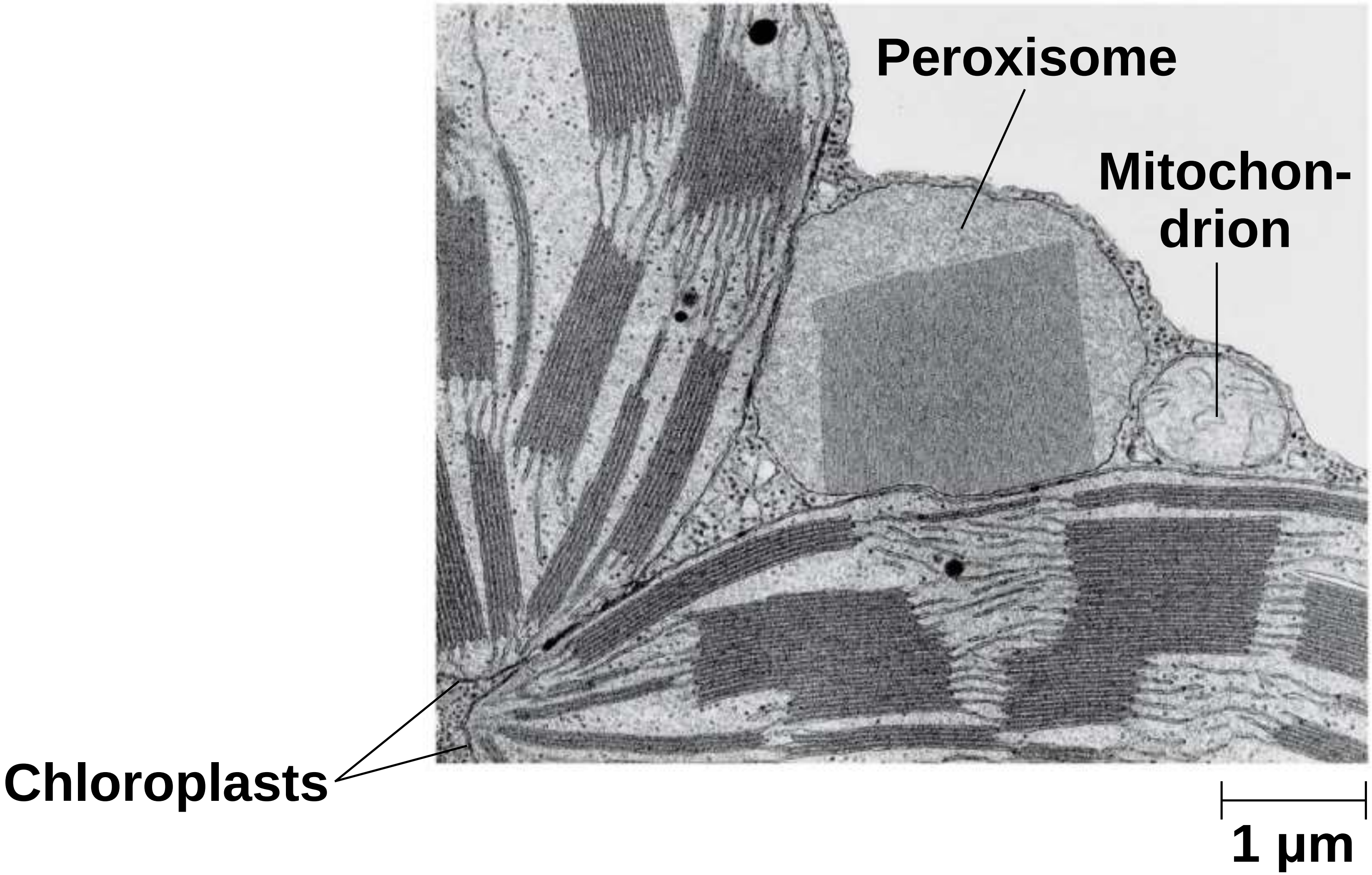
d) How peroxisomes are related to other organelles is still unknown

a) Some peroxisomes use oxygen to **break fatty acids** down into smaller molecules that are transported to mitochondria and used as fuel for cellular respiration.

b) Peroxisomes in the liver **detoxify alcohol** and other harmful compounds by transferring hydrogen from the poisons to oxygen. The H_2O_2 formed by peroxisomes is itself toxic, but the organelle also contains an enzyme that converts H_2O_2 to water

c) Glyoxysomes are found in the fat-storing tissues of plant seeds. These organelles contain enzymes that initiate the conversion of **fatty acids to sugar**

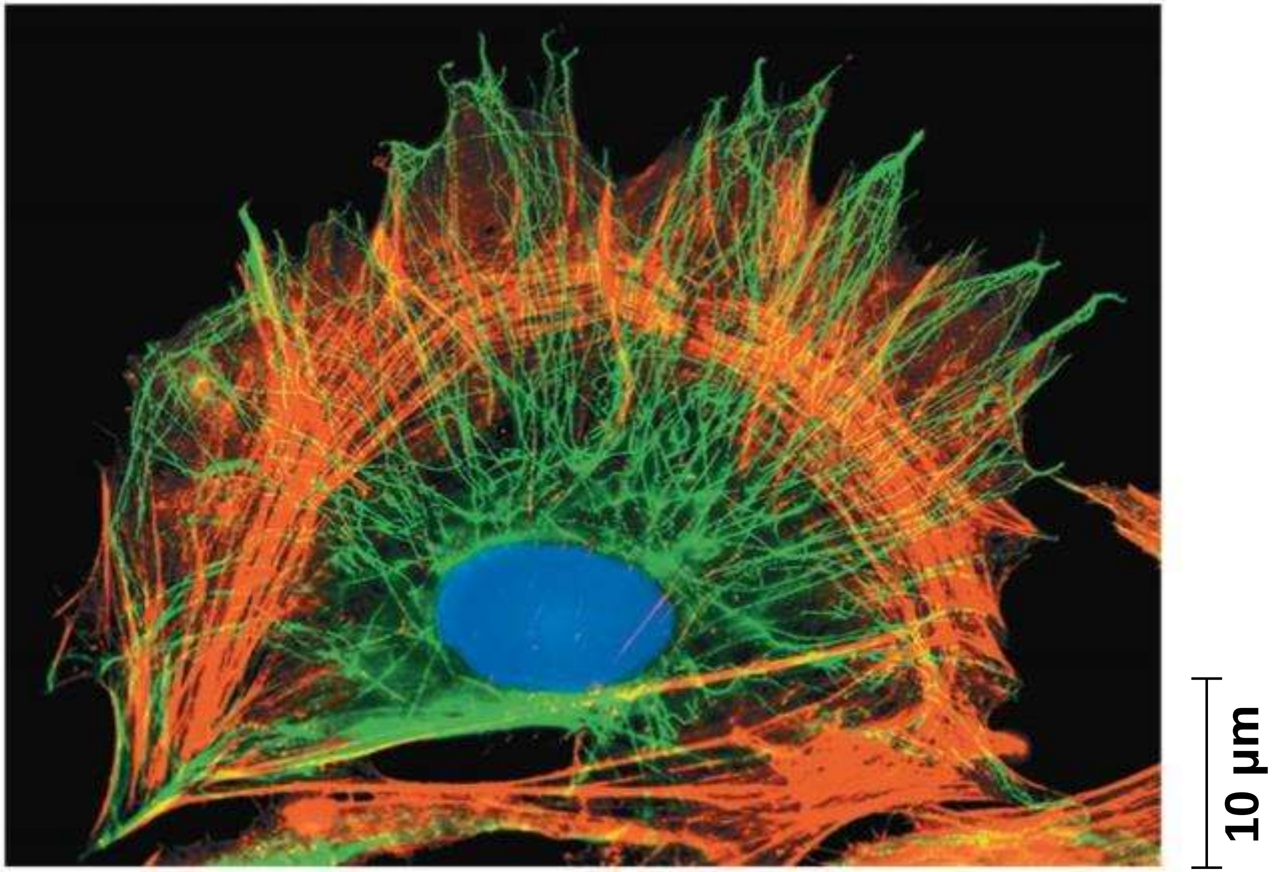
Figure 6.19



Concept 6.6: The cytoskeleton is a network of fibers that organizes structures and activities in the cell

a) The **cytoskeleton** is a network of fibers extending throughout the cytoplasm
b) It organizes the cell's structures and activities, anchoring many organelles
c) It is composed of three types of molecular structures
    a) Microtubules
    b) Microfilaments
    c) Intermediate filaments

Figure 6.20



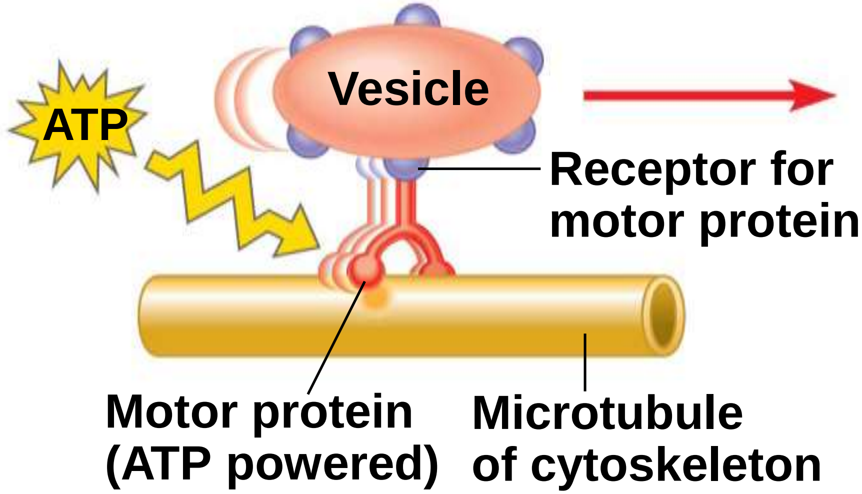
Roles of the Cytoskeleton: Support and Motility

a) The cytoskeleton helps to support the cell and maintain its shape

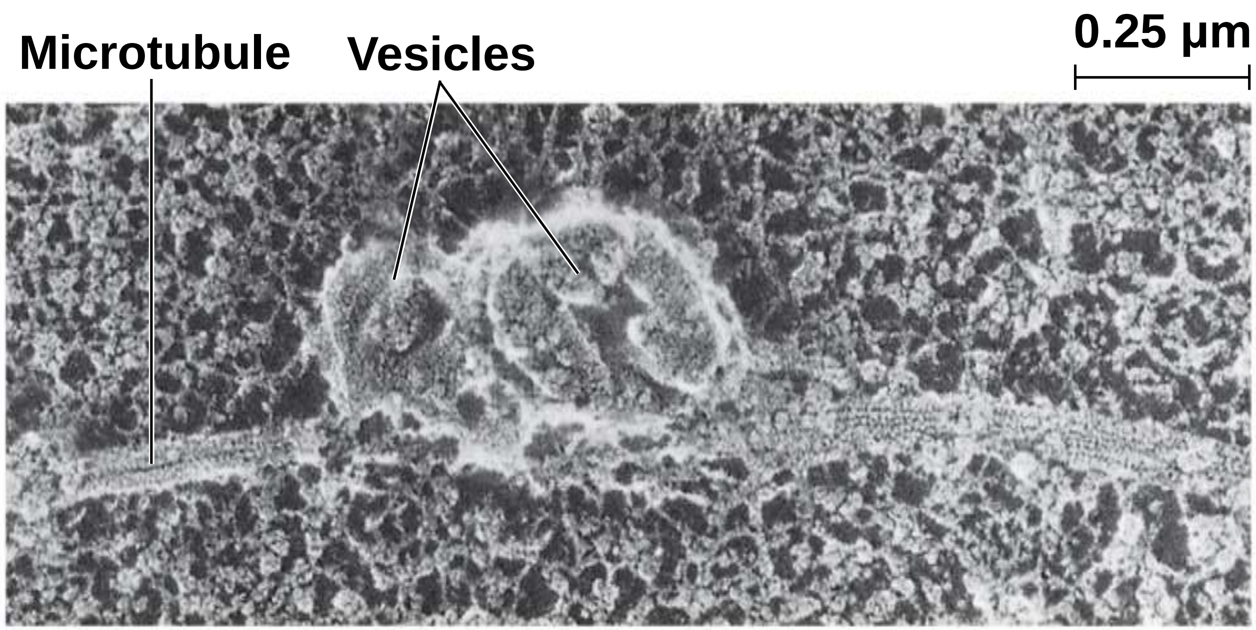
b) It interacts with **motor proteins** to produce motility

c) Inside the cell, vesicles can travel along tracks provided by the cytoskeleton

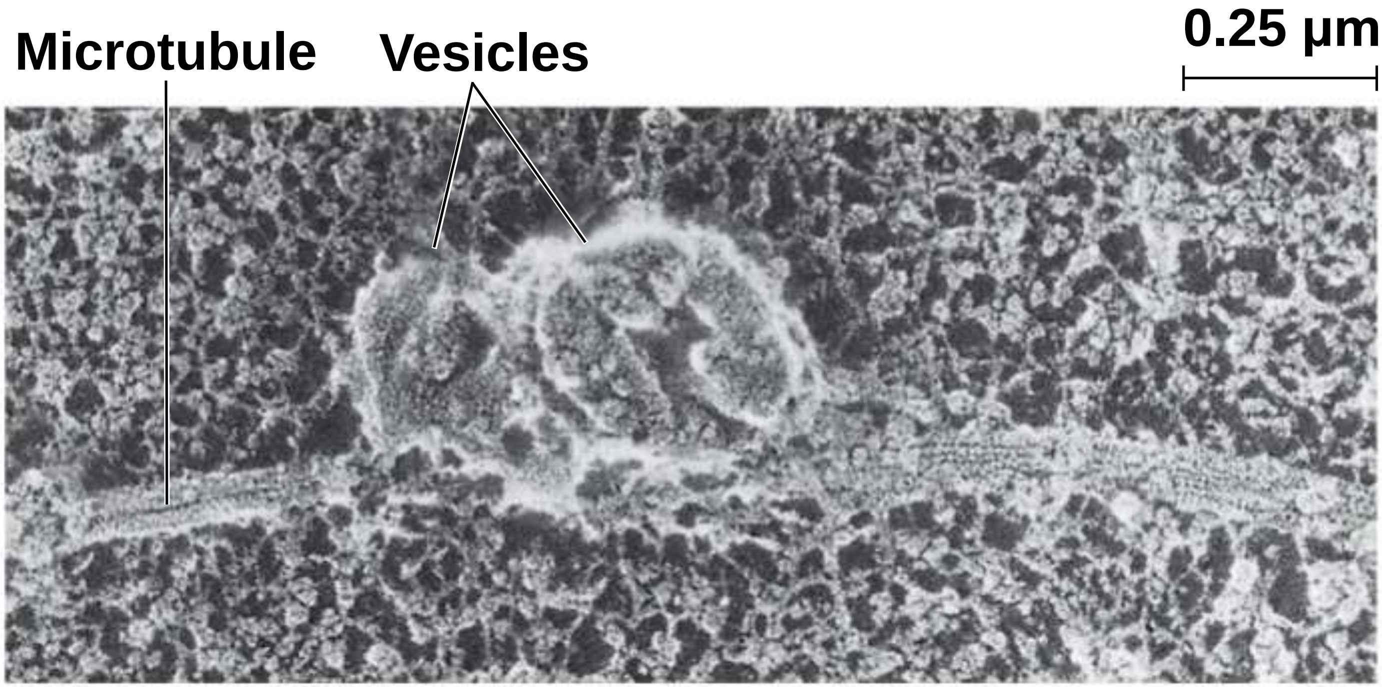
Figure 6.21



(a) Motor proteins “walk” vesicles along cytoskeletal fibers.



(b) SEM of a squid giant axon



(b) SEM of a squid giant axon

Components of the Cytoskeleton

a) Three main types of fibers make up the cytoskeleton

  a) *Microtubules* are the thickest of the three components of the cytoskeleton

  b) *Microfilaments*, also called actin filaments, are the thinnest components

  c) *Intermediate filaments* are fibers with diameters in a middle range

Table 6.1 The Structure and Function of the Cytoskeleton

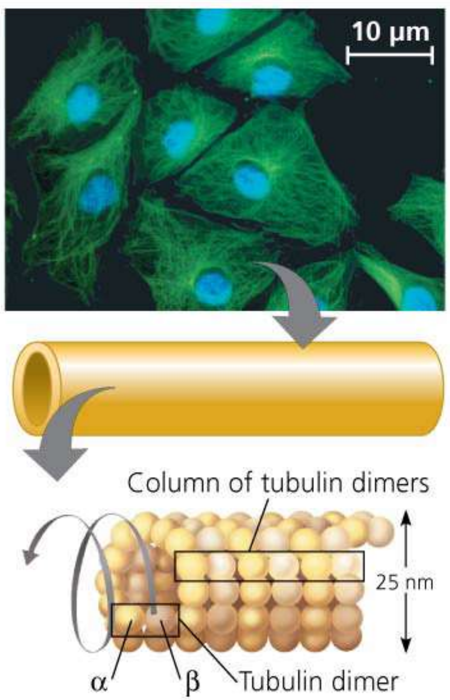
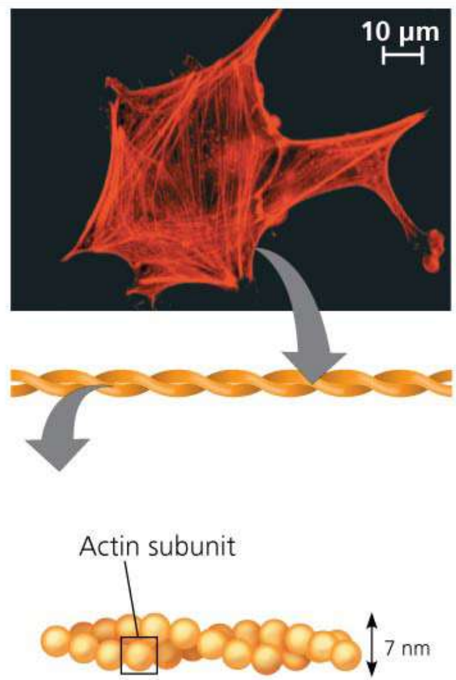
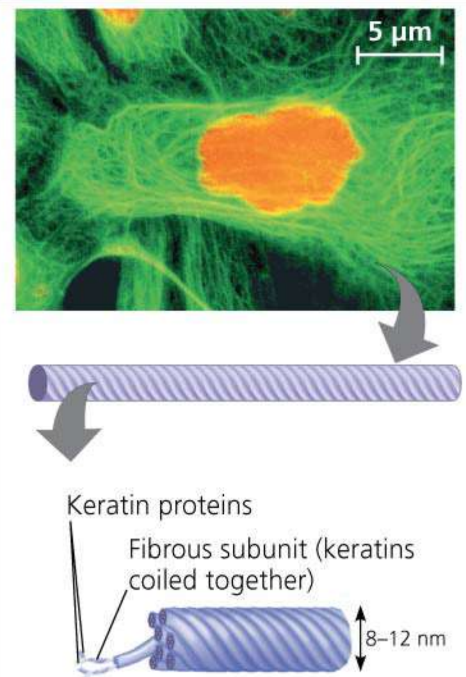
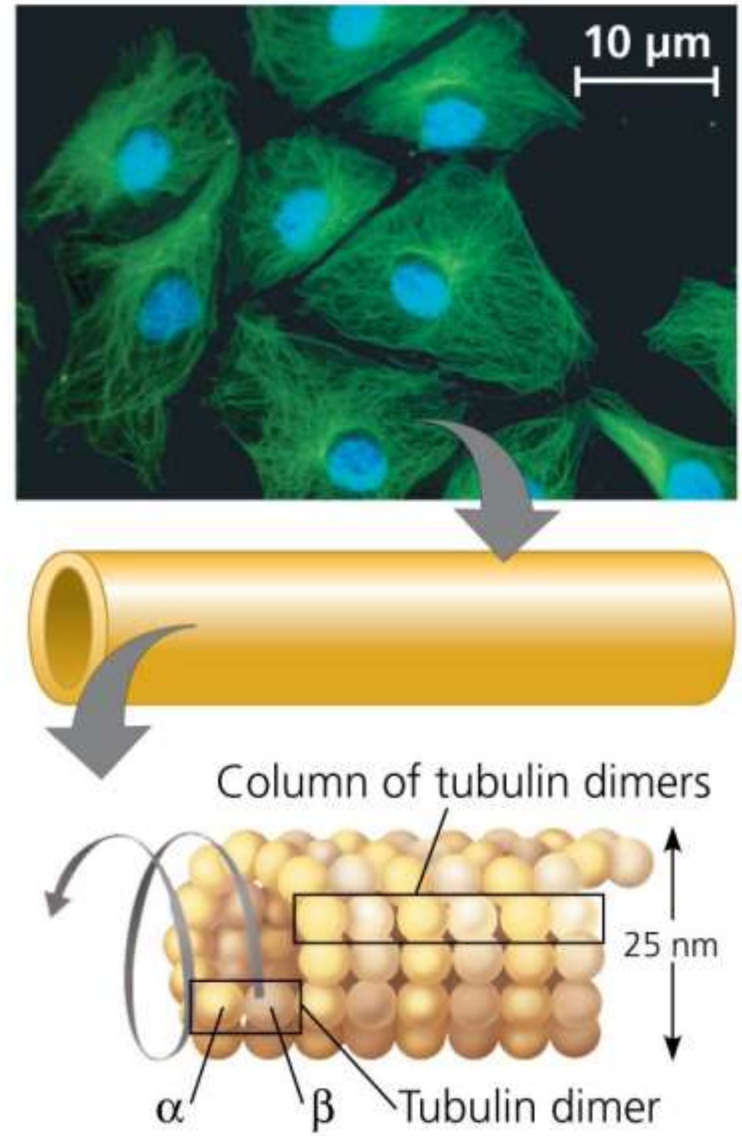
Property	Microtubules (Tubulin Polymers)	Microfilaments (Actin Filaments)	Intermediate Filaments
Structure	Hollow tubes	Two intertwined strands of actin	Fibrous proteins coiled into cables
Diameter	25 nm with 15-nm lumen	7 nm	8–12 nm
Protein subunits	Tubulin, a dimer consisting of α-tubulin and β-tubulin	Actin	One of several different proteins (such as keratins)
Main functions	Maintenance of cell shape (compression-resisting “girders”); cell motility (as in cilia or flagella); chromosome movements in cell division; organelle movements	Maintenance of cell shape (tension-bearing elements); changes in cell shape; muscle contraction; cytoplasmic streaming in plant cells; cell motility (as in amoeboid movement); division of animal cells	Maintenance of cell shape (tension-bearing elements); anchorage of nucleus and certain other organelles; formation of nuclear lamina
Fluorescence micrographs of fibroblasts. Fibroblasts are a favorite cell type for cell biology studies. In each, the structure of interest has been tagged with fluorescent molecules. The DNA in the nucleus has also been tagged in the first micrograph (blue) and third micrograph (orange).			

Table 6.1a

Table 6.1 The Structure and Function of the Cytoskeleton

Microtubules (Tubulin Polymers)	Microfilaments (Actin Filaments)	Intermediate Filaments
Hollow tubes	Two intertwined strands of actin	Fibrous proteins coiled into cables
25 nm with 15-nm lumen	7 nm	8–12 nm
Tubulin, a dimer consisting of α-tubulin and β-tubulin	Actin	One of several different proteins (such as keratins)
Maintenance of cell shape (compression-resisting "girders"); cell motility (as in cilia or flagella); chromosome movements in cell division; organelle movements	Maintenance of cell shape (tension-bearing elements); changes in cell shape; muscle contraction; cytoplasmic streaming in plant cells; cell motility (as in amoeboid movement); division of animal cells	Maintenance of cell shape (tension-bearing elements); anchorage of nucleus and certain other organelles; formation of nuclear lamina

Microtubules (Tubulin Polymers)
Hollow tubes
25 nm with 15-nm lumen
Tubulin, a dimer consisting of α-tubulin and β-tubulin
Maintenance of cell shape (compression-resisting "girders"); cell motility (as in cilia or flagella); chromosome movements in cell division; organelle movements



Microfilaments (Actin Filaments)
Two intertwined strands of actin
7 nm
Actin
Maintenance of cell shape (tension-bearing elements); changes in cell shape; muscle contraction; cytoplasmic streaming in plant cells; cell motility (as in amoeboid movement); division of animal cells

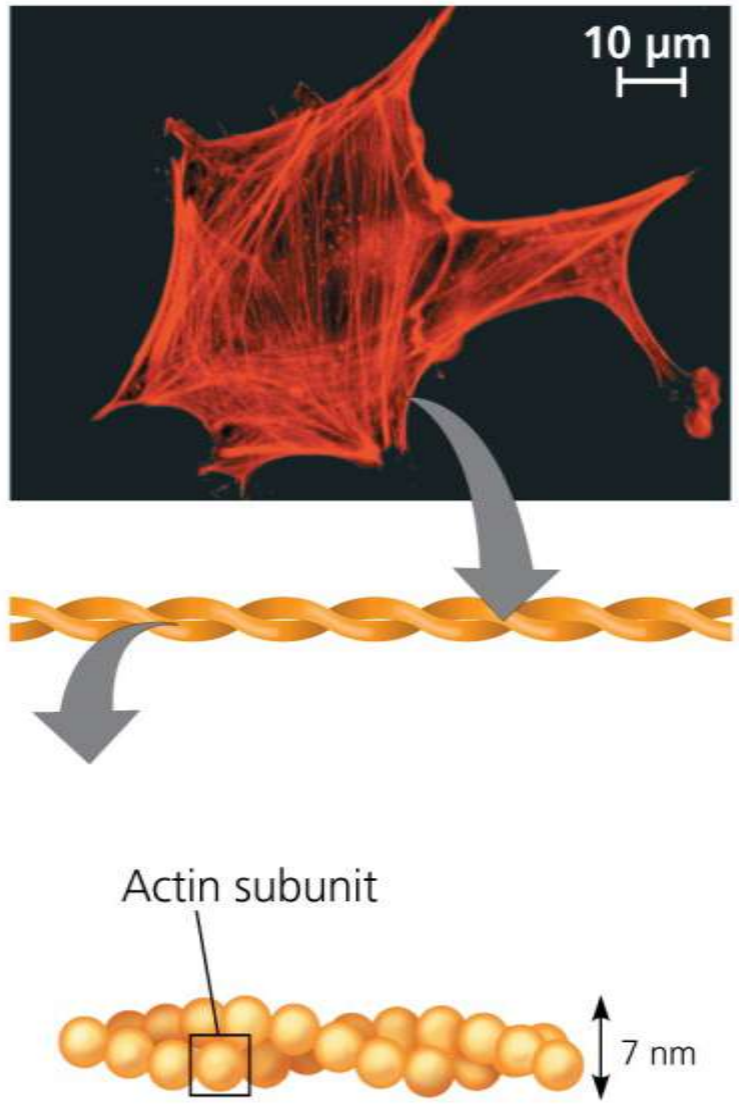
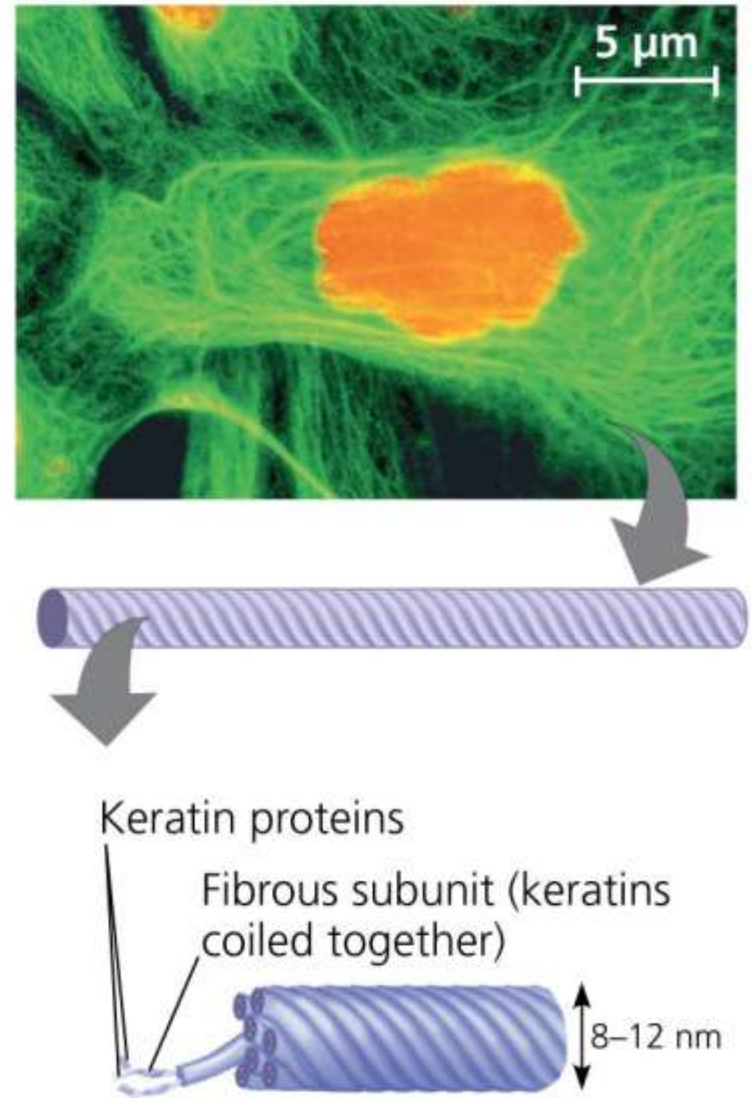


Table 6.1d

Intermediate Filaments
Fibrous proteins coiled into cables
8–12 nm
One of several different proteins (such as keratins)
Maintenance of cell shape (tension-bearing elements); anchorage of nucleus and certain other organelles; formation of nuclear lamina



Microtubules

a) **Microtubules** are hollow rods about 25 nm in diameter and about 200 nm to 25 microns long

b) Functions of microtubules

   a) Shaping the cell

   b) Guiding movement of organelles

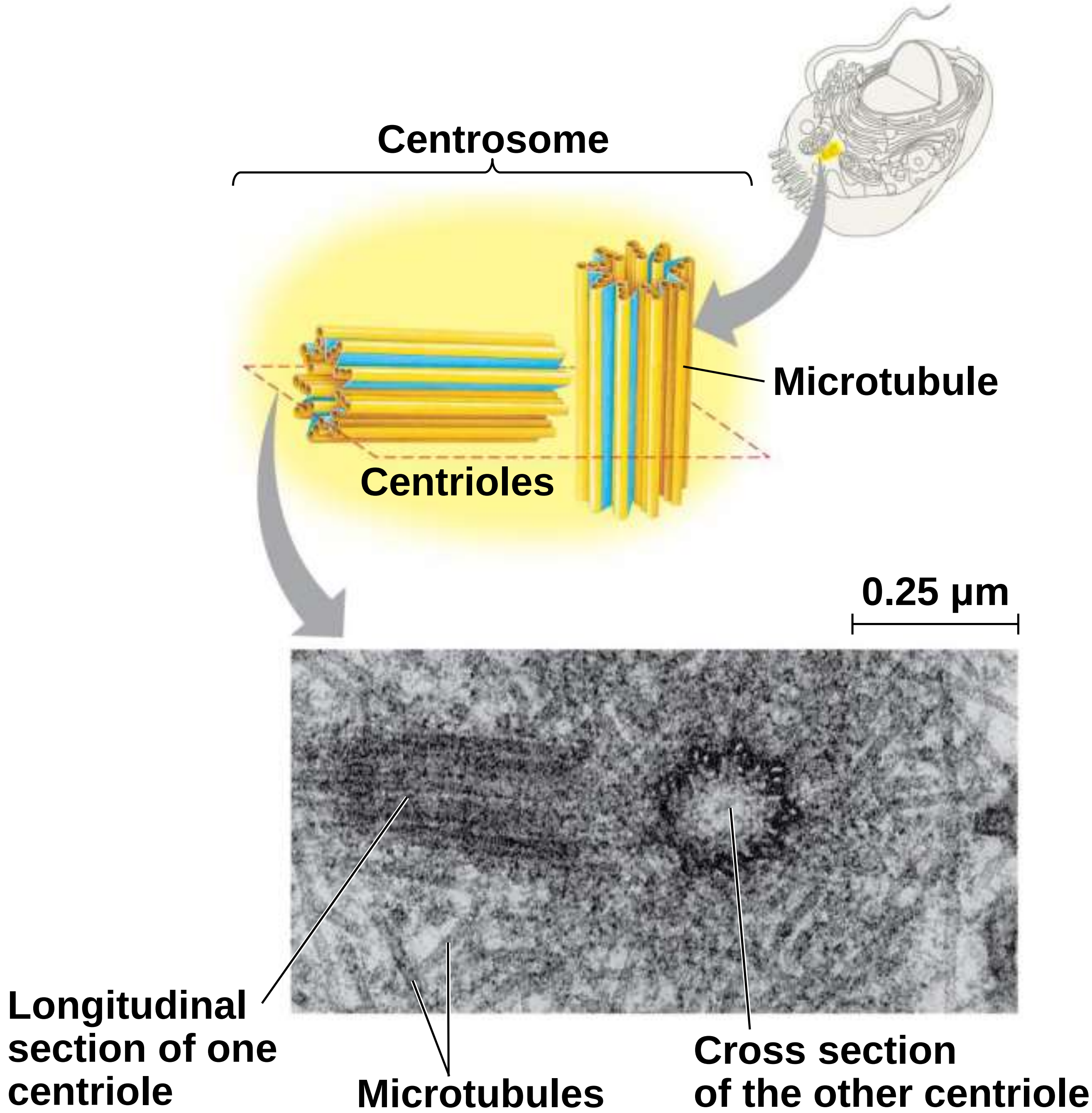
   c) Separating chromosomes during cell division

Centrosomes and Centrioles

a) In animal cells, microtubules grow out from a **centrosome** near the nucleus

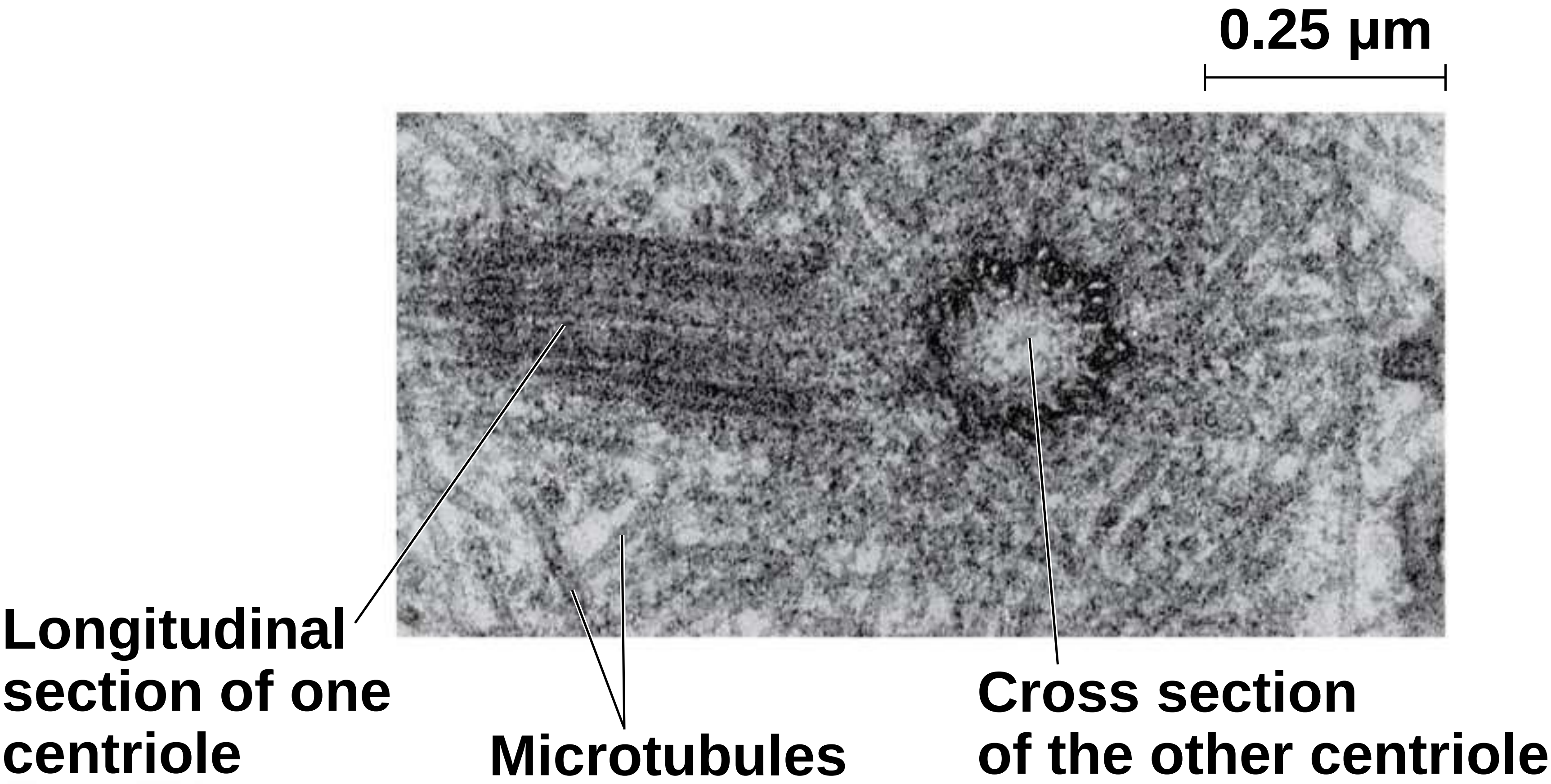
b) In animal cells, the centrosome has a pair of **centrioles**, each with nine triplets of microtubules arranged in a ring

Figure 6.22



Centrosome
Microtubule
Centrioles
0.25 µm
Longitudinal section of one centriole
Microtubules
Cross section of the other centriole

Figure 6.22a


0.25 µm
Longitudinal section of one centriole
Microtubules
Cross section of the other centriole

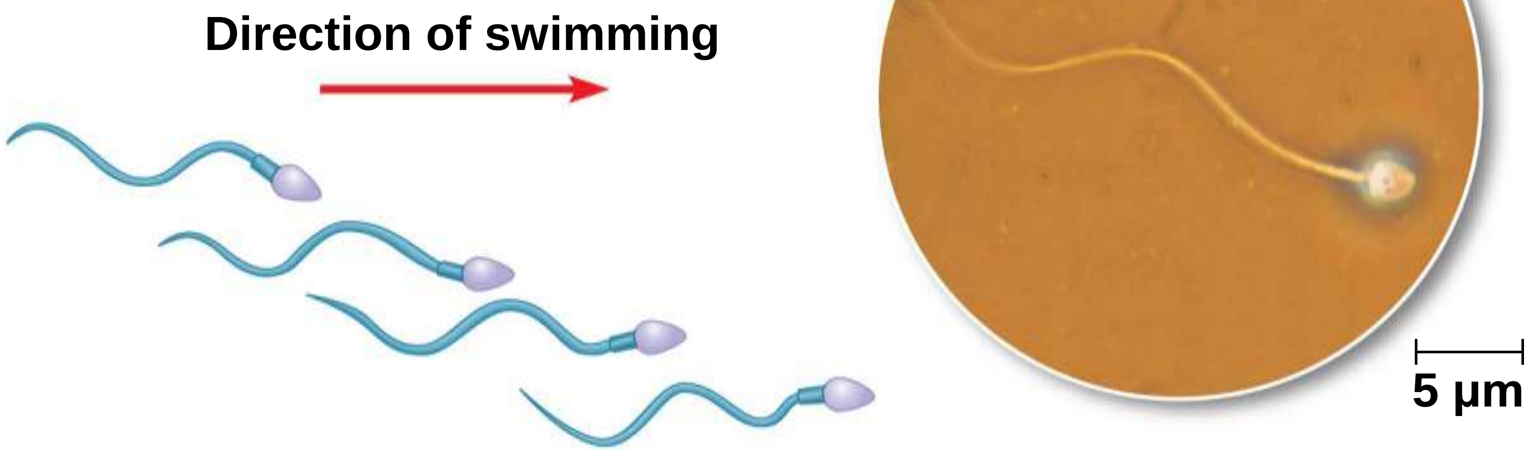
Cilia and Flagella

a) Microtubules control the beating of **flagella** and **cilia**, microtubule-containing extensions that project from some cells

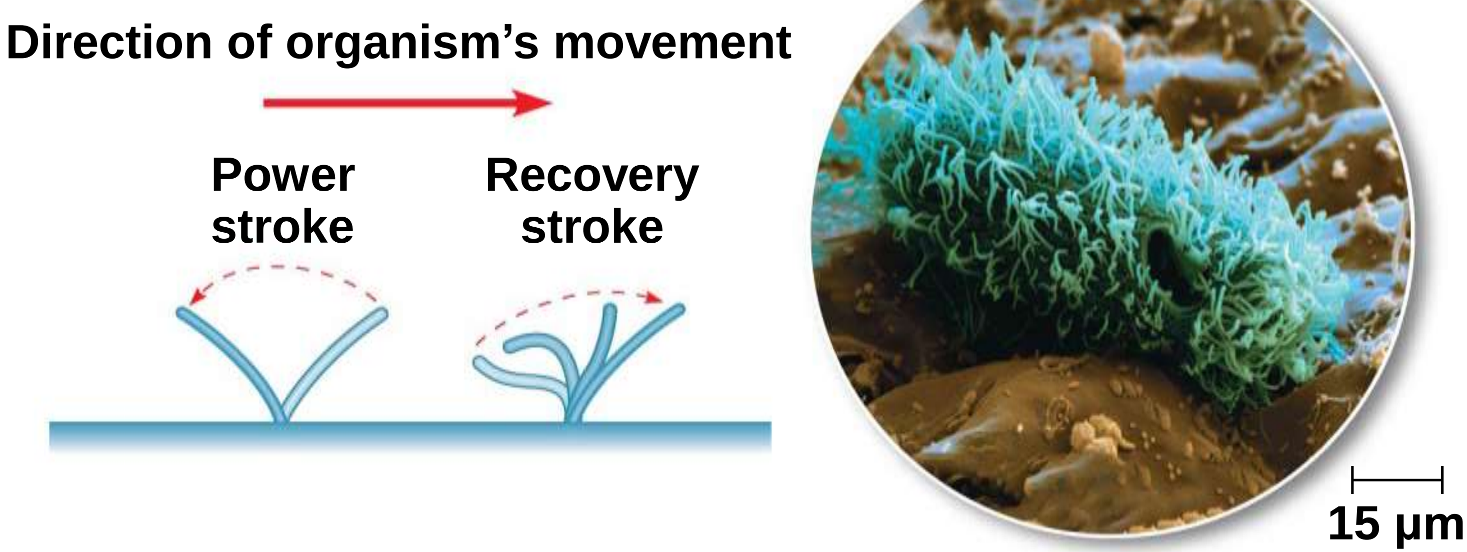
b) Cilia and flagella differ in their beating patterns

Figure 6.23

(a) Motion of flagella

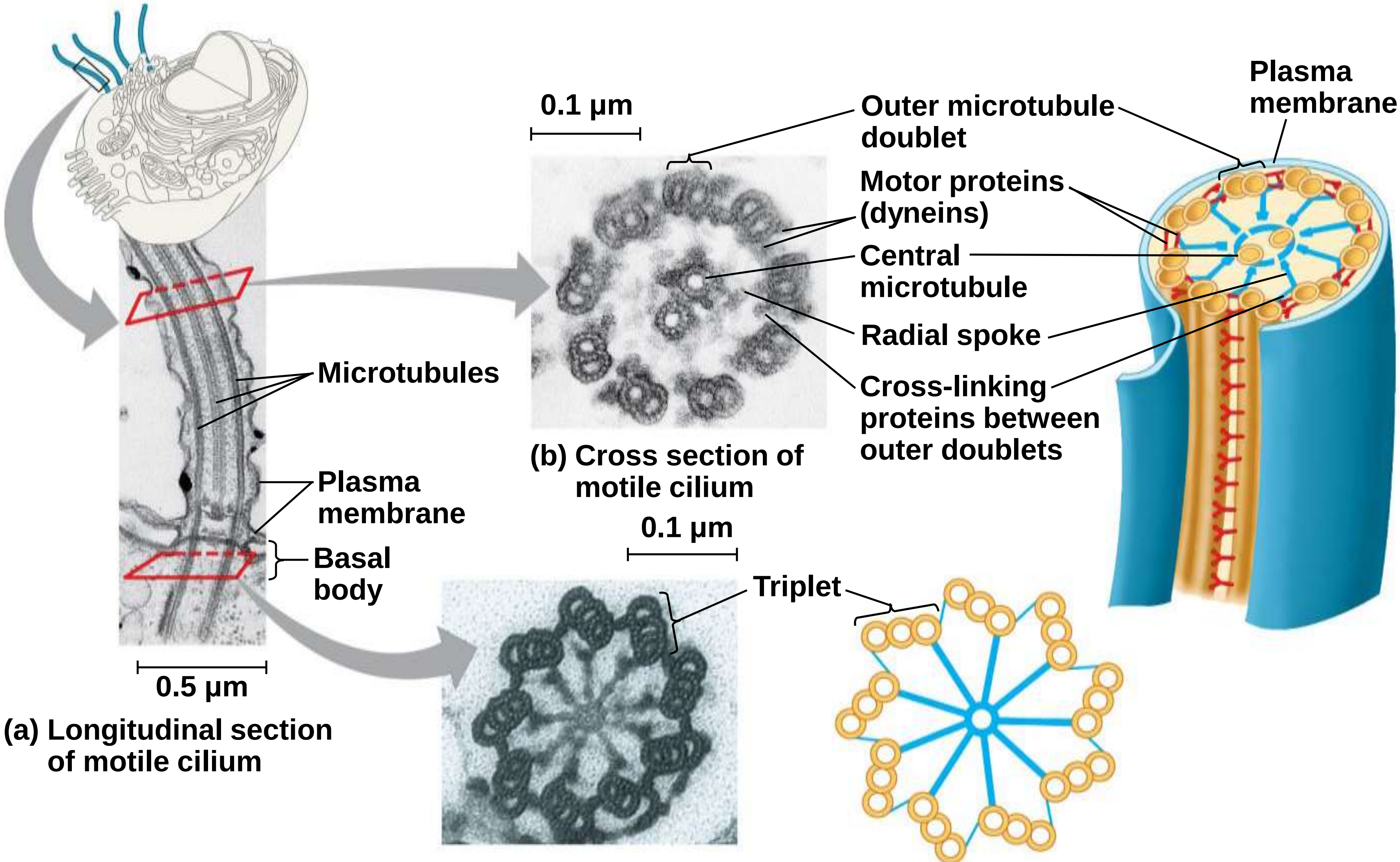


(b) Motion of cilia



a) Cilia and flagella share a common structure
   a) A core of microtubules sheathed by the plasma membrane
   b) A **basal body** that anchors the cilium or flagellum
   c) A motor protein called **dynein**, which drives the bending movements of a cilium or flagellum

Figure 6.24

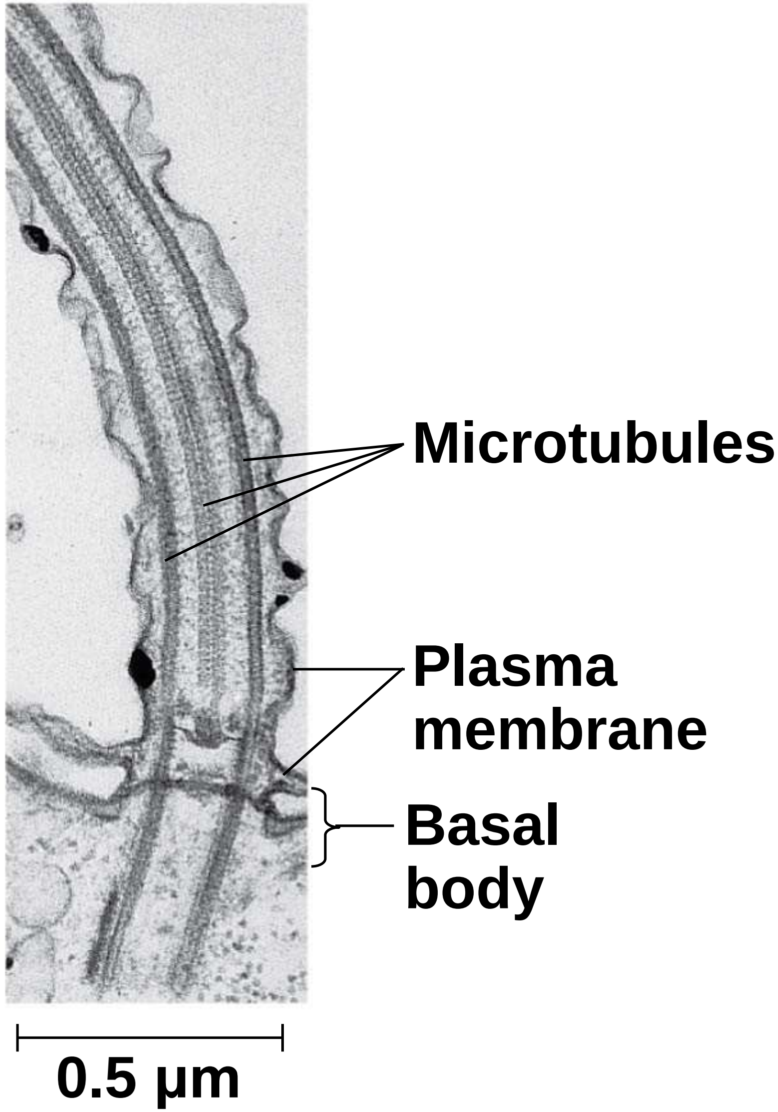


(a) Longitudinal section of motile cilium

(b) Cross section of motile cilium

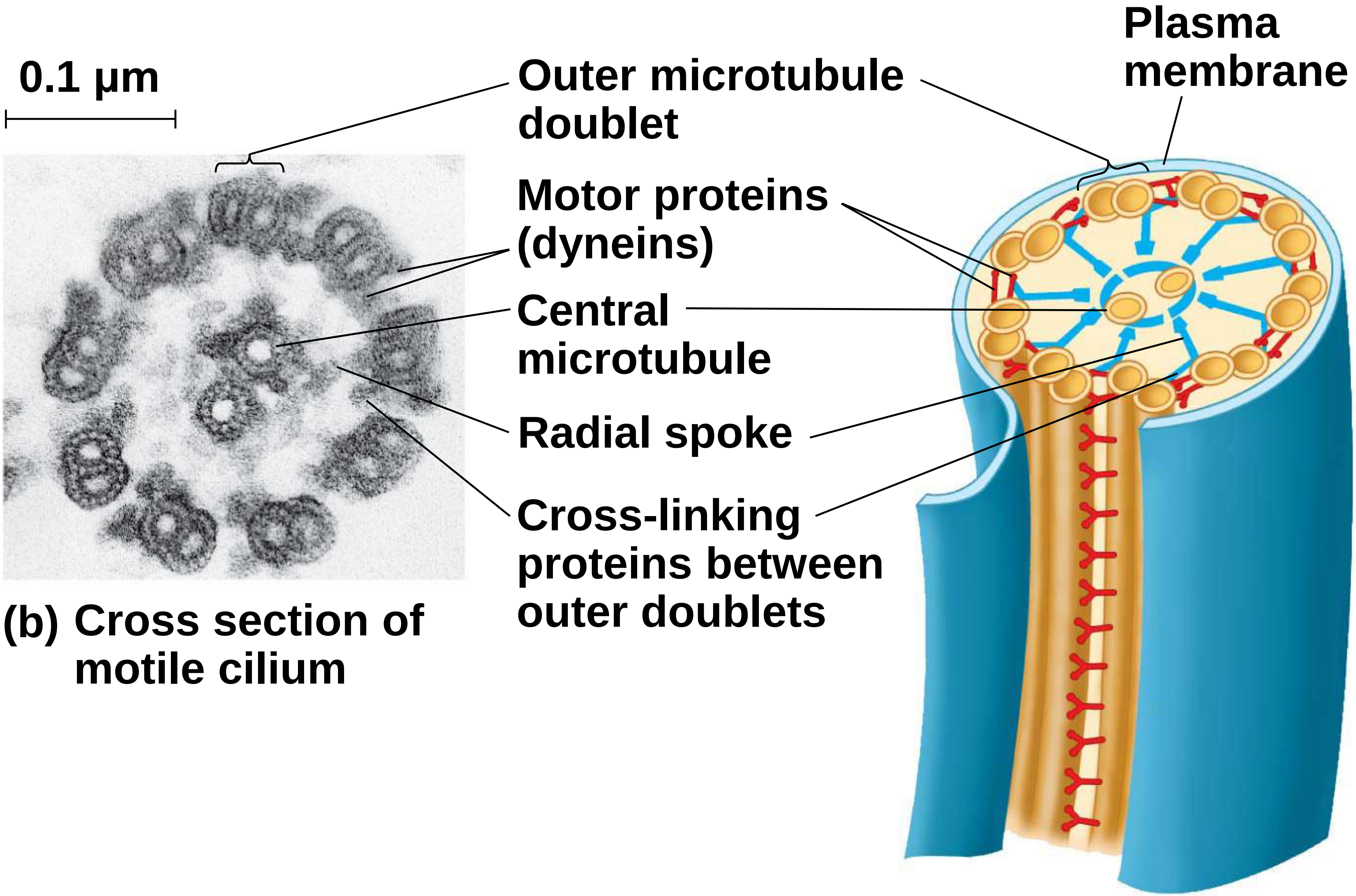
(c) Cross section of basal body

Figure 6.24a



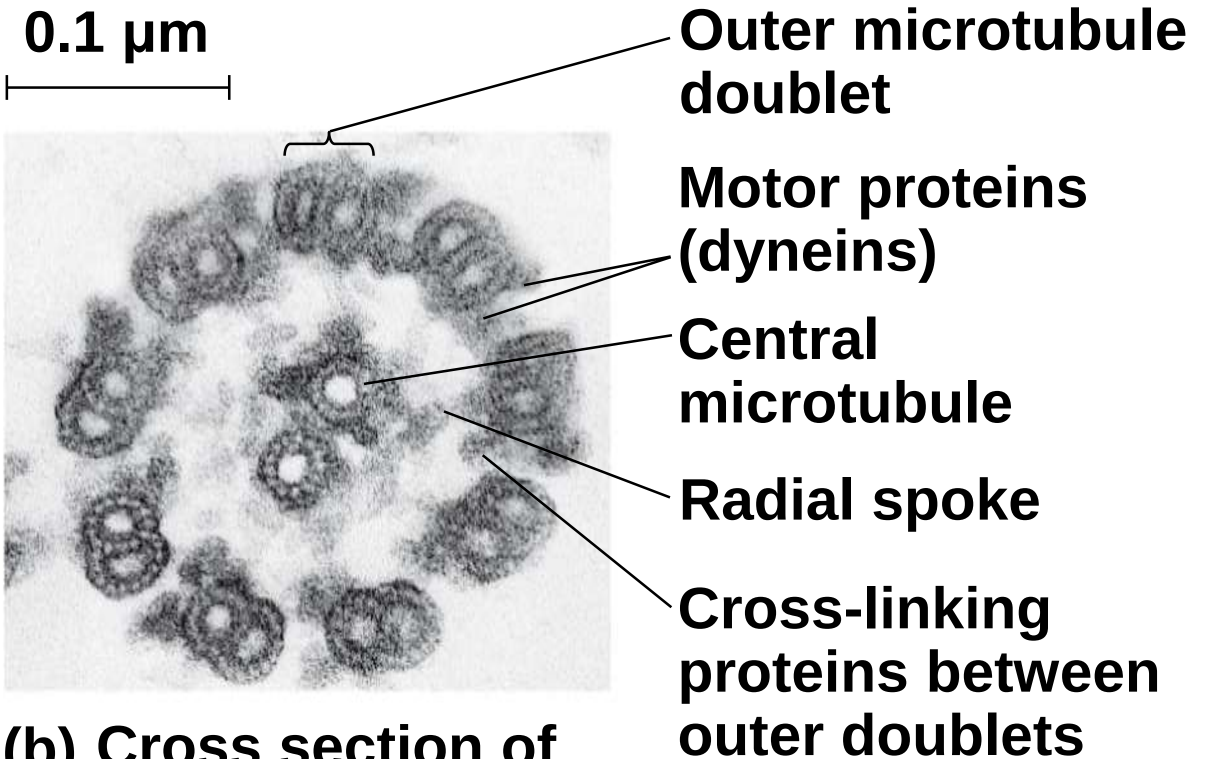
(a) Longitudinal section of motile cilium

Figure 6.24b



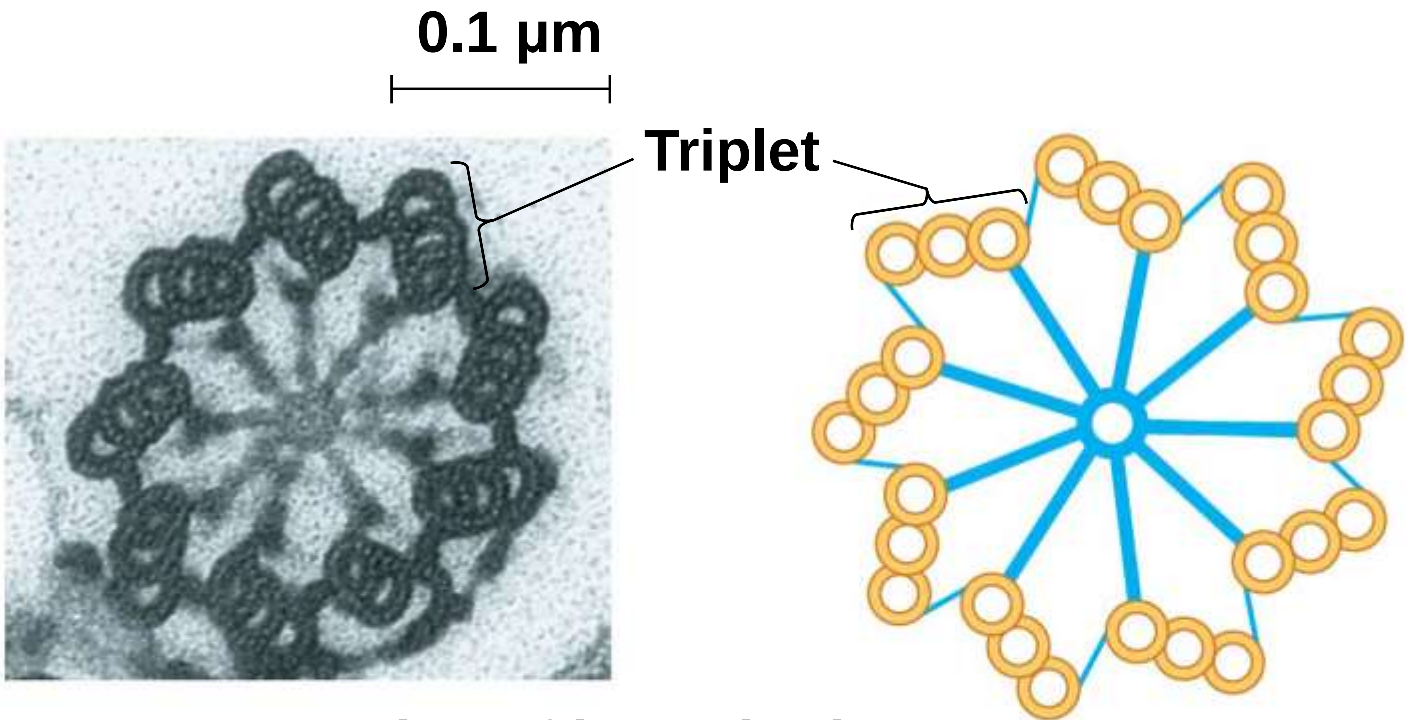
(b) Cross section of motile cilium

Figure 6.24ba



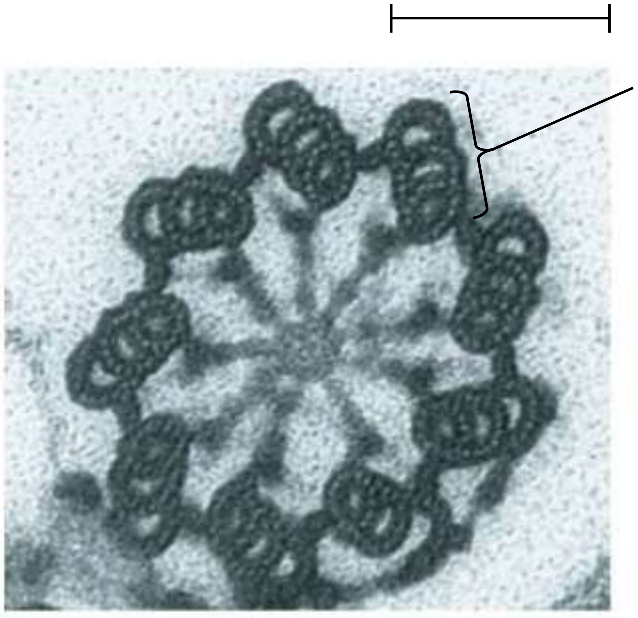
(b) Cross section of motile cilium

Figure 6.24c



(c) Cross section of basal body

Figure 6.24ca



(c) Cross section of basal body

a) How dynein “walking” moves flagella and cilia

   a) Dynein arms alternately grab, move, and release the outer microtubules

   b) Protein cross-links limit sliding

   c) Forces exerted by dynein arms cause doublets to curve, bending the cilium or flagellum

Microfilaments (Actin Filaments)

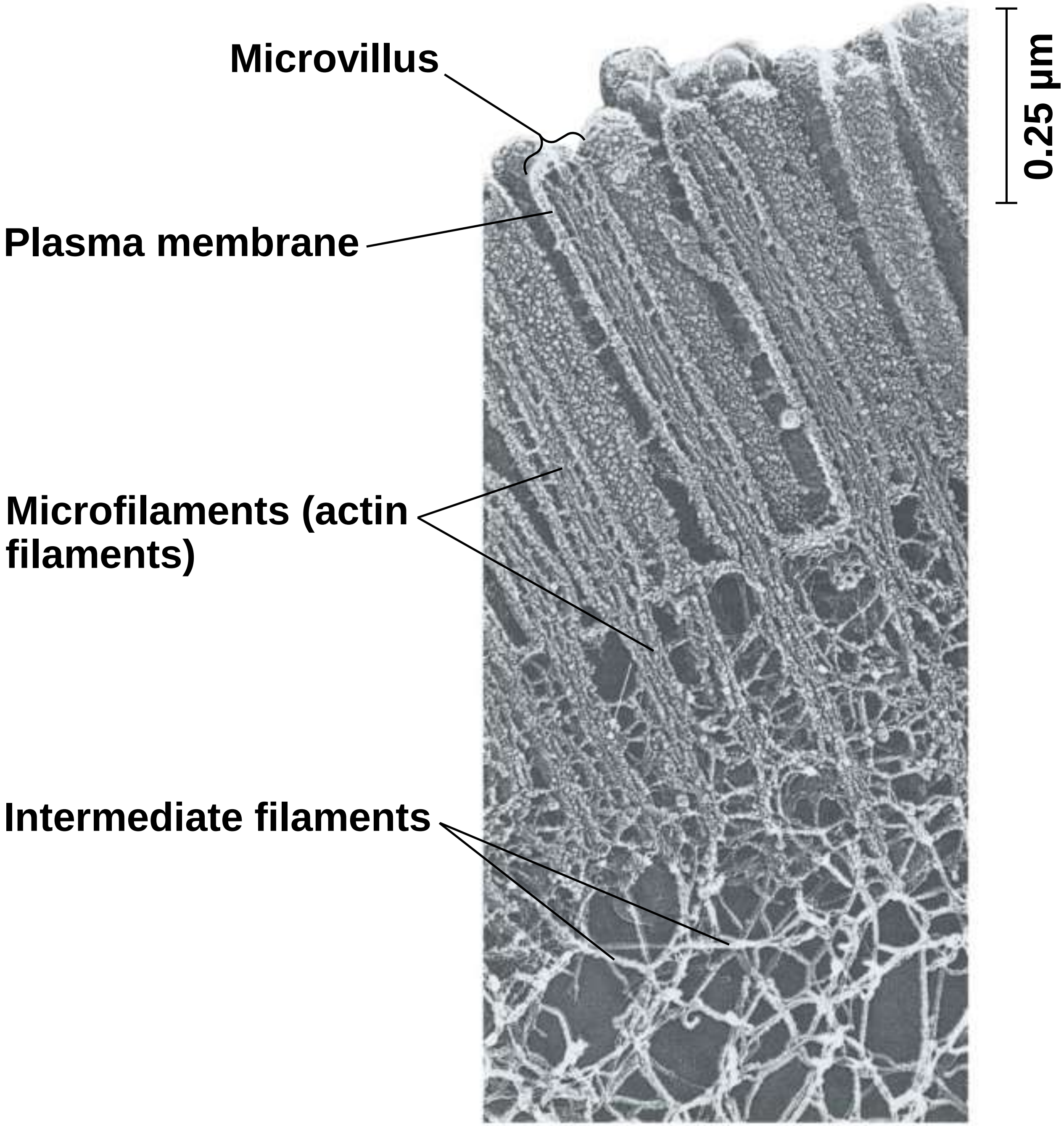
a) **Microfilaments** are solid rods about 7 nm in diameter, built as a twisted double chain of **actin** subunits

b) The structural role of microfilaments is to bear tension, resisting pulling forces within the cell

c) They form a 3-D network called the **cortex** just inside the plasma membrane to help support the cell’s shape

d) Bundles of microfilaments make up the core of microvilli of intestinal cells

Figure 6.25



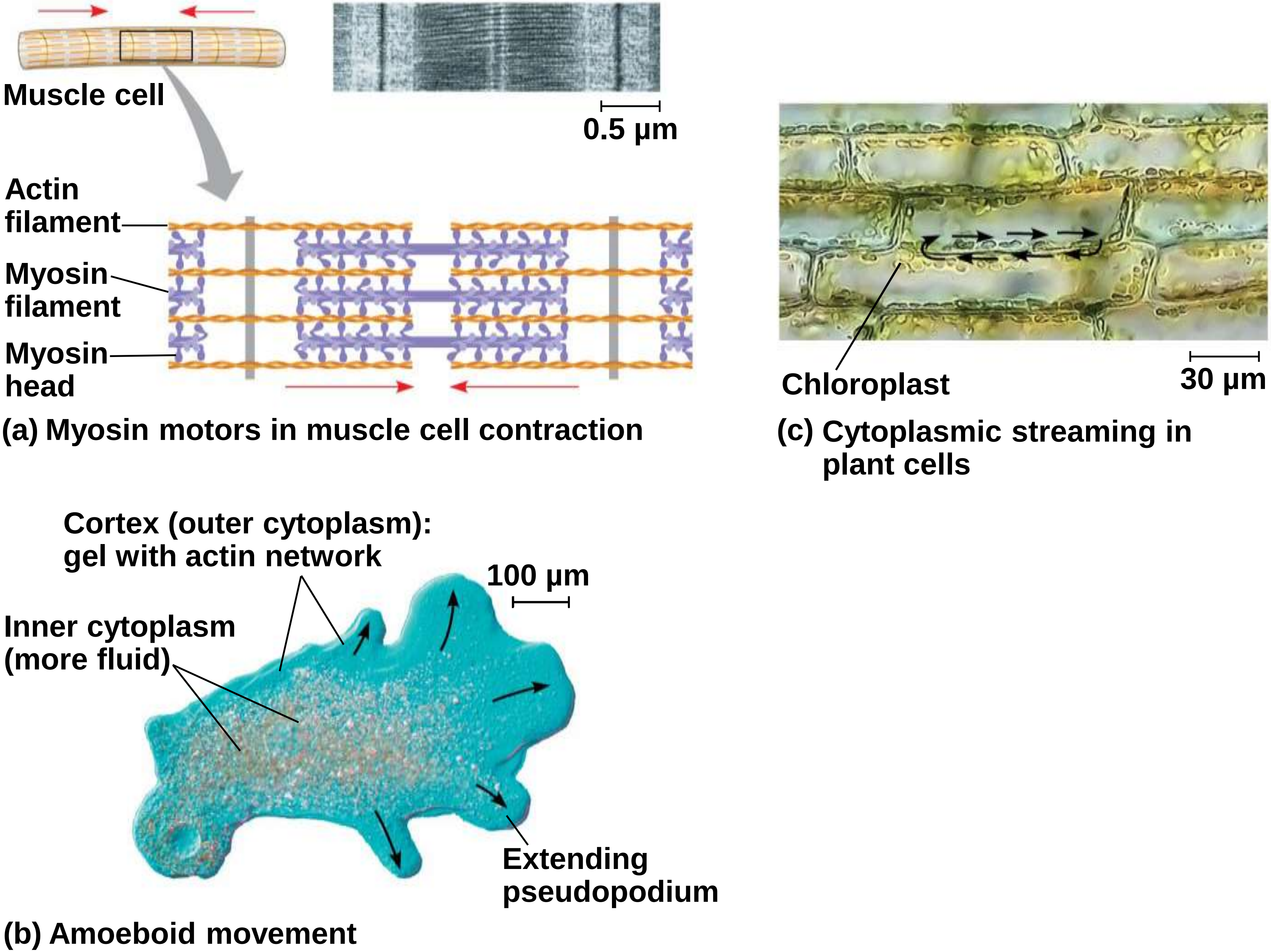
Microvillus
Plasma membrane
Microfilaments (actin filaments)
Intermediate filaments
0.25 µm

a) Microfilaments that function in cellular motility contain the protein **myosin** in addition to actin

b) In muscle cells, thousands of actin filaments are arranged parallel to one another

c) Thicker filaments composed of myosin interdigitate with the thinner actin fibers

Figure 6.26

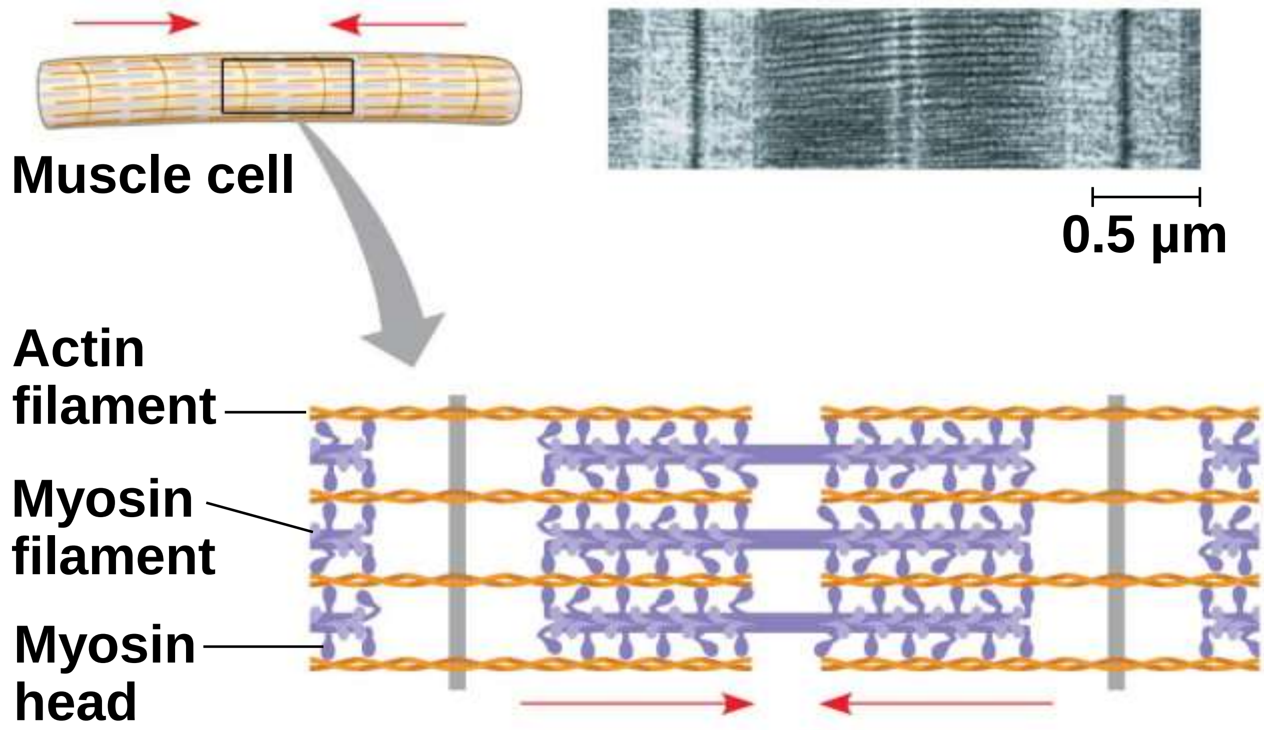


(a) Myosin motors in muscle cell contraction

(b) Amoeboid movement

(c) Cytoplasmic streaming in plant cells

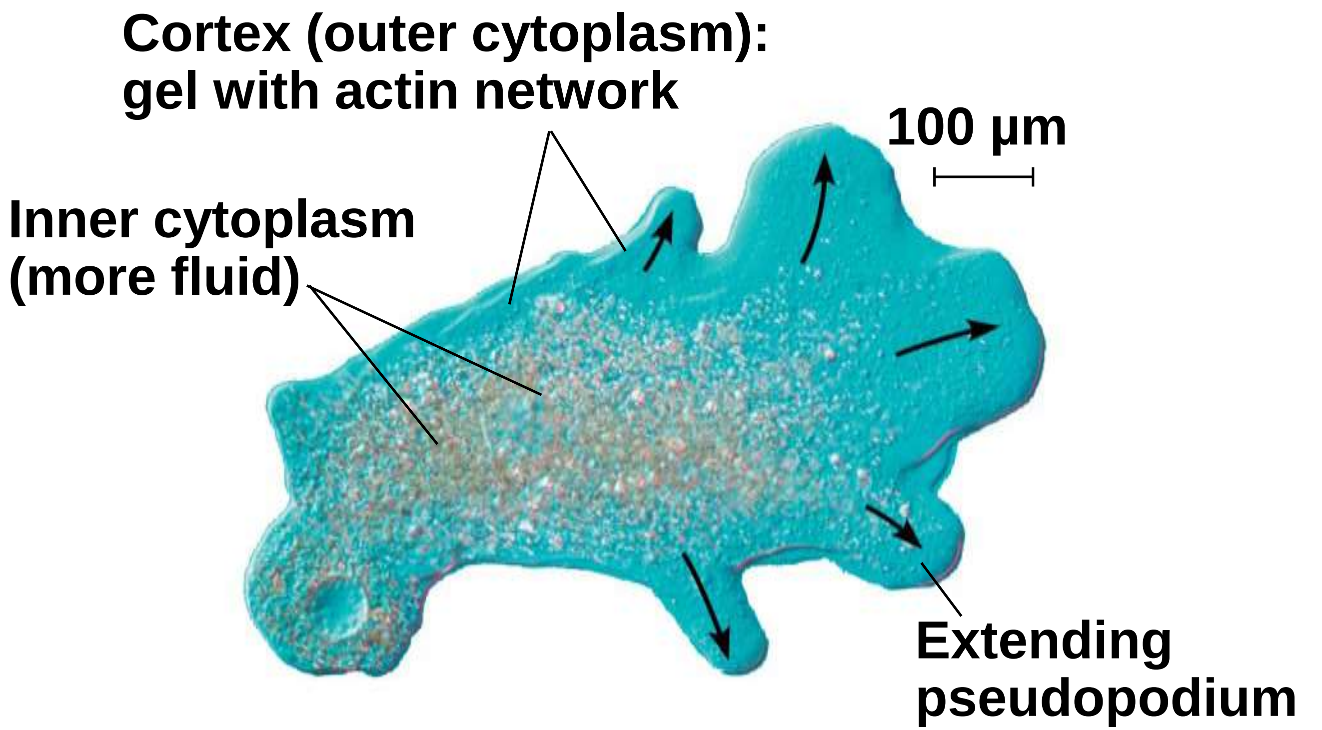
Figure 6.26a



(a) Myosin motors in muscle cell contraction

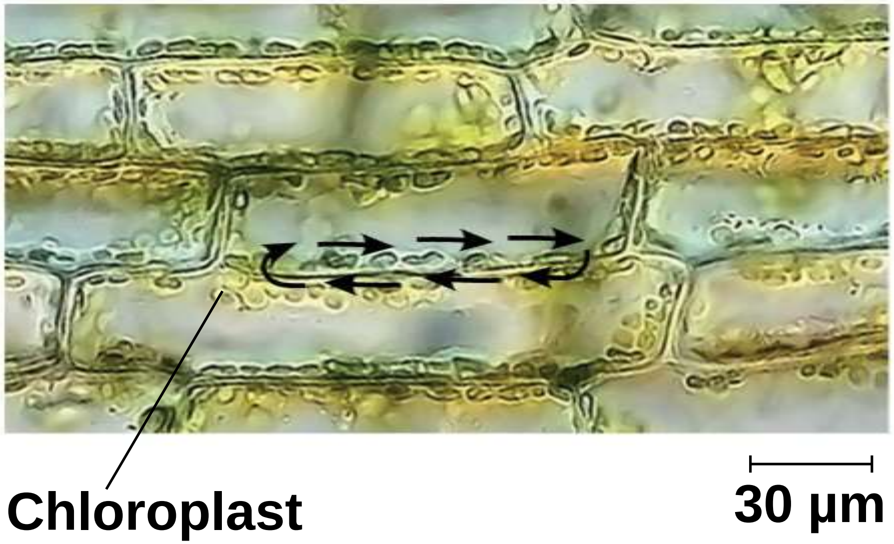
Figure 6.26aa

Figure 6.26b



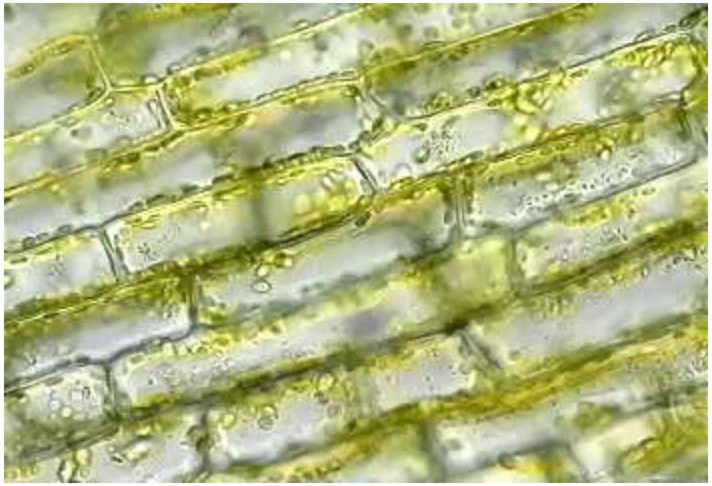
(b) Amoeboid movement

Figure 6.26c



(c) Cytoplasmic streaming in plant cells

Video: Cytoplasmic Streaming



a) Localized contraction brought about by actin and myosin also drives amoeboid movement

b) Cells crawl along a surface by extending **pseudopodia** (cellular extensions) and moving toward them

a) **Cytoplasmic streaming** is a circular flow of cytoplasm within cells

b) This streaming speeds distribution of materials within the cell

c) In plant cells, actin-myosin interactions and sol-gel transformations drive cytoplasmic streaming

Intermediate Filments

a) **Intermediate filaments** range in diameter from 8–12 nanometers, larger than microfilaments but smaller than microtubules

b) They support cell shape and fix organelles in place

c) Intermediate filaments are more permanent cytoskeleton fixtures than the other two classes

Concept 6.7: Extracellular components and connections between cells help coordinate cellular activities

a) Most cells synthesize and secrete materials that are external to the plasma membrane

b) These extracellular structures are involved in a great many cellular functions

Cell Walls of Plants

a) The **cell wall** is an extracellular structure that distinguishes plant cells from animal cells

b) Prokaryotes, fungi, and some unicellular eukaryotes also have cell walls

c) The cell wall protects the plant cell, maintains its shape, and prevents excessive uptake of water

d) Plant cell walls are made of cellulose fibers embedded in other polysaccharides and protein

a) Plant cell walls may have multiple layers

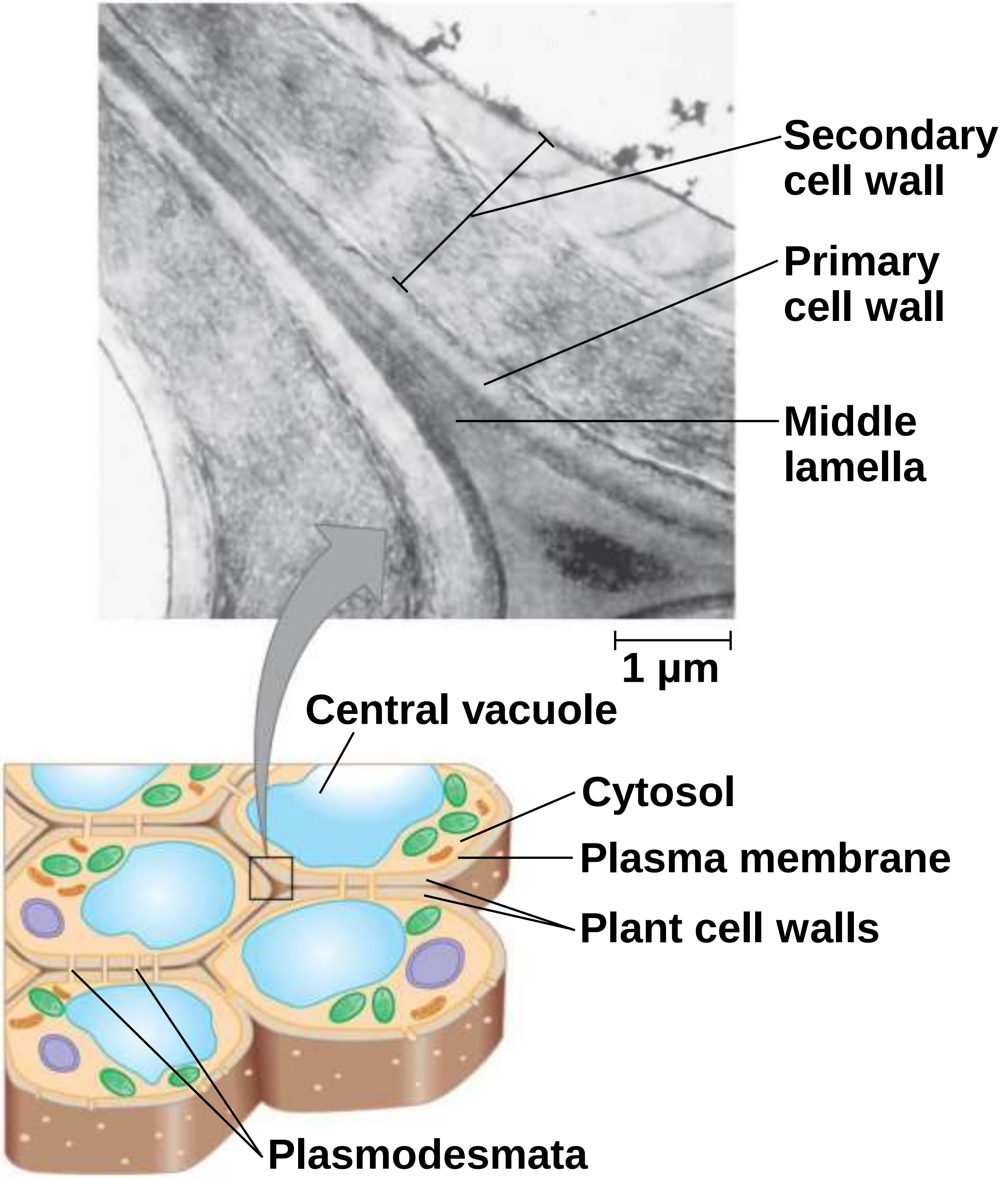
   a) **Primary cell wall**: Relatively thin and flexible

   b) **Middle lamella**: Thin layer between primary walls of adjacent cells

   c) **Secondary cell wall** (in some cells): Added between the plasma membrane and the primary cell wall

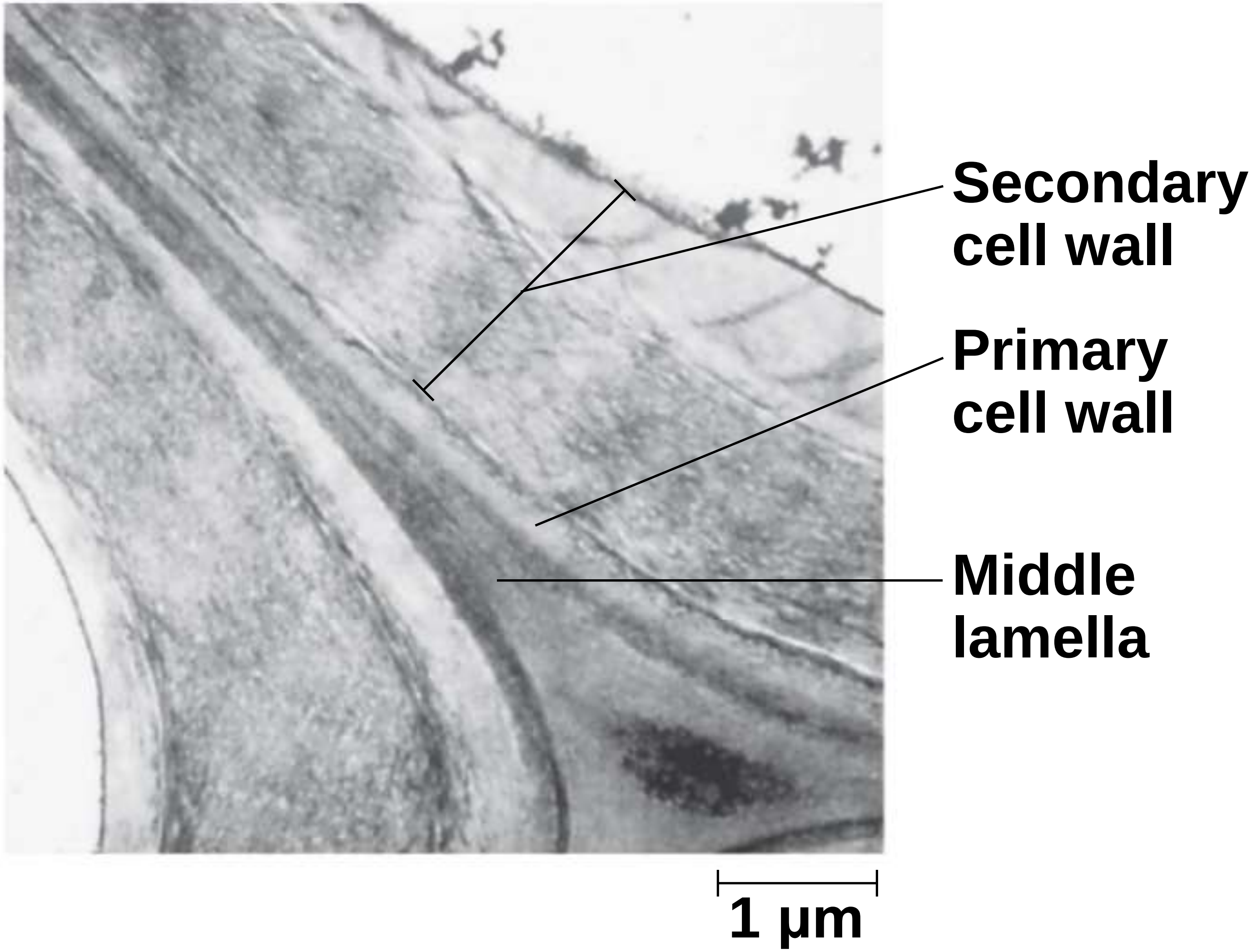
b) Plasmodesmata are channels between adjacent plant cells

Figure 6.27



Secondary cell wall
Primary cell wall
Middle lamella
1 µm
Central vacuole
Cytosol
Plasma membrane
Plant cell walls
Plasmodesmata

Figure 6.27a



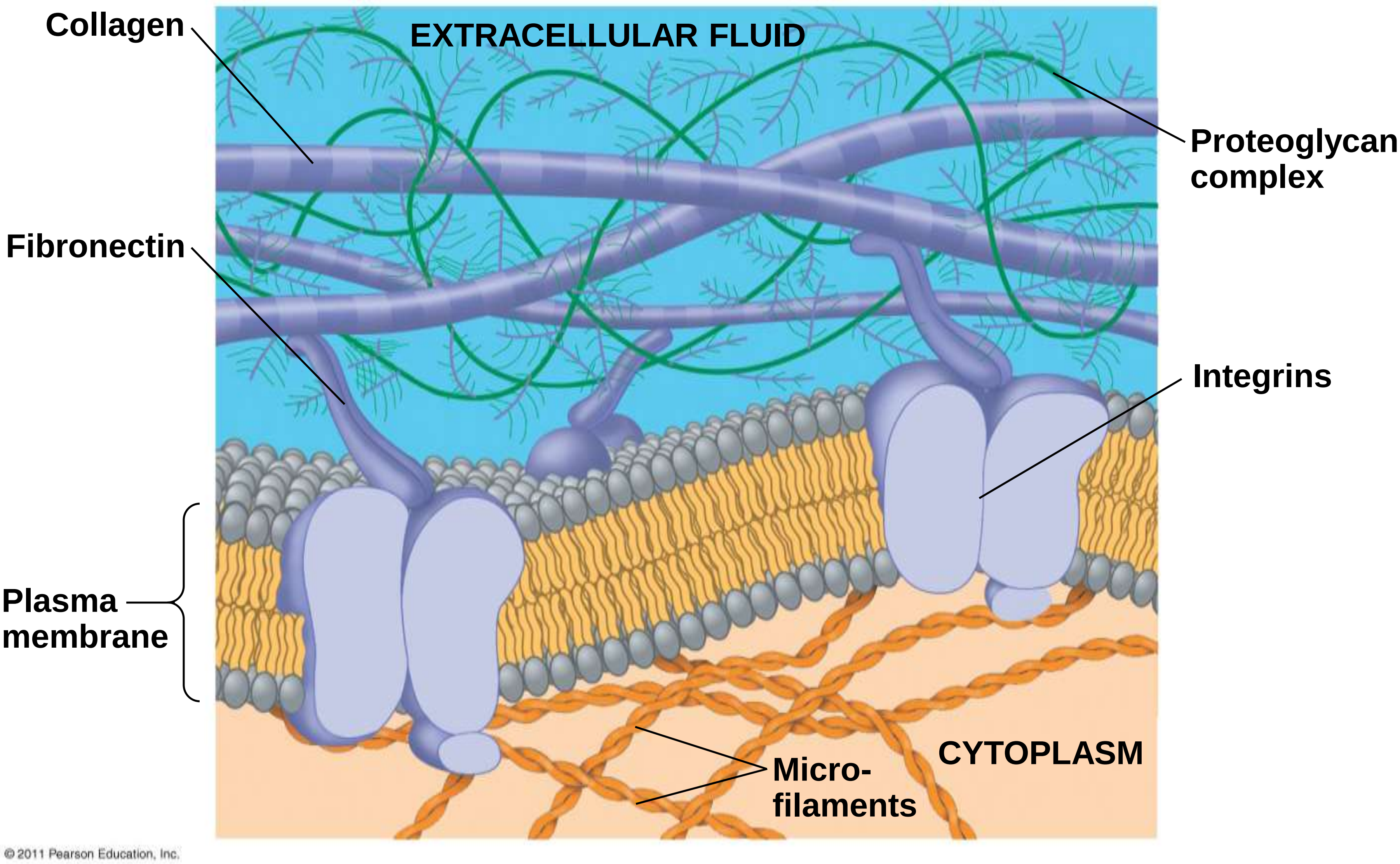
The Extracellular Matrix (ECM) of Animal Cells

Animal cells lack cell walls but are covered by an elaborate **extracellular matrix (ECM)**

The ECM is made up of glycoproteins such as **collagen**, **proteoglycans**, and **fibronectin**

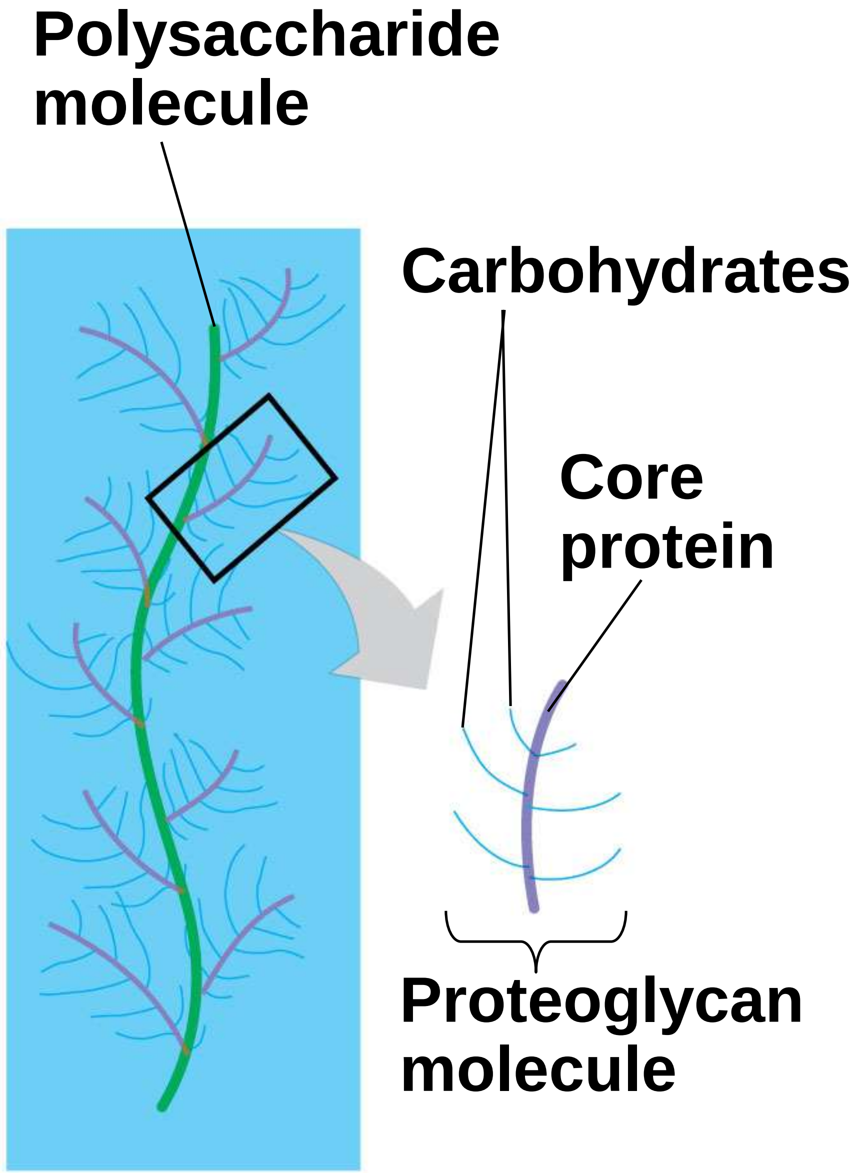
ECM proteins bind to receptor proteins in the plasma membrane called **integrins**

Figure 6.30a



Collagen
EXTRACELLULAR FLUID
Proteoglycan complex
Fibronectin
Integrins
Plasma membrane
Micro-filaments
CYTOPLASM
© 2011 Pearson Education, Inc.

Figure 6.30b



Proteoglycan complex

Functions of the ECM

- Support
- Adhesion
- Movement
- Regulation

Cell Junctions

Neighboring cells in tissues, organs, or organ systems often adhere, interact, and communicate through direct physical contact

Intercellular junctions facilitate this contact

There are several types of intercellular junctions

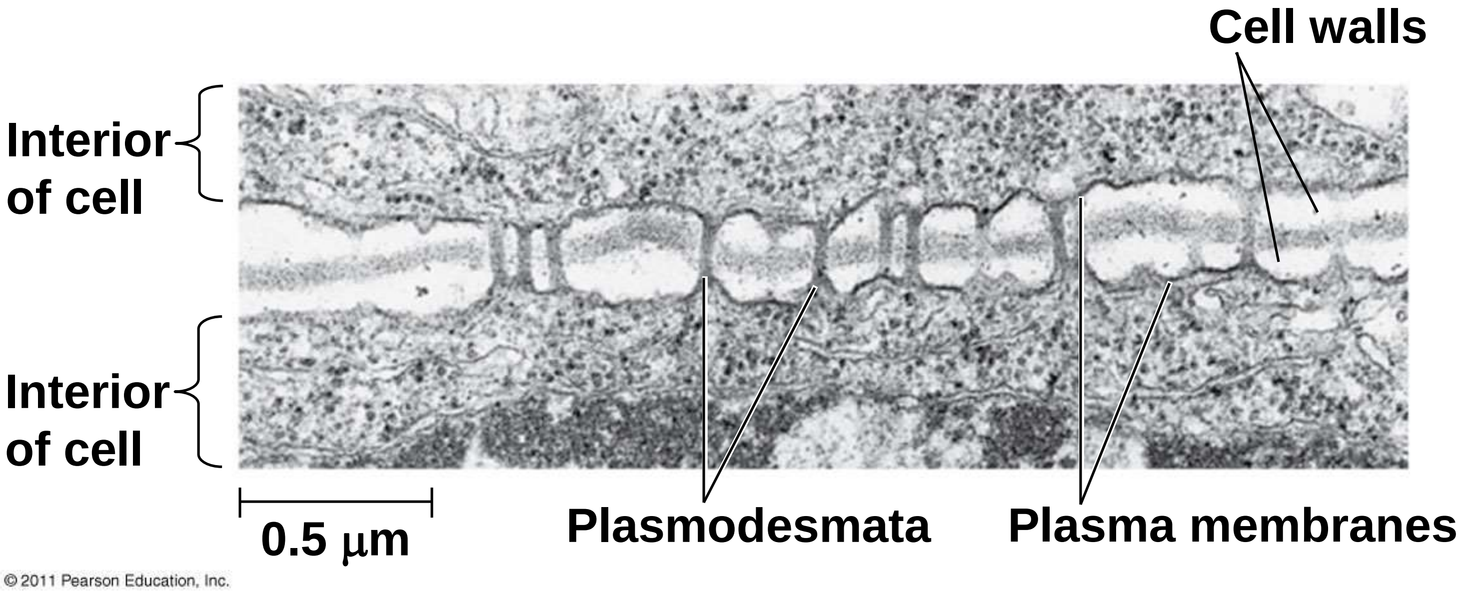
- Plasmodesmata
- Tight junctions
- Desmosomes
- Gap junctions

Plasmodesmata in Plant Cells

Plasmodesmata are channels that perforate plant cell walls

Through plasmodesmata, water and small solutes (and sometimes proteins and RNA) can pass from cell to cell

Figure 6.31



Cell walls
Interior of cell
Interior of cell
0.5 μm
Plasmodesmata
Plasma membranes
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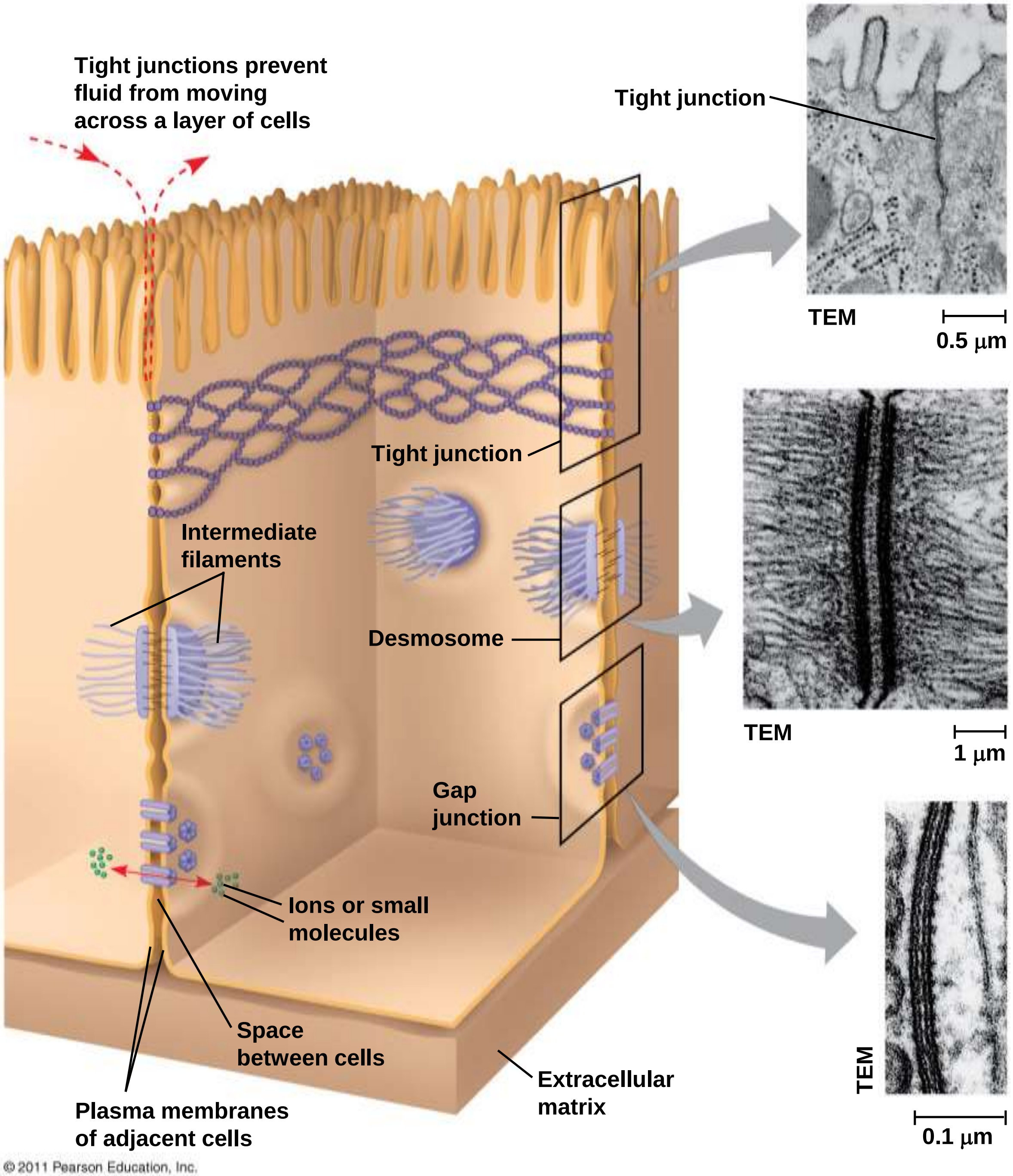
Tight Junctions, Desmosomes, and Gap Junctions in Animal Cells

At **tight junctions**, membranes of neighboring cells are pressed together, preventing leakage of extracellular fluid

Desmosomes (anchoring junctions) fasten cells together into strong sheets

Gap junctions (communicating junctions) provide cytoplasmic channels between adjacent cells

Figure 6.3?



Tight junctions prevent fluid from moving across a layer of cells
Tight junction
Tight junction
TEM
0.5 µm
Intermediate filaments
Desmosome
TEM
1 µm
Gap junction
Ions or small molecules
Space between cells
Extracellular matrix
TEM
0.1 µm
Plasma membranes of adjacent cells
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The Cell: A Living Unit Greater Than the Sum of Its Parts

- Cells rely on the integration of structures and organelles in order to function
- For example, a macrophage’s ability to destroy bacteria involves the whole cell, coordinating components such as the cytoskeleton, lysosomes, and plasma membrane